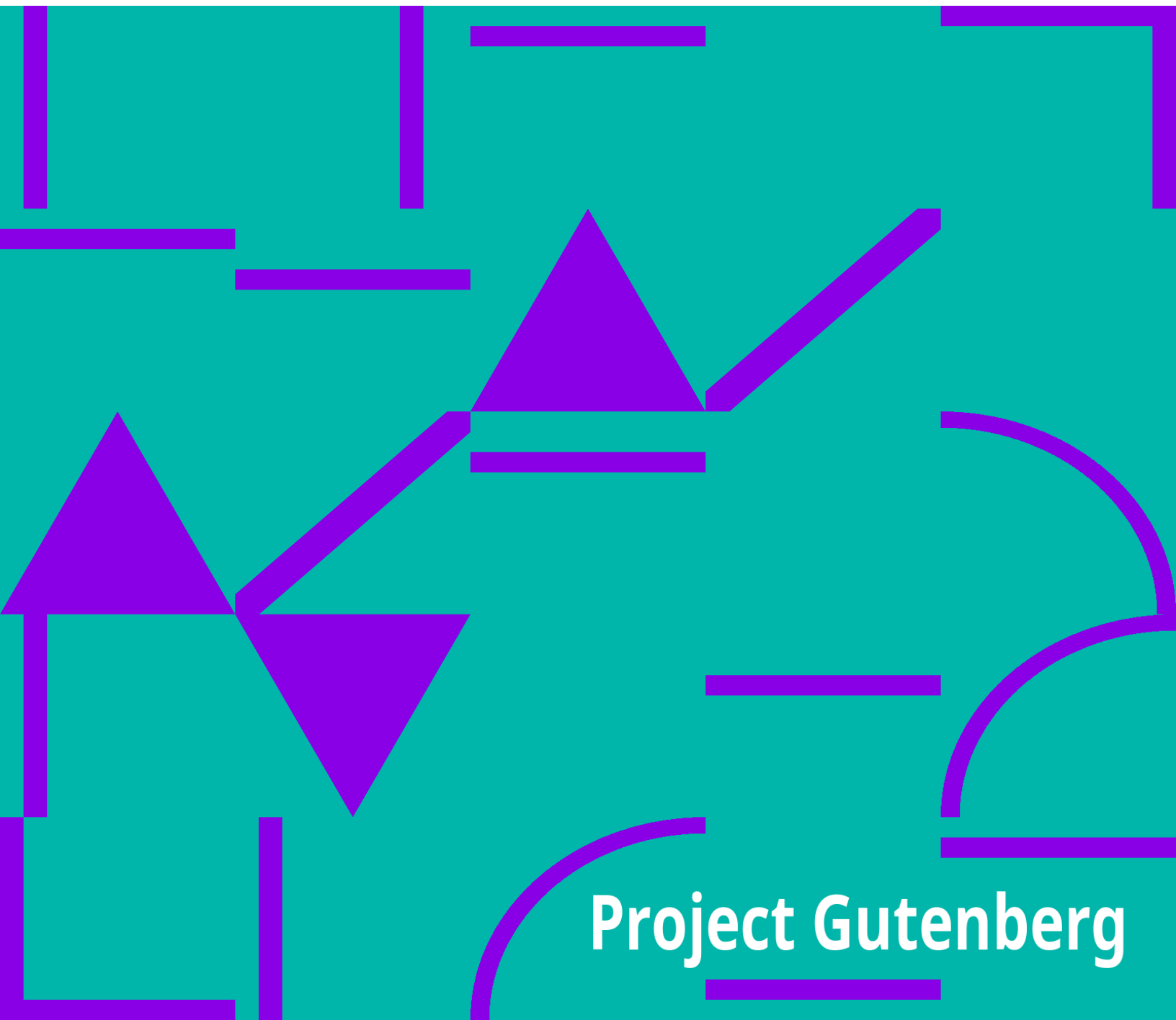


The Pecan and its Culture

H. Harold Hume

The background of the lower half of the image is a solid teal color. Overlaid on this are various geometric shapes in bright yellow and orange. These include several triangles of different sizes and orientations, some horizontal and some vertical lines, and two large, thick, curved lines that resemble stylized parentheses or brackets. The shapes are scattered across the teal background, creating a modern, abstract design.

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The Pecan and its Culture

H. Harold Hume



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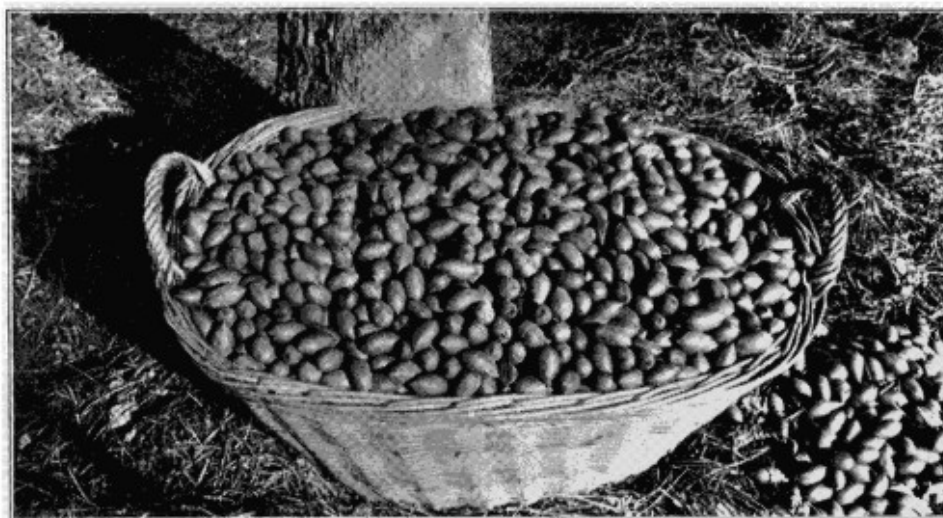
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*** START OF THE PROJECT GUTENBERG EBOOK THE PECAN AND
ITS CULTURE ***



**Pecan Nuts—uniform in size, color and shape. Variety,
Curtis.**

THE PECAN

AND

ITS CULTURE

BY

H. HAROLD HUME

PETERSBURG, VIRGINIA
THE AMERICAN FRUIT AND NUT JOURNAL
1906

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By
H. HAROLD HUME.

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PREFACE.

In the horticultural development of the country, new fruits, new groups of fruits, new fruit industries are coming into prominence. Our native fruits in particular are now receiving, in many parts of the country, a larger share of the attention which they have always merited, and none has proven itself more worthy of careful study and painstaking care than the pecan.

Within the last ten or fifteen years it has rapidly emerged from a wild or semi-wild condition to the status of an orchard nut. The foundations of its culture were laid a considerable time ago, but only now is it coming to its own, its well merited standing among the fruits of the country.

In any horticultural industry many questions must be asked of the plant, the soil, the climate, in short, of the plant in its environment. They must be answered aright, if the industry is to succeed. The newer the plant in cultivation, the more numerous the questions are, the more difficult to answer.

In an endeavor to aid in solving some of the problems connected with the culture of the pecan this small volume has been prepared. Pecan culture has been the subject of careful study, observation and experimentation on the part of the author for a number of years and the results of these studies are presented in the following pages.

To the many who have so kindly and willingly assisted in its preparation, my thanks are herein expressed.

H. HAROLD HUME.

Raleigh, N. C.,
Aug. 1, 1906.

PART I.

Introduction. Botany.

THE PECAN AND ITS CULTURE.

CHAPTER I.

COMMERCIAL AND ORNAMENTAL IMPORTANCE OF THE PECAN.

In all-around excellence, the pecan is equalled by none of the native American nut-bearing trees and certainly it is surpassed by no exotic species. It stands in the list of nut trees with but few equals and no superiors. With this fact known and admitted by all, it seems reasonable to suppose that the pecan will be grown and cultivated much more extensively than it now is. Its intrinsic worth deserves a large share of attention, more than it has received. At present it is gaining a position of so much importance as an orchard tree, that, ere long, it will become an extremely important item in the horticultural wealth of the Southern and Southwestern States.

Large quantities of pecans are sold in the American markets. These are the product of uncultivated or forest trees. Many orchards of considerable size, planted with meritorious budded and grafted varieties, are now in bearing, but the product of these plantings is entirely used by what may be termed a private trade, either by seedsmen, or by private individuals for dessert purposes. Some day, varieties of pecans will become known in the markets just as varieties of grapes, apples or pears are known. People ask for Niagara or Concord grapes, Northern Spy or Greening apples, Bartlet or Seckel pears—ask for what they want, and know what they are getting. The day is far distant when Frotscher, Schley, San Saba, Curtis, Georgia or other varieties of pecans will be known by name by the purchasing public, asked for in the markets and recognized when procured. But that time must and will come, and until then there is no danger of the industry being overdone, and not even then, because our population is constantly growing; because the pecan nut is being put to a variety of new uses, and as yet the export trade is comparatively undeveloped. (See table, page 15.) It would seem then that the pecan might reasonably be expected to replace to a certain extent the foreign nuts in our own markets.

According to the investigations of Woods and Merrill,[A] the pecan has a higher food value than either the walnut, filbert, cocoanut, almond or peanut. The results of their analyses are as follows:

EDIBLE PORTION.

	Edible Portion.	Water.	Protein.	Fat.	Carbohydrates.	Ash.	Fuel Value per Pound. ^[A]
	per cent.	pr ct.	pr ct.	pr ct.	pr ct.	pr ct.	Calories
Pecans, kernels	100.0	2.9	10.3	70.8	14.3	1.7	3445
Walnuts, kernels	100.0	2.8	16.7	61.4	14.8	1.3	3305
Filberts, kernels	100.0	3.7	15.6	65.3	13.0	2.4	3290
Cocoanuts, shred'd	...	3.5	6.3	57.3	31.6	1.3	3125
Almonds, kernels	100.0	4.8	21.0	54.9	17.3	2.0	3030
Shelled Peanuts	100.0	1.6	30.5	49.2	16.2	2.5	2955



Plate II. An Avenue Shaded by Pecan Trees.

It is a fact worthy of note that the average man requires 3,500 calories of energy each day, an amount which must be secured from food consumed. One pound of pecan kernels, according to the above analysis, would supply 3,445 calories, or only 55 calories less than the amount required per day. We are not, be it understood, pointing out this fact because we believe that the pecan alone would be a satisfactory food, though it is wholesome, nourishing and palatable and should be used in larger quantities than is usually the case, but simply to emphasize its high food value.

According to the foregoing analysis, the pecan is richer in fat than any of the other nuts. Seventy per cent. of the kernels is fat. The pecan may at some time be in requisition as a source of oil—an oil which would doubtless be useful for salad purposes—but it is never likely to be converted into oil until the present prices of the nuts are greatly reduced.

If we turn from the dietary value of the nut to the ornamental value of the tree, we cannot but be forcibly impressed with its value as a shade and ornamental tree. For these purposes it may be planted far outside the area in which fruit may be reasonably expected. If given good soil and sufficient food supply, it grows quite rapidly, making a stately, vigorous, long-lived tree. In its native forests it is a giant tree, sometimes reaching a height of upwards of two hundred feet with a trunk of six feet. Isolated specimens, grown in the open, come to maturity with wide-spreading branches and the whole tree has an exceedingly graceful appearance. Wherever it will succeed, no other shade tree is so worthy of attention as the pecan, and in the fruiting area, beauty and healthful shade may be combined with utility.

As an orchard tree it is well worth planting. The ground in which the trees are planted may be cultivated in other crops for a number of years, thus reducing to a minimum the cost of maintaining the planting, and when the trees have come into bearing, the same area in trees will yield more in net returns than the same area in cotton or corn at the usual market prices.

On the whole, considered from whatever standpoint we may choose, the pecan is a valuable tree, whether cultivated for its nuts or planted for shade or ornamental effect.

Exports of Nuts from United States for Years 1900-1904 inclusive.

1900	1901	1902	1903	1904
------	------	------	------	------

VALUE. VALUE. VALUE. VALUE. VALUE.
 \$156,490 \$218,743 \$304,241 \$299,558 \$330,366

Importations of Nuts into the United States for the Years 1899 to 1904 inclusive, according to the most authoritative statistics. ^[B]

		1899		1900		1901	
VARIETY	Quant'y	Value.	Quant'y	Value.	Quant'y	Value.	
OF NUTS.	lbs.		lbs.		lbs.		
Almonds	9,957,427	\$1,222,587	6,317,633	\$949,083	5,140,232	\$946,138	
Cocoanuts...		625,789	...	702,947	...	804,233	
Walnuts	(a)	(a)	(a)	(a)	(a)	(a)	
Other	...	879,166	...	1,326,804	...	1,518,184	
Total Nuts	...	\$2,727,542	...	\$2,978,834	...	\$3,268,255	
		1902		1903		1904	
VARIETY	Quant'y	Value.	Quant'y	Value.	Quant'y	Value.	
OF NUTS.	lbs.		lbs.		lbs.		
Almonds	9,868,982	\$1,240,886	8,142,164	\$1,337,717	9,838,852	\$1,246,474	
Cocoanuts...		832,383	...	908,242	...	971,852	
Walnuts	(a)	(a)	12,362,567	1,106,033	23,670,761	1,729,378	
Other	...	1,971,072	...	1,514,406	...	1,523,462	
Total Nuts	...	\$4,044,341	...	\$4,866,398	...	\$5,471,166	

FOOTNOTES:

^[A] Calculated from analysis.

^[B] Yearbook U.S. Dept. of Agr., 1903, page 686, and 1904, page 728.

CHAPTER II.

NATIVE AND CULTIVATED RANGE.

The pecan is found as a forest tree in the moist bottom lands along the Mississippi river and its tributaries, from Indiana southward to Mississippi, and from Iowa to Texas and Mexico.

This region (see Fig. 1) in which the pecan is, or has been found, native, reaches its northern limit at Davenport, Iowa. It skirts the Wabash as far north as Terre Haute, Indiana, and along the Ohio river nearly to Cincinnati, Ohio. From thence its range extends south to Chattanooga, Tenn., and on to Vicksburg, Miss. From Vicksburg it skirts the Gulf of Mexico at a distance of seventy-five to one hundred miles to Laredo, Texas; thence along the Salado river into Mexico. The western boundary embraces the headwaters of the Colorado river and returns more or less directly to Davenport, Iowa. On the outskirts of this area, it extends farthest in all directions along the streams and rivers, while on the drier intervening ground the line does not extend so far from the center of the region. Particularly is this true in Southwestern Texas, where the pecan is confined almost solely to river bottoms.

CULTURAL AREA.

The area in which the pecan is cultivated as an orchard tree is not confined to the limits of its native range. Plantings have been made outside its native home in New Mexico, California and Oregon in the West, and in Virginia, North Carolina, South Carolina, Georgia, Florida, Southern Alabama and the Gulf regions of Louisiana, Mississippi and Texas. In many other States experimental plantings have been made. Leaving these out of consideration, however, it will be seen that in about twenty States the pecan is either found as a native tree in the forests or is cultivated in orchard form. The area corresponds in some measure with that in which cotton is grown, though it extends farther north and west than the cotton region.



Fig. 1. Approximate Pecan areas. Native areas within solid line. Cultural area within dotted line.

The attempts which have been made from time to time to cultivate the pecan in the more northerly States have not proved successful. The tree has, in many cases, grown well, but fruit has not been produced. The pistils and stamens of the pecan are not found in the same flower but in different flowers borne some distance apart on new and one-year-old wood, respectively. Consequently, it frequently happens that the flowers are not matured at the same time, as a result of which pollination cannot take place. Moreover, late spring frosts often destroy one or both sets of flowers, and the result, as far as fruit is concerned, is the same in either case. As a result of these experiences, the pecan cannot be recommended as a nut-bearing tree north of its natural range in the Mississippi Valley, neither will it succeed at the high elevations in the Alleghany mountains. It reaches its most northerly cultural extension in the Mississippi valley and in the coastal plain of the Atlantic seaboard. But it grows well and makes a good shade tree farther north, and at elevations far above its native range. Even then, however, the nuts from which these seedling shade trees are grown should be brought from the northern sections of its natural distribution. They are much more likely to withstand the rigorous cold of winter.

Frequently the question is asked as to whether the pecan can be grown in a certain given locality. Such a question can be answered only in the most general way. The presence of the larger species of hickories in the vicinity may be used in some parts of the country as an indication of the success which

might attend the planting of pecan trees, but such a guide should not be followed too implicitly, and even if the pecan tree should grow well, fruit might not be secured.

The presence of pecan trees, single specimens perhaps, or two or three, in yards or about buildings here and there throughout a region, may be taken as a guide in the matter of planting, and no better can be had. Nothing will take the place of a practical demonstration in the way of a vigorous fruiting tree.

CHAPTER III.

PECAN BOTANY.

The aborigines of the country used hickory nuts of different kinds as food, and in the region in which the pecan grows as a native tree, it was valued by them above all its relatives.

Penicaut found in his travels that the Indians stored large amounts of pecans for winter use. The scientific name of the pecan is appropriately derived from two Indian words, "powcohiccora" and "pacan."

In 1785, the pecan was described under the name *Juglans Pecan*, by Marshall in his *Arboretum Americanum*. In 1818, Thomas Nuttall, an English botanist, separated the hickories from the walnuts and butternuts, putting them under a new genus which he called *Carya*, naming the pecan *Carya olivaeformis*. Nuttall's classification was followed for many years until it was found that in the year previous to the publication of his work, 1817, C. S. Rafinesque, a French naturalist, had separated the hickories along the same lines as Nuttall and published them under the name *Hicoria*. In accordance with the laws of priority, Rafinesque's name, *Hicoria*, takes precedence over *Carya*.

The family *Juglandaceæ*, embraces but two genera, *Juglans* and *Hicoria*, the former including the walnuts and butternuts, and the latter the pecan and other hickories. With the exception of the Shellbark hickory, *Hicoria ovata* Britton, and the Big shellbark, *Hicoria laciniosa* Sargent, the pecan is the only one of the genus worthy of cultivation.

Family. Juglandaceæ Lindl. Nat. Syst. Ed. 2, 180. 1836. Trees with alternate pinnate leaves and monœcious bracted flowers. Staminate flowers in long, drooping catkins, provided with three or more stamens and occasionally with an irregular-lobed perianth adnate to the bractlet and a rudimentary ovary. Anthers erect, with short filaments, two-celled; dehiscent longitudinally. Pistillate flowers bracted with a three to five, normally four-lobed calyx and sometimes with petals. Ovule solitary, erect, styles two, stigmatic along the inner surface. Fruit a bony nut, incompletely two to four-celled. Seed large, two to four-lobed, cotyledons corrugated, oily, without endosperm.

Genus. *Hicoria* Raf. Med. Rep. (II) 5:352. 1808. Trees, with close or scaly bark, odd-pinnate leaves and serrate leaflets. Staminate flowers in slender drooping catkins, borne in groups of three, occasionally on the new shoots, but usually from buds just back of the terminal buds on last year's shoots, calyx naked, adherent to the bract, unequally two-third lobed or cleft; stamens with short filaments, three to ten in number. Pistillate flowers, two to eight, produced on a terminal peduncle, calyx four-parted, petals none, styles two to four, short, papillose. Fruit oblong, or obovoid, the husk separating into four parts; nut smooth or angled, bony, incompletely two to four-celled. Seed oily, sweet, edible or bitter and astringent. Natives of eastern North America and Mexico.

Species. *H. Pecan* (Marsh.) Britton. Bull. Torr. Club, 15:282. 1888. Pecan, Illinois nut, a large tree, 75 to 170 feet in height and a diameter reaching 6 feet, with rough-broken bark. Young twigs and leaves pubescent, later nearly or quite glabrous; leaflets seven to fifteen, falcate, oblong—lanceolate, sharp-pointed, serrate, green and bright above, lighter below; staminate catkins five to six inches long, sessile or nearly so, sometimes borne near the base on the young shoots but usually from the uppermost lateral buds on last year's shoots; pistillate flowers terminal on shoots of the current season's growth, produced singly or in clusters of two to nine; fruit oblong cylindrical; husk four-valved; nut $\frac{3}{4}$ to 2- $\frac{1}{2}$ inches in greatest diameter, roundish, or cylindrical and pointed, two-celled at the base, partition thin, bitter, seed deliciously sweet. Found native on the moist bottom lands along streams from Indiana south to Kentucky and from Iowa south to Texas, principally along the Mississippi and its tributaries, the Colorado river in Texas, and along some of its tributaries into Mexico.



Plate III. Pecan Flowers. Pistillate enlarged below.

POLLINATION.

Since two kinds of flowers are produced on the pecan, one bearing the pistils, the other stamens, the pollen must be transferred from the latter to the former in order that pollination may take place. In many plants the pollen is transferred from one plant or flower to another by means of insects; but in the pecan there are no bright colors, no nectar, no scent to attract insects to carry pollen, but, instead, the wind is the carrying agent and it needs no attractions.

Pollen is produced in large quantities, necessarily so, since much of it is wasted.

Unfavorable weather conditions at time of blooming may, however, interfere seriously with pollination. Heavy winds or wind-storms, and rains of several days duration, may prevent the necessary and desired distribution of the pollen, as a result of which no fruit is formed.

Sometimes the staminate blooms are destroyed by frost while the pistillate ones escape. It makes little difference which is destroyed, however, as in either case the result is the same—no fruit sets.

The staminate flowers push out from the lateral buds at the same time the new shoot develops from the terminal one. The pistillate blossom does not appear until the terminal shoot has grown six or eight inches, and in the meantime it is protected by the unfolded leaves. The staminate bloom, on the contrary, is exposed from the first, having no leaves to protect it. In consequence it is much more likely to be cut off by frost. Dr. Trelease refers to several observations on proterandry (maturing of the pollen before the stigmas of the pistils) in the pecan. This, together with the unprotected condition of the staminate blooms, we believe, accounts in a large measure for the non-setting of fruit on the northern boundaries of the pecan area.

The artificial or hand pollination of the pecan is an easy matter and offers an inviting field for those interested in plant breeding. Emasculation, or the removal of the stamens from the flowers necessary in breeding so many plants, is not necessary in the pecan. All that is needed is to cover the pistillate blossoms with a sack until they are matured. At this time the inner or stigmatic surfaces of the pistils will be exposed and ready for the pollen. The pollen, collected from adjoining trees in bloom or brought from a distance, can then be placed upon the stigmas and the sack replaced. When the fruit is set, the paper sack should be replaced by one of mosquito netting. Some careful work has already been done along this line, and it is hoped that many more will take up the work. Much yet remains to be desired, and varieties may be better adapted to different sections. The ideal, large, full-meated, thin-shelled, prolific and precocious variety of pecan has not yet been brought forward. It may be accidentally discovered; it may be produced and can be produced by systematic, painstaking work in breeding. It is hoped that the number of workers in this inviting field may be increased. Some may be deterred by the fact that it will take the seedlings so long to come into bearing. But scions may

be taken from the seedlings raised from cross-bred nuts, top-worked on large trees, and fruit could be obtained in many cases in a period not exceeding five or six years from the seed. Those which would not produce fruit in six years in this way might perhaps as well be discarded.

PART II.

Varieties.

CHAPTER IV.

VARIETIES OF PECANS.

While the list of varieties of pecans is comparatively small, yet a surprisingly large number of names has been used. The attempt has been made to collect all the names which have appeared in different publications. These have presumably all been applied to some pecan at some time or other, but many of them have never been propagated by budding or grafting and a very large proportion of them have been lost track of entirely. In short, they are now represented by names only. However, they are all given, for the reason that it would be well not to apply any of these names to other varieties. It might be well to emphasize the fact that many meritorious varieties would be the better for re-naming.

In the original descriptions, it will be noted that the thickness of the shell is given in millimeters. A piece of the shell about the center of the side covering the back of the half kernels, was accurately measured. These measurements must not be regarded as absolute, but they are comparative. All nut illustrations are natural size.

For the origin and synonymy of many varieties credit must be given to the excellent work of Mr. William A. Taylor, of the United States Department of Agriculture, who has probably done more than any one else to straighten out the tangled nomenclature of the pecan.

CLASSIFICATION OF VARIETIES.

Heretofore, no attempt has been made to group or classify the different varieties of pecans. Classification does not become necessary until the number of varieties has increased sufficiently. The following classification of the varieties with which the author is acquainted, is based entirely upon the shape of the nuts. No classification of those varieties of which descriptions are copied has been attempted, as the descriptions are frequently so meagre as to render it impossible:

1. *Varieties*: Hound or roundish oblong. Types—Post, Hollis, Money-maker.

Bacon, Bolton, Extra Early, Georgia, Hollis, Money-maker, Post, Randall, San Saba, Thomas.

2. *Varieties*: Oblong, rounded at the base, blunt and quadrangular at the apex. Types—Pabst, Success.

Frotscher, Pabst, Pegram, Perfection, Success, Sweetmeat.

3. *Varieties*: Oblong in general outline, rounded, blunt and abruptly tipped at the base, and abruptly short-pointed at the apex. Types—Russell, Stuart.

Alley, Carman, Capital, Franklin, Havens, Jacocks, James No. 1, Kincaid, Lewis, Moore, Morris, Russell, Stuart.

4. *Varieties*: Oblong cylindrical to almost conical, rounded at the base, sloping from the middle or above to the sharp-pointed apex. Types—Jewett, Curtis, Schley.

Clarke, Curtis, Daisy, Dalzell, Dewey, Hume, James' Giant, Jewett, Kennedy, Mammoth, Rome, Schley, Young.

5. *Varieties*: Usually long in proportion to thickness, more or less pointed at both base and apex. Types—Atlanta, Ideal, Schaifer.

Atlanta, Centennial, Delmas, Domestic, Ideal, James' Paper-shell, Ladyfinger, Longfellow, Louisiana, Monarch, Money, Schaifer, Van Deman.

6. *Hybrid Varieties*: Nussbaumer, McCallister, Schneck, Pooshee, Westbrook.

ALBA. Size below medium, cylindrical, with pointed apex; cracking quality good; shell of medium thickness; corky shell lining thick, adhering to the kernel; kernel plump, light colored; quality good. (Report Sec. Agr., 1893: 295, 1894).

ALLEY. Size medium, 1-5/8 x 7/8; form ovate; color grayish-brown with a few purplish-black markings about the apex; base rounded, tipped; apex abruptly short-pointed, slightly four-angled; shell

brittle, thin, .8 mm.; partitions thin; cracking quality excellent; kernel full, plump, bright straw-colored, sutures narrow, moderately deep, secondary sutures slightly marked; texture firm, compact fine grained; flavor sweet, delicate, pleasant; quality very good and a good keeper.

Described from specimens received from Mr. Theo. Bechtel, Ocean Springs, Miss.

ATLANTA. Size medium, 1-7/8 x 7/8 x 11/16 inches; ovate, compressed; color dull gray liberally specked with small, dark dots, splashed with purplish markings from middle to apex; base sloping, blunt-pointed; apex sloping, short-pointed; shell brittle, moderately thin; partitions rather thick, corky; cracking quality quite good; kernel full, plump, sutures narrow of medium depth, secondary sutures lacking; color light yellowish-brown, bright; texture solid, compact; flavor sweet, good; quality very good.

Originated by G. M. Bacon, DeWitt, Ga., and first catalogued about 1900.

BACON. (Syn.: *Bacon's Choice*.) Size small, 1-1/4 x 7/8 inches; rounded, compressed toward the apex; color dull brownish-gray, thickly dotted with dark specks, liberally splashed with purplish-brown markings toward the apex; base rounded; apex abruptly blunt-pointed; shell thin, .85 mm.; cracking quality excellent; partitions thin, papery; kernel roundish, bright, light brownish-yellow, plump, full, smooth, sutures broad, of medium depth; flavor sweet, nutty, good; quality very good.

A small pecan of good quality, originated by G. M. Bacon, DeWitt, Ga., and introduced by him in 1900.

BARTOW. Medium size, thin shell and fine flavor. (Bacon's Cat., page 29, 1904.)

BEAUTY. Illustrated in "The Pecan and How to Grow It." (Stuart Pecan Co., 1893, p. 59, fig. 5.)

BELLE. Medium, ovate, quality very good. (J. V. Munson, Farm and Ranch, Dec. 3, 1904, p. 2.)



Fig. 2.
Money-
maker, Post,
San Saba,
Half Kernel
of Bacon

BIEDIGER. Listed in "Nut Culture in the United States," U. S. Dept. Agr., Div. Pomology, 64, 1896.

BILOXI. (W. R. Stuart, Ocean Springs, Miss.) Medium size, cylindrical, pointed at each end; surface quite regular, light brown; shell thin; cracking quality medium; kernel plump, with yellowish-brown surface; free from astringency, of good quality, and keeps well without becoming rancid. Introduced several years ago by W. R. Stuart as Mexican Paper-shell, but the name has since been changed to Biloxi. (Report Sec. Agr., 1893. 295, 1894).

BLACK JACK. Listed in "Nut Culture in the United States," U. S. Dept. Agr., Div. Pomology, 64, 1896.

BOLTON. Size medium, 1-3/8 x 1 inches; ovate conical; color dull gray marked with purplish-brown blotches about the apex; base rounded; apex angled, blunt, sloping gradually from the center; shell thick, 1.9 mm.; partitions thick; cracking quality medium; kernel brownish-yellow, somewhat wrinkled; sutures broad, deep, inner surface wrinkled, broadly oval in outline, texture rather open; flavor sweet, nutty; quality good.

Originated in Jefferson county, Florida. Described from specimens received from J. H. Girardeau, Monticello, Fla.

BRACKETT. Named for our U. S. Pomologist. It is a very fine market pecan, unexcelled in richness of flavor, and has a thin shell. Trees are fine growers, heavy bearers, and with proper care and attention come into bearing at six years old. (Bacon's Cat., 1900).

BRADLEY. Large, oblong, ovoid, shell thin, kernel plump, best. (J. V. Munson, Farm and Ranch, Dec. 3, 1904, p. 2.)

BRIDEX. Listed in "Nut Culture in the United States," U. S. Dept. Agr., Div. Pomology, 64, 1896.

BULLETS. A decided novelty in pecans. As its name indicates, it is of bullet shape, being almost perfectly round. It has a fine flavor, shell is very thin. (Bacon's Cat., 1900).

CAPITAL. Size medium to large, 1-7/8 x 7/8 x 3/4 inches; ovate oblong, compressed with well-marked sutures; color light-brown streaked and splashed with purplish-brown markings from center to apex; base rounded, blunt-tipped; apex abruptly short-pointed, nipped; shell brittle, of medium thickness, 1.3 mm.; partitions of medium thickness; cracking quality very good; kernel plump, filling the shell, brownish-yellow in color, primary sutures broad and fairly deep, secondary ones well defined, running almost the length of the kernel; texture rather open; flavor good; quality good.

Described from specimens received from Mr. Theo. Bechtel, Ocean Springs, Miss.

CARMAN. Size medium, 1-7/8 x 3/4 inches; oblong, compressed; color light yellowish-brown marked with splashes and blotches of brownish-black about the apex; base rounded, blunt-tipped; apex abruptly-pointed, shouldered and four-angled; shell brittle of medium thickness, 1.2 mm.; partitions thin; cracking quality very good; kernel long, slender, plump, straw-colored, sutures straight, narrow, shallow; texture firm, compact; flavor sweet, pleasant; quality very good.

Described from specimens received from Prof. F. H. Burnette, Baton Rouge, La. Originated and introduced by Mr. S. H. James, Mound, La.

CENTENNIAL. Size large, 2 x 7/8 x 3/4 inches; oblong, compressed, constricted in the middle, with well marked sutures; color grayish-brown, bright, marked with a few purplish markings in the grooves at the apex; base tapering to a blunt point; apex tapering, pointed, wedge-shaped, sometimes curved; shell medium thick, 1.5 mm.; partitions thin; cracking quality medium; kernel plump, full, brownish-yellow, bright, sutures rather small, straight, secondary ones marked by a line, surface rather wrinkled; flavor sweet, delicate; quality very good.

Described from specimens received from Mr. J. F. Jones, Monticello, Fla. "The original tree stood on the Anita plantation of Mr. Amant Bourgeois, on the east bank of the Mississippi river in St. James Parish, La."^[C] It was destroyed March 14, 1890, by the Anita Crevasse. Sixteen trees were grafted in 1846 and 1847 by the slave gardener, Antoine, of Mr. Telesphore J. Roman, owner of Oak Alley plantation. Two of these earlier trees are still standing. Nuts were exhibited at the Centennial Exposition, Philadelphia, in 1876, by Hubert Bonzano. Under the name Centennial, it was probably first catalogued by the late Richard Frotscher, of New Orleans, in 1885.

CHIQUITA. Small, ovate, shell medium, best, long keeper. (T. V. Munson, Farm and Ranch, December 3, 1904, p. 2.)

CLARK. Size medium to large, 1-3/4 x 7/8 inches; ovate oblong; color dull gray, with a few purplish spots about the apex; base rounded; apex blunt; shell brittle of medium thickness, 1.3 mm.; cracking quality medium; partitions thick, corky; kernel full and plump with narrow sutures of medium depth, light yellow in color and marked here and there with black dots; texture rather open; flavor good; quality good.

Obtained of J. H. Girardeau, Monticello, Fla.

COLORADO. Mentioned by Andrew Fuller in "The Nut-Culturist," 1896, p. 169.

CURTIS. (Syn.: *Curtis No. 2*). Medium, 1-5/8 x 7/8 inches; ovate, conical, compressed; color brownish-gray, marked throughout with dark specks and a few purplish specks about the apex; base rounded; apex sloping, pointed; shell thin, .7 mm.; cracking quality excellent; partitions thin, smooth; kernel bright straw-colored, plump, full, with narrow sutures of medium depth; texture compact, firm; flavor sweet, rich, nutty; quality excellent.

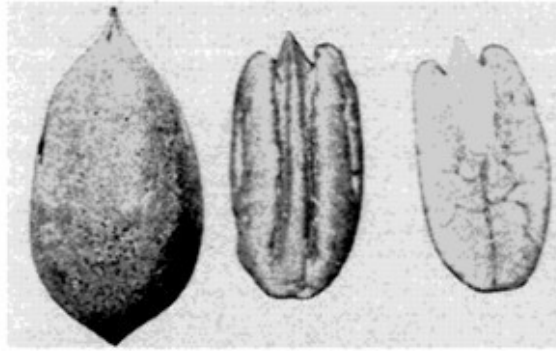


Fig. 3. Curtis Pecan.

The original tree of this variety is to be found in the grove of Dr. J. B. Curtis, Orange Heights, Fla. It was raised from seed secured from Arthur Brown, Bagdad, Fla., and planted in 1886. It is a meritorious variety, being prolific, of good appearance and excellent quality.

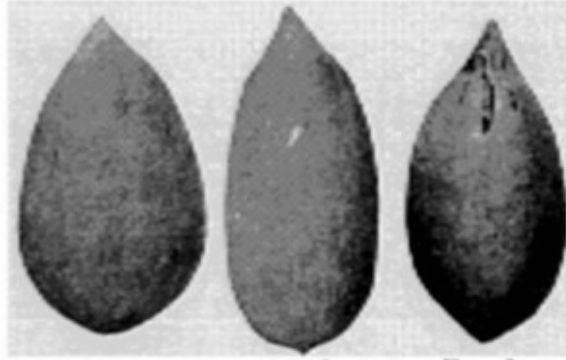
DAISY. Medium to large, $1\frac{7}{8} \times \frac{13}{16} \times \frac{3}{4}$ inches; oblong cylindrical; color reddish-brown marked with a few purplish-brown spots about the apex; base rounded; apex abruptly tapering, rather short; shell brittle, thin .93 mm.; cracking quality fairly good; partitions thick; kernel light brownish-yellow, full, plump, with broad and very shallow sutures; texture firm and compact; flavor sweet, good; quality good.

Obtained of S. W. Peek, Hartwell, Ga.

DALZELL. Large, $2 \times \frac{7}{8} \times \frac{3}{4}$ inches; cylindrical flattened; dull grayish-brown, pebbled, marked with narrow splashes of purplish-brown from center to apex; base rounded; apex abruptly sharp-pointed, four-angled and shouldered; shell rather thick, brittle, 1.4 mm.; cracking quality medium; partitions thin; kernel long, narrow with deep sutures, yellowish-brown in color, texture firm and compact; flavor sweet, good; quality good.

Obtained of S. H. Graves, Gainesville, Fla. The original tree^[D] stands in a 14-acre grove, four miles south of Gainesville. The grove was planted in 1888, by Mr. J. R. Zetrour, now of Rochelle, Fla.

DELMAS. Size large, 1-7/8 x 1 inches; ovate, marked with four distinct ridges; color dull dark gray marked with dark specks and blotches with purplish-black from center to apex; base sloping, rounded, blunt; apex abruptly short-pointed, four-angled; shell thick, brittle, 1.4 mm.; partitions thick, corky; cracking quality good; kernel bright light yellow, sutures broad, open, shallow, secondary ones almost lacking, sometimes slack at bottom end; texture rather open; flavor sweet; quality good.



**Fig. 4. Mammoth. Dalzell.
Kennedy.**

Described from specimens received from Mr. Theo. Bechtel, Ocean Springs, Miss. A large nut of fairly good quality, said in some cases to have been substituted for Schley, from which it is very distinct.

DEWEY. Medium to large, 1-7/8 x 3/4 inches; ovate pointed; color dull gray, marked with splashes of purplish-brown; base rounded; apex sharp; shell brittle and thin, .88 mm.; cracking quality very good; partitions thin; kernel full, plump, smooth, bright light straw-colored, with narrow sutures of medium depth; texture firm and solid; flavor sweet, rich, good; quality very good.

Specimens for description obtained of H. K. Miller, Monticello, Fla. Originated in Jefferson county, Fla.

DEWITT. An oddity, having the shape of a spinning top. Shell is thin, and its rich meat is easily extracted on account of its peculiar shape. (Bacon's Cat. 1900.)

DOMESTIC. Large, 2 x 3/4 inches; oblong ovate, compressed toward the base; color light reddish-brown, with splotches of purplish-

brown throughout; base sloping, pointed; apex four-angled, abruptly blunt-pointed; shell brittle, thin, .95 mm.; cracking quality good; partitions thick, red, corky; kernel brownish-yellow, plump, full, wrinkled on the sides with straight, narrow, deep sutures and secondary ones fairly well developed; texture compact and fine grained; flavor sweet, good; quality very good.

Specimens for description obtained from Frank H. Lewis, Scranton, Miss.

EARLY TEXAN. (Louis Biediger, Idlewild, Tex.) Size above medium, short, cylindrical, with rounded base and blunt conical crown; shell quite thick, shell lining thick, astringent; cracking quality medium; kernel not very plump, of mild nutty flavor; quality good. (Report Sec'y Agr., 1893: 295, 1894.)

EGG. (Syn.: *Eggshell*.) Medium; ovate; shell thin; partitions thin; kernel plump; quality good. D. L. Pierson, Monticello, Fla. Grown from seed procured from Louisiana in 1889. (Hume, Bul. 54, Florida Exp. Station, 203, 1900.)

EXCELSIOR. A variety reported by Ladd Bros., Stonewall, Miss. (Listed in "Nut Culture in the United States," U. S. Dept. Agr., Div. Pomology, 64, 1896.)

EXTRA EARLY. Size medium to large, 1-3/8 x 1 inch; oblong ovoid abruptly-pointed; color grayish-yellow with small purplish blotches more or less over the whole surface; base rounded; apex abruptly-pointed, blunt; shell of medium thickness, 1.15 mm.; partitions of medium thickness; cracking quality good; kernel filling the shell, plump, smooth, sutures broad, open, deep, not clasping the shell, color brownish-yellow, texture open; flavor very good, quality fair.

Described from specimens received from E. E. Risien, San Saba, Texas; not catalogued, so far as we are aware.

FAUST. (O. D. Faust, Bamberg, S. C.) A pecan of large size; very long in shape; quite thin shell; kernel separating readily from shell; quality best. (Report Sec'y Agr., 1891, p. 395: 1892.)

FAVORITA. A variety named and grown at one time by Arthur Brown, Bagdad, Fla. (Listed in "Nut Culture in the United States," U. S. Dept, Agr., Div. Pomology, 64, 1896.)



Fig. 5. Frotscher Pecan.

FRANKLIN. Size medium large, 1-5/8 x 7/8 inches; ovate; color dull grayish-brown splashed about the apex with purplish-black; base rounded, blunt-tipped; apex blunt-pointed, four-angled; shell brittle, of medium thickness, 1.32 mm.; partitions thick; cracking quality good; kernel full, plump, bright brownish-yellow, primary sutures of medium width, deep, secondary ones almost lacking; texture rather coarse, fairly firm and compact; flavor sweet, good; quality very good.

Described from specimens received from S. W. Peek, Hartwell, Ga.

FROTSCHER. (Syn.: *Frotscher's Eggshell*, *Eggshell*, *Olivier*, *Majestic*.) Large, 1-5/8 x 1-7/8 inches; cylindrical, ovate; color bright yellowish-brown, with a few black splashes about the apex; base broad, rounded; .9 mm.; partitions thin; cracking quality excellent; kernel brownish-yellow, dark veined, frequently slack at one end; sutures of medium depth, rather narrow, secondary sutures well marked; texture dry, rather coarse; flavor good; quality fair to medium.

The above description was made from specimens received from the J. Steckler Seed Co., New Orleans, La. The original tree stands in the garden of H. J. Pharr, Olivier, La.; the place was formerly owned by Oscar Olivier. The variety was first propagated by William Nelson, and catalogued as *Frotscher's Eggshell*, by Richard Frotscher, in 1885. The variety is precocious, productive, and succeeds over a wide range of country.

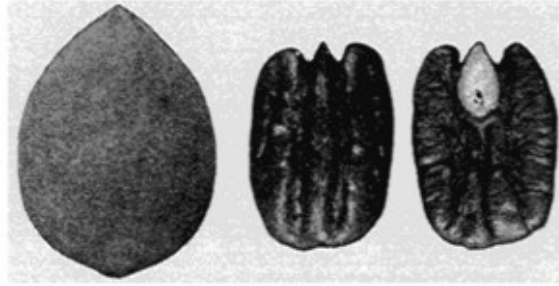


Fig. 6. Georgia Pecan.

GEORGIA. (Syn.: *Georgia Giant*.) Size large, 1-1/2 x 1/8 x 1 inches; rounded ovate; color brownish-gray marked with splashes and dots of dark brown covering a good part of the surface; base rounded; apex tapering, blunt; shell brittle, medium in thickness, 1.3 mm.; cracking quality medium; partitions thick, corky, red; kernel bright reddish-brown, plump, full, rather deeply sutured, two secondary sutures fairly well developed; texture compact, fine grained; flavor sweet, good; quality very good.

Originated and introduced by G. M. Bacon, DeWitt, Ga. Said to be a precocious and prolific bearer.

GEORGIA MELON. Size above medium, short, rather blunt at apex; cracking quality medium, shell thick; kernel plump, brown; meat yellow, moderately tender, pleasant, good. (Report Sec. Agr., 1893, 295: 1894.)

GIANT. Named, and at cue time propagated, by Louis Biediger, Idlewild, Tex. (Listed in "Nut Culture in the United States," 64, 1896.)

GONZALES. (T. V. Munson, Denison, Tex.) Above medium size, with firm, clean shell; quality excellent. Originated in Gonzales county, Tex. (Report Sec. Agr. 1893, 295: 1894.)

GRAFF. Named, and at one time propagated, by Louis Biediger, Idlewild, Tex. (Listed in "Nut Culture in the United States," 64, 1896.)

HALBERT. Very large, oval, shell thick, fair quality. (T. V. Munson, Farm and Ranch, p. 2, Dec. 3rd, 1904.)

HAMILTON. (Syn.: *R. Hamilton*.) Illustrated in Farm and Ranch, Vol. 23, No. 49, p. 1, Dec. 3rd, 1904.

HARCOURT. (Syn.: *Helen Harcourt?*) Size medium, short, slightly acorn-shaped; cracking qualities medium; Shell rather thick, but very smooth inside; kernel short, very plump; meat yellow; very tender; rich; very good. (Report Sec. Agr., 1893, 295: 1894.)

HAVENS. Large. 1-7/8 x 1 x 7/8 inches; ovate, compressed; color dull gray specked and splashed with purplish-brown; base rounded, blunt-tipped; apex abruptly short-pointed, four-angled; shell brittle, thin, .85 mm.; partitions of medium thickness; cracking quality excellent; kernel very plump, full, brownish-yellow marked with dark specks, primary sutures narrow, deep, secondary ones very slightly marked, bottom ends of halves of kernel divided; texture solid, compact, fine grained; flavor sweet, good; quality very good.

Described from specimens received from Frank H. Lewis, Scranton, Miss.

HOLLIS. (Syn.: *Post's Select in part*.) Size medium, 1-3/8 x 1 inches; form roundish ovate, marked with four more or less prominent longitudinal ridges; color dull brownish-yellow, slightly splashed with purplish-brown about the apex; base rounded; apex roundish, blunt; shell thick, 1.6 mm.; partitions thick; cracking quality medium; kernel plump, filling the shell, quite smooth, broadly and deeply grooved, oval in outline, light brownish-yellow in color; texture fine grained; flavor delicate, good; quality good.

Described from specimens received from Herbert Post, Fort Worth, Tex. The seed nuts of this variety have been sold under the name, "Post's Select." It originated at Bend, San Saba county, Texas.

HUME. (Syn.: *Curtis No. 5*.) Size medium, 1-1/2 x 7/8 inches; short, oblong cylindrical, marked with two longitudinal ridges; color grayish-brown marked with a number of short, narrow purplish-brown splashes; base rounded, very blunt-tipped; apex abruptly-pointed, flattened on two sides; shell thin, .8 mm.; partitions medium, corky; cracking quality very good; kernel full, plump, light yellowish-brown, marked and dotted with dark spots, sutures

straight, narrow, of medium depth; texture firm, compact; flavor sweet, pleasant, quality very good.

The original tree of this variety stands in the grove of Dr. J. B. Curtis, Orange Heights, Fla. It was grown from seed secured from Arthur Brown, Bagdad, Fla., in 1886. It is a shy bearer.

IDEAL. Medium, 1-7/8 x 3/4 x 5/8 inches; oblong, somewhat compressed, slightly constricted in the middle; color bright grayish-brown marked with narrow strips of purplish-brown at the apex; base sloping, pointed; apex sloping, pointed; shell thin, brittle, .9 mm.; partitions medium thick; cracking quality good; kernel full, plump, smooth, bright straw-colored, sutures very narrow, shallow; texture compact, firm; flavor sweet, good; quality very good.

Described from specimens received from S. W. Peek, Hartwell, Ga.

IDLEWILD. Medium size, thick shell, kernel good. Louis Biediger, Idlewild, Tex. (Thomas' American Fruit Culturist, 21st ed. 452, 1903.)

JACOCKS. (Syn.: *Jacocks' Mammoth*.) Size large or very large, 1-7/8 x 1 inches; ovate, long; color bright yellowish-brown; base rounded, abruptly blunt-pointed; apex blunt, four-angled, slightly wedged; shell brittle, of medium thickness, 1.3 mm.; partitions very thick, corky, red; cracking quality medium; kernel light yellowish-brown, full or sometimes shrunken, sutures broad, of medium depth, secondary sutures well developed and fairly deep; texture open, rather coarse; flavor sweet, rather dry; quality fairly good.

Introduced by Mrs. C. W. Jacocks, Formosa, Fla., from whom specimens were received.

JAMES GIANT. Medium to large, 2 x 7/8 inches; ovate cylindrical; color brownish-gray, marked with a few purplish splashes about the apex; base rounded; apex abruptly sharp-pointed with four rather prominent ridges; shell thin, 1. mm.; cracking quality good; partitions medium thickness; kernel bright light yellow, with

narrow deep sutures and well defined secondary sutures; texture firm, compact; flavor sweet, good; quality very good.

Obtained of Prof. F. H. Burnette, Baton Rouge, La.

JAMES No. 1. Size large, 2 x 13/16 x 3/4 inches; oblong, ovate, compressed; brownish-yellow in color with a few brownish streaks about the apex; base rounded, blunt-tipped; apex abruptly blunt-pointed, four-angled, nipped; shell thin, .8 mm.; partitions thin; cracking quality very good; kernel straw-colored, usually full and plump, though sometimes shrunken at one end, primary sutures broad, shallow, secondary ones well defined; texture solid, fine grained; flavor very good, sweet; quality very good.

Described from specimens received from Prof. F. H. Burnette, Baton Rouge, La. Originated and introduced by S. H. James, Mound, La.

JAMES PAPER-SHELL. Medium to large, 1-7/8 x 3/4 inches; cylindrical or slightly quadrangular, slender; color yellowish-brown marked with purplish splashes from center to apex; base rounded; apex abruptly-pointed, four-angled; shell thin, .96 mm.; partitions thin; cracking quality very good; kernel sometimes slack at one end, usually plump, smooth, bright brownish-yellow; sutures narrow, shallow; texture firm, compact; flavor very good, sweet; quality very good.

Originated by S. H. James, Mound, La., and described from specimens received from Prof. F. H. Burnette, Baton Rouge, La.

JEWETT. Large, 1-7/8 x 7/8 inches; obovate, flattened, angular, frequently constricted at the middle; color dull reddish-brown, marked with large purplish splashes; base rounded; apex blunt four-angled, frequently curved; shell brittle, thick; cracking quality very good; partitions of medium thickness; kernel bright straw-colored, plump, smooth, somewhat triangular, with broad, open, shallow sutures; texture firm, compact; flavor fair; quality medium.

Obtained of Chas. E. Pabst, Ocean Springs, Miss.

JUMBO. Size large, 1-5/8 x 7/8 inches; ovate, slightly tapering; color grayish-brown marked with a few narrow streaks about the apex; base rounded; apex four-angled, wedged, blunt-pointed; shell brittle, of medium thickness, 1.3 mm.; partitions thick, corky; cracking quality medium; kernel full, plump, straw-yellow in color, primary sutures broad, deep, secondary sutures almost lacking; texture fairly solid, fine grained; flavor sweet, good; quality very good.

Described from specimens received from Summit Nurseries, Monticello, Fla.

KENNEDY. Large, 1-3/4 x 7/8 inches; ovate-conical, flattened; color dull brownish-gray, marked with a few narrow streaks of purplish-black about the apex; base rounded; apex sharp-pointed, flattened on two sides; shell of medium thickness, .98 mm.; cracking quality very good; partitions thin; kernel bright, plump, full, smooth with narrow sutures of medium depth and secondary ones marked by a line; texture firm and compact, flavor rich, sweet; quality excellent.

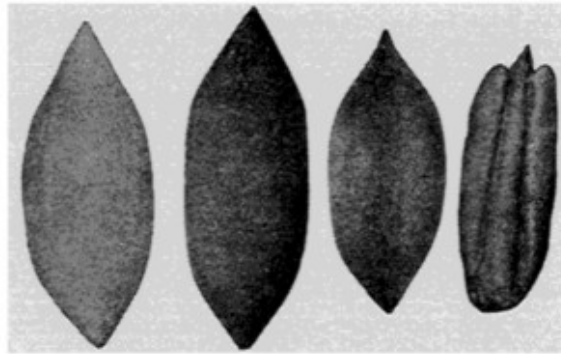
Described from specimens received from Dr. J. B. Curtis, Orange Heights, Fla. Origin similar to Curtis.

KENTUCKY GEM. Listed. (Burnette, F. H., Bul. La. Exp. Station, sec. ser. No. 69, 1902, p. 875.)

KIDD. Illustrated in Farm and Ranch, Vol. 23, No. 49, Dec. 3, 1904, p. 1.

KINCAID. Size medium to large, 1-5/8 x 1 inches; ovate compressed with well defined sutures; color light brownish-yellow, bright, marked with narrow splashes of purplish-black at the apex; base almost flattened, blunt-tipped; apex blunt-pointed, slightly wedged, four-angled; shell brittle, compact, thin, .98 mm.; partitions thick, corky; cracking quality very good; kernel very full and plump, smooth, bright, light straw-colored, primary sutures broad and deep, secondary sutures creased and very shallow; texture fine grained, solid, compact; flavor sweet, rich, good; quality excellent; a good keeper.

Described from specimens received from E. E. Rislen, San Saba, Texas. This apparently is a very good variety of pecan.



**Fig. 7. Shaifer. Ideal. Ladyfinger.
Kernel of Atlanta.**

KRACK-EZY. Medium, ovoid, very thin shell, full of meat, best (T. V. Munson, Farm and Ranch, Dec. 3, 1904, p. 2.)

LADYFINGER. Size small, 1-1/2 x 5/8 inches; ovate pointed at both ends; color grayish-brown marked with a very few small narrow streaks about the apex; base pointed; apex pointed; shell thin, 1. mm.; partitions of medium thickness; cracking quality excellent; kernel small and narrow, plump full, smooth, sutures narrow and shallow; flavor sweet, good; quality very good.

Described from specimens received from the Summit Nurseries, Monticello, Fla. Originated on the grounds of this nursery company in Jackson county, Fla. A small nut of very fine quality, but too small to be recommended for extensive planting.

LAMAR. Large, oblong, pointed, medium shell, full, best. (T. V. Munson, Farm and Ranch, Dec. 8, 1904, p. 2).

LEWIS. Large, 1-5/8 x 7/8 inches; ovate, compressed; color bright yellowish-brown marked with purplish-brown blotches three-quarters of the distance back from apex; base rounded, blunt-tipped; apex blunt-pointed, slightly wedged; shell thin, .98 mm.; cracking quality good; partitions thick; kernel plump or sometimes shrunken at lower end, wrinkled on the sides, bright, light yellow in color, primary sutures broad, of medium depth, secondary ones very shallow, wrinkled; texture fine grained, solid; flavor sweet, pleasant; quality very good.

Described from specimens received from Frank H. Lewis, Scranton, Miss.

LONGFELLOW. Large, 1-7/8 x 7/8 inches; obovate, angular, sutured; color light yellowish-brown strongly marked with purplish-black splashes throughout; base sloping, rounded; apex shouldered, abruptly-pointed, flattened and quadrangular; shell of medium thickness, 1.15 mm.; partitions very thin; cracking quality good; kernel full, plump, somewhat wrinkled; light straw-colored, sutures narrow of medium depth; texture fine grained, compact; flavor sweet, rich, nutty; quality excellent.

Described from specimens received from E. E. Risien, San Saba, Texas. A pecan of good quality and an excellent keeper.

LOUISIANA. Size medium, 1-7/8 x 7/8 x 3/4 inches; oblong cylindrical; color grayish-brown, marked with splashes of purplish-black towards the apex; base rounded, sloping; apex sloping, pointed; shell rather thick, 1.4 mm.; partitions of medium thickness; cracking quality very good; kernel full, plump, dark yellow, sutures broad, shallow; texture firm, compact; flavor sweet, good; quality very good.

Described from specimens received from Summit Nurseries, Monticello, Fla.

MAGNUM BONUM. Medium, ovate; shell thin; partitions thin; kernel plump, sweet; quality very good. (Hume, Bul. 54, Fla. Exp. Station, 1900, 207).

MAMMOTH. (Syn.: *Steckler's Mammoth*.) Large to very large, 2 x 1 inches; form ovate; color dull gray, pebbled, with a very few dark lines at the apex; base rounded; apex flattened, four-angled, blunt; shell thick, 1.4 mm.; cracking quality very poor; partitions corky, very thick; kernel bright yellowish-brown with broad, deep sutures and fuzzy lining adhering to kernel; texture coarse; flavor sweet and good; quality quite good.

Obtained of J. Steckler Seed Company.

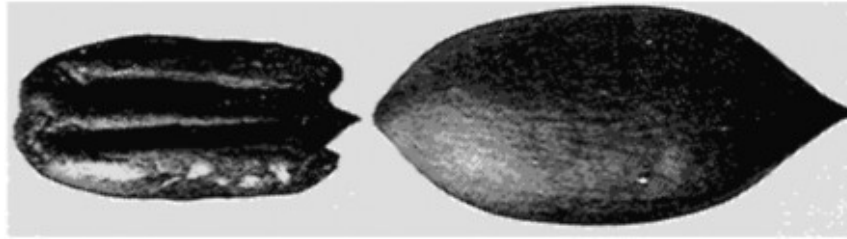


Fig. 7a. The Mantura Pecan.

MANTURA. Size large, 2 x 13/16, 1-7/8 x 7/8 inches; oblong, oval; color dull reddish-brown liberally marked with large, irregular black splashes; base taper-pointed, blunt; apex sharp-pointed, nipped; shell very thin, .78 mm.; brittle, dense; cracking quality very good; partitions thin; kernel dark straw-colored, plump, smooth, oval, with open sutures of medium depth; texture firm, solid; flavor sweet, nutty; quality very good indeed.

Described from specimens received from Wm. N. Roper, Petersburg, Va., by whom it was named and introduced in 1906.

The original tree of this variety stands on the Mantura homestead, in Surry county, Va., two miles south of the James river, now owned by W. P. Wilson. Mr. Wilson's mother planted four trees from nuts secured from a tree at Surry Courthouse, Va., the Mantura being one of the four, The parent tree measures about fourteen feet around the body, and bears crops of good sized nuts. It stands about ten miles from the site of the Mantura tree.

The Mantura tree is a large, symmetrical specimen with wide-spreading branches. It is about eighty feet high and measures about eleven feet around the trunk. It has been bearing for the past fifteen years, and in 1905 yielded 275 pounds of nuts.

This variety will doubtless prove a valuable acquisition for planters on the northern limits of the pecan area, as the particular strain from which it comes has been growing in Virginia for more than sixty years.

MEXICAN PAPER-SHELL. Reported by Ladd Bros., Stonewall, Miss. Listed in "Nut Culture in the United States," 1906, p. 64. (See Biloxi.)

MEYERS. The fruit of a variety of this name was distributed by Judge Samuel Miller, Bluffton, Mo. (Andrew Fuller, in *The Nut Culturist*, p. 170, 1896.)

MONARCH. (Syn.: *De Witt Mammoth*.) Large, 2 x 7/8 inches; ovate, sloping to base and apex; color dull gray strongly marked with purplish-black splashes; base pointed; apex pointed, wedged; shell medium thick, 1.1 mm.; partitions thick, corky; cracking quality poor; kernel frequently badly filled at base, sutures of medium width and depth, color yellowish-brown; texture firm; flavor good, rather dry; quality good.

Originated by G. M. Bacon, DeWitt, Ga. (of the G. M. Bacon Pecan Co.), and introduced about the year 1900. Owing to the preemption of the name Mammoth, by another variety introduced by the late Richard Frotscher, of New Orleans, La., the name DeWitt Mammoth was changed to Monarch.^[E]

MONEY. (Syn.: *Senator Money*.) Size large, 1 x 7/8 x 7/8 inches; ovate, somewhat four-angled, color light brown marked with blotches of purplish-brown sometimes throughout; base abruptly blunt-pointed; apex wedged, pointed; shell brittle, medium to thick, 1.3 mm.; partitions medium; kernel plump, full, bright light yellow, sutures broad, shallow, secondary ones indistinct; texture rather open, of medium grain; flavor sweet, good; quality very good.

A large, plump-meated pecan of very good quality, described from specimens received from Frank H. Lewis, Scranton, Miss.

MONEY-MAKER. Size medium, 1-5/16 x 1 inches; ovate, oblong; color light yellowish-brown with a few purplish-brown marks about the apex; base rounded; apex abruptly rounded, slightly wedged; small nipples; shell of medium thickness, 1.1 mm.; partitions medium thick, corky; cracking quality very good; kernel full, plump, broadly oval, sutures straight, broad, shallow, secondary ones small; texture firm, solid; flavor sweet, good; quality very good.

Described from specimens received from Prof. F. H. Burnette, Baton Rouge, La. This pecan was originated and introduced by S.

H. James, Mound, La.; the quality is very good and the variety is precocious, prolific and hardy.

MOORE. Size small, 1-3/8 x 3/4 inches; ovate; color light yellowish-brown marked with a few small purplish spots about the apex; base rounded; apex abruptly nipped, short; shell brittle, thin, 1.1 mm.; partitions rather thin; cracking quality very good; kernel dark yellow, plump, full, sutures narrow, shallow; texture firm, compact, solid; flavor sweet and good; quality very good.

Described from specimens received from J. H. Girardeau, Monticello, Fla. The variety is so small that we deem it scarcely worthy of propagation.

MORRIS. Size medium, 1-5/8 x 3/4 inches; ovate; color light brown, bright, clean, base sloping, rounded; apex tapering abruptly to a blunt point; shell brittle, of medium thickness, 1.45 mm.; partitions thick; cracking quality very good; kernel plump, filling the shell, straw-colored, primary sutures broad and deep, secondary ones shallow; texture firm, compact; flavor sweet, good; quality very good.

Described from specimens received from Summit Nurseries, Monticello, Fla.

NELSON. Nut the largest of all known; some specimens weighing nearly one ounce; elliptical-oblong in shape; medium thin shell, clean, bright in color; kernel plump, sweet and rich; quality very best, a quick grower; early bearer, very prolific; habit of growth like the Frotscher, forming a round-headed tree. (Catalogue J. Steckler Seed Co., 1905, p. 172.)

NIGGER. Medium, short oval, thin shell, full, excellent. (T. V. Munson, Farm and Ranch, Dec. 3, 1904, p. 2.)

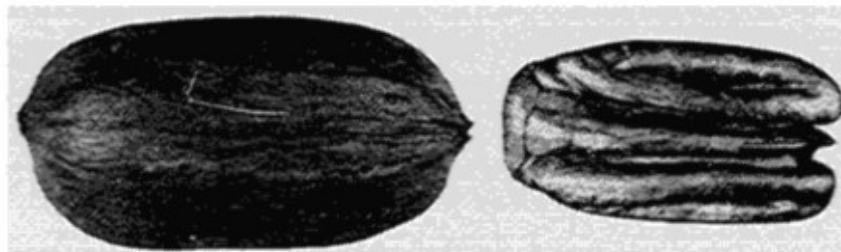


Fig. 8. Pabst Pecan.

PABST. Size large, 1-5/8 x 7/8 inches; oblong cylindrical; color dull gray marked with broad splashes of purplish-black; base rounded; apex blunt, four-angled, grooved; shell of medium thickness, 1.22 mm.; partitions rather thick; cracking quality fair; kernel plump, large, thick with broad, shallow sutures, secondary sutures short, shallow, bright yellow in color; texture fine; flavor good; quality very good.

Described from specimens received from Wm. A. Taylor, United States Department of Agriculture. The original tree, according to Mr. Taylor, is one of a number of seedlings on the grounds of the late William E. Schmidt at Ocean Springs, Miss. The original tree is now about thirty years old. Quite productive and recommended for planting by those who know it.

PAN-AMERICAN. Large, oblong, thick shell, full, best. (T. V. Munson, Farm and Ranch, Dec. 3, 1904, p. 2.)

PEARL. (E. E. Risien, San Saba, Tex.) Medium size, thin shell, sweet kernel; no corky growth inside. A choice nut for family use, but said to be too small for market. (Thomas' Am. Fruit Culturist, 21st Ed., 1903.)

PEARL. This is a very productive pecan, originated by Mr. James. It is distinct from the Pearl which originated in Texas. (Burnette, Bul. Sec. Series, 69, La. Exp. Station, 874, 1902.)

PEGRAM. Size medium, 1-1/2 x 7/8 inches; oblong; color light grayish-brown marked with a few purplish-brown markings at the apex; base rounded; apex blunt, quadrangular; shell creased, roughened, brittle, of medium thickness, 1.15 mm.; partitions medium thick, corky; cracking quality medium; kernel plump, full, quite smooth, sutures narrow and of medium depth; texture firm, compact, solid; flavor sweet and good; quality good.

Described from specimens received from Prof. F. H. Burnette, Baton Rouge, La. Originated by S. H. James, Mound, La.

PERFECTION. (Syn.: *James' Perfection*.) Size medium, 1-3/8 x 7/8 inches; oblong; color grayish-brown marked well down the sides from the apex with purplish-black splashes; base flattened, rounded; apex abrupt, blunt; shell slightly ridged, of medium thickness, 1.3 mm.; partitions rather thick, corky; cracking quality medium; kernel full, plump, brownish-yellow, narrow and moderately deep, sutures narrow, of moderate depth, secondary ones well defined; texture fairly solid; flavor sweet, good; quality very good.

Originated by S. H. James, Mound, La.

PETITE. Small and plump; white hull; very desirable. (Helen Harcourt, Florida.)

PRESIDENT. Large, oblong, pointed, thin shell, full, best. (T. V. Munson, Farm and Ranch, Dec. 3, 1904, p. 2).

PRIMATE. (W. R. Stuart, Ocean Springs, Miss.) Of medium size, slender, rather long; shell thin; quality good; ripens in September, thirty days before the other nuts. (Report Sec. Agr., 1893, 295: 1894.)

POST. (Syn.: *Port's Select in part*.) Size medium, 1-3/4 x 1 inches; short, obovate, compressed on the upper half color light brownish-yellow, marked with a few purplish splashes about the apex; base rounded; apex blunt, abruptly shouldered; shell of medium thickness, 1.35 mm.; partitions thick; cracking quality medium; kernel plump, bright straw-colored, deeply grooved and wrinkled, texture firm, solid; flavor sweet, delicate; quality good.

Described from specimens from the original tree, received from Wm. A. Taylor, U. S. Department of Agriculture. The original seedling tree stands on H. B. Freeman's farm on the Colorado river bottom, San Saba county, Texas. It took its name from Mr. Post, a former owner of the place.^[F]

RANDALL. (Syn.: *Curtis No. 3*.) Small, 1-3/8 x 1 inches; ovate-oblong; color grayish-brown splashed with broad marks of purplish-brown, and covered with small dots throughout; base rounded; apex abruptly blunt-pointed; shell rough, of medium

thickness; cracking quality very good; partitions corky, of medium thickness, 1.25 mm.; kernel medium size, smooth, roundish sutures, reddish-yellow in color; texture firm and compact; flavor sweet and good; quality very good.

Specimens for description obtained of Dr. J. B. Curtis, Orange Heights, Fla. Origin similar to Curtis.

REPTON. Large, shell rather whitish one end round, the other decidedly pointed; black points; meat sweet and tender; tree remarkably beautiful. From one Repton tree, said to be forty years old, over five hundred pounds of nuts were gathered the season of 1904. (Helen Harcourt, "Florida Fruits and How to Grow Them," 1886, p. 212.)

RIBERA. Size above medium; oblong-ovate; cracking quality good; shell thin; kernel plump, light brown, free from the bitter, red, corky growth which adheres to the shell; meat yellow; tender, with rich, delicate, pleasant flavor, (Report Sec. Agr., 1893, 295: 1894.)

RISIEN. Large ovate; quality excellent. E. K. Risien, San Saba, Texas. (Thomas' American Fruit Culturist, 21st Ed., 453, 1903.)

ROBSON. A medium-sized, very thin-shelled nut, oblong ovoid in shape. A comparatively new variety, but of considerable merit. (Bacon's Cat., 1904, p. 28.)

ROME. (Syn.: *Century*, *Columbia*, *Columbian*, *Mammoth*, *Pride of the Coast*, *Southern Giant*, *Twentieth Century*.) Size large to very large, 1-7/8 x 1 to 2 x 1 inches; oblong cylindrical or cylindrical ovate; color grayish, dirty, much splashed and spotted with dirty, black marks sometimes throughout; base rounded; apex abruptly-pointed, flattened on two sides; shell hard, brittle, thick, 1.6 mm.; cracking quality poor; partitions thick, corky; kernel frequently shrunken, bright yellowish in color, sutures of medium depth, secondary ones well marked, fuzzy material often adhering to lower end; texture coarse, rather dry; flavor dry, lacking in character; quality fair.

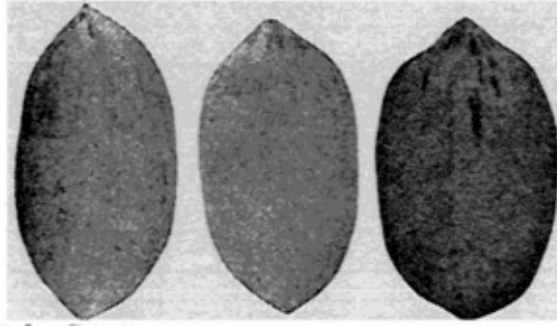


Fig. 9 Russell. Franklin. Kincaid.

Described from specimens received from J. Steckler Seed Co., New Orleans, La. This much-named variety, according to Taylor, was originated by the late Sebastian Rome, at Convent, St. James Parish, La., about 1840. Catalogued by the late Richard Frotscher, under the name "Rome," in 1885. It cannot be recommended for planting.

RUSSELL. Size medium to large, 1-5/8 x 7/8 inches; form ovate, slightly compressed; color grayish-brown with small specks and splashes of purplish-black; base rounded, blunt-pointed; apex abruptly sloping; shell very thin, brittle, .74 mm.; partitions very thin: cracking quality excellent; kernel usually plump though sometimes shrunken at the base, sutures broad and shallow; texture fairly compact; flavor dry, sweet; quality good.

Described from specimens received from Chas. E. Pabst, Ocean Springs, Miss. The original tree stands in the yard of Mrs. H. F. Russell, at Ocean Springs, and is one of a lot of seedlings raised by the late Col. W. R. Stuart, about 1875. The tree was planted where it now stands by Peter Madsen. It was named by Mr. Pabst, and propagated by him in 1894.

RUSSELL No. 1. Large, long-ovoid, shell thin, plump, good. (T. V. Munson, Farm and Ranch, Dec. 3, 1904, p. 2.)

RUSSELL No. 2. Very large, ovoid, shell rather thick, very good. (T. V. Munson, Farm and Ranch, Dec. 3rd, 1904, p. 2.)

SAN SABA. Size small, 1-3/8 x 7/8 inches; ovate, slightly compressed toward the apex; color bright reddish-yellow, marked with purplish-brown splashes extending from about the middle of

the apex; shell very thin and brittle; partitions thin; cracking quality excellent; kernel very plump, smooth, deeply and broadly grooved, bright straw-colored, oval in outline; texture solid, fine grained; flavor rich, sweet, delicate; quality excellent.

The San Saba may be regarded as a standard of quality among pecans, as the Seckel is among pears. Described from specimens received from E. E. Risien, San Saba, Texas. The variety was introduced by Mr. Risien about 1893. The original tree stands on the San Saba river near its intersection with the Colorado river in Texas.

SCHAFER. (Syn.: *Kate Schaifer*.) Size medium, 1-3/4 x 3/4 inches; cylindrical, slender; color light yellowish-brown, marked with a few narrow, purplish splashes at the apex; base sloping, pointed; apex sloping, sharp-pointed; shell rather thick, 1.35 mm.; partitions thick, corky; cracking quality quite good; kernel bright yellowish, plump, filling the shell, smooth, sutures shallow of medium width; texture fine grained; flavor sweet, good; quality very good.

Described from specimens received from Prof. F. H. Burnette, Baton Rouge, La. Originated by S. H. James, Mound, La. Said to be prolific.

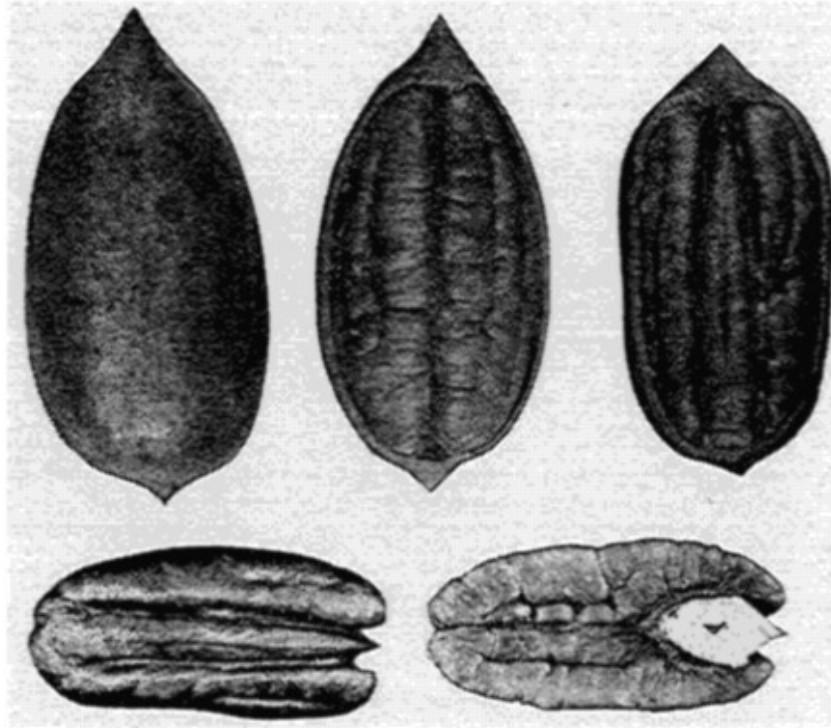


Fig. 10. The Schley Pecan.

SCHLEY. Size large, $1\frac{7}{8} \times \frac{7}{8} \times \frac{3}{8}$ inches; oblong, oval, flattened; color light reddish-brown, marked with small specks about the base and small splashes of purplish-brown about the apex; base rounded, abruptly short nipped; apex abrupt, flattened on two sides and rather sharp pointed; shell brittle, dense, thin, .75 mm.; cracking quality excellent, shell breaking easily and readily separating from the kernel; kernel very full and plump, smooth, with shallow sutures and almost entirely free from wrinkles, bright light yellowish-brown in color; texture very firm; flavor rich, sweet, nutty; quality best; season early.

Obtained from Summit Nurseries, Monticello, Fla. Not as prolific as some varieties, but, in point of quality, unsurpassed.

SENATOR. Medium; ovate; shell and partitions thin; kernel full and plump; quality excellent. G. M. Bacon, DeWitt, Ga. (Hume, Bul. 54, Fla. Exp. Station, 204, 1900.)

SOVEREIGN. Origin, San Saba, Texas. A seedling of San Saba, grown and introduced by E. E. Risien, of San Saba, Tex. Cylindrical, medium to large, with very thin shell and full kernel of fine quality.

A new variety of very much promise. (Taylor, Wm. A., Cyclopedia Am. Hort., 1256, 1901.)

STEVENS. Named for Hon. O. B. Stevens. Commissioner of Agriculture. Not very Large, but bright, pretty and neatly shaped. Very thin shell and always full of nice, rich meat, whether the seasons are wet or dry. Trees medium bloomers, and full bearers of nuts uniform in shape and size. (Bacon's Cat., 1900.)

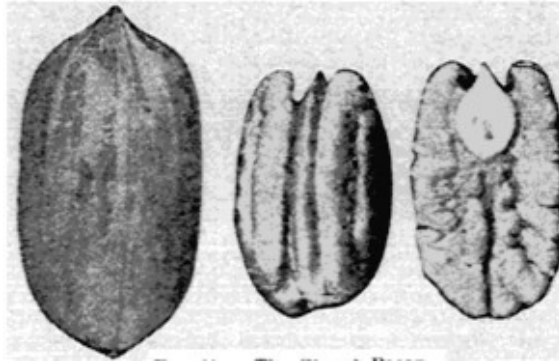


Fig. 11. The Stuart Pecan.

STUART. (Syn.: *Castanera*.) Size large to very large, 1-7/8 x 1 inches; ovate cylindrical; color grayish-brown splashed and dotted with purplish-black; base rounded, tipped; apex blunt, abrupt, somewhat four-angled; shell medium in thickness, 1.1 mm.; partitions thin; cracking quality very good; kernel plump, full, bright straw-colored, sutures moderately broad and deep, secondary sutures not well defined; texture solid, fine grained; flavor rich, sweet; quality very good.

Described from specimens received from the Stuart Pecan Co., Ocean Springs, Miss. This variety has been tested and found to succeed over a wide range of country. The original tree,^[G] grown from a nut planted by John R. Lassabe, about 1874, stood in the garden now owned by Capt. E. Castanera, Pascagoula, Miss. It was blown down in October, 1893, but a new shoot, now in bearing, has sprung up from the roots.



Fig. 12. Success Pecan.

SUCCESS. Size large, 1-9/16 x 1 inches; oblong-ovate tapering from near base to apex; color light yellowish-brown strongly marked with purplish-brown splashes about the apex; base flattened, roundish; apex blunt, four-angled; shell thin, .93 mm.; cracking quality very good; partitions thin; kernel large, full, plump, filling the shell, light yellow in color, sutures broad of medium depth, inner surface wrinkled, oval in outline; texture firm, solid, compact; flavor sweet, rich; quality very good.

The original tree was found "growing in a crowded row of seedlings planted at Ocean Springs, Miss., by the late W. B. Schmidt, about ten years previously. The original Success tree first attracted attention in the fall of 1901." Described from specimens received from Theo. Bechtel, Ocean Springs, Miss.

SWEETMEAT. Size medium, 1-1/4 x 7/8 inches; color bright grayish-brown marked with small streaks of purplish-brown about the apex; abruptly blunt; shell thin, .8 mm.; partitions of medium thickness, corky; cracking quality good; kernel plump, full, light yellow, sutures broad, shallow; texture fine grained, compact; flavor sweet; quality good.

Described from specimens received from Summit Nurseries, Monticello, Fla.

TEXAS. Quite large, some very long; white hull; black points. (Helen Harcourt, "Florida Fruits and How to Grow Them," 1886,

p. 212.)

TEXAS PROLIFIC. Large, oblong, shell thin, cream, clean, plump, best. (T. V. Munson, Farm and Ranch, Dec. 3rd, 1904, p. 2.)

THOMAS. Size small, 1-1/8 x 1 inches; short, roundish oblong; color brownish-gray dotted with small specks throughout, marked with dark purplish splashes from middle to apex; base rounded; apex abruptly short, pointed, nipped; shell of medium thickness, 1.2 mm.; partitions thick, corky, reddish; cracking quality quite good; kernel plump, filling the shell, sutures of medium depth, narrow, texture compact, fine grained, solid; flavor good; quality good.

Described from specimens received from Walter Thomas, Palatka, Fla.

TURKEY EGG, JR. Smaller and shorter than the above; cracking quality medium; shell of medium thickness; kernel plump, light colored; tender, oily, rich; good. (Report Sec. Agr., 1893, 296: 1894.)

TURKEY EGG, SR. Large, long, pointed; cracking quality very good; shell of medium thickness; kernel long, plump; brownish-yellow; separates readily from the shell; meat yellow, a little tough; not of highest quality. (Report Sec. Agr., 1893, 296: 1894.)

TURNER. Medium; elliptical oblong; shell thin; partitions slightly corky; kernel plump, sweet; quality excellent. G. L. Taber, Glen St. Mary, Fla. (Hume, Bul. 54, Fla. Exp. Station, 203, 1900.)

VAN DEMAN. (Syn.: *Bourgeois*, *Duminie Mire*, *Southern Beauty*, *Paragon in part*.) Large to very large, 2-1/8 x 1 x 7/8 inches; oblong cylindrical; color reddish-brown with splashes and streaks of purplish-brown; base sloping, blunt-pointed; apex tapering, sharp-pointed; shell of medium thickness; cracking quality fine; partitions thick; kernel light brownish-yellow, sutures rather deeply and narrowly grooved with secondary sutures forming a mere line; kernel fine grained and compact, sometimes slack at the end; flavor sweet and delicate; quality very good.

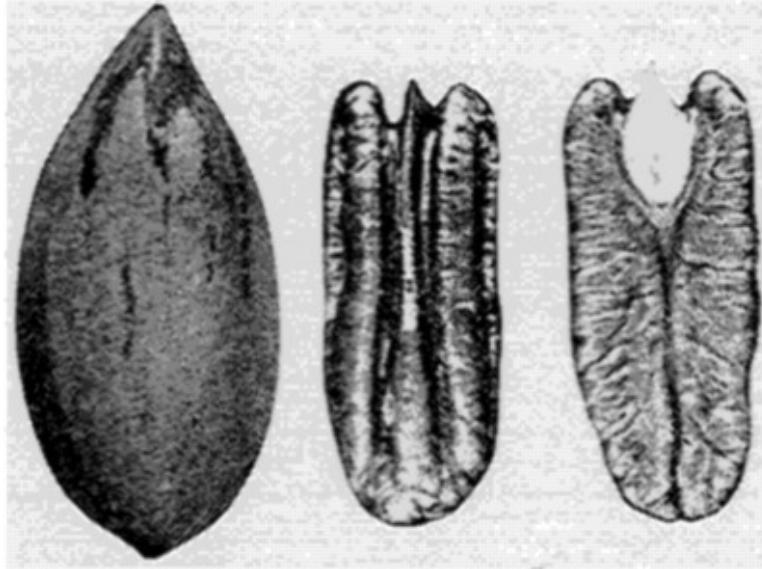


Fig. 13. Van Deman Pecan.

Specimens for description obtained of Dr. J. B. Curtis, Orange Heights, Fla. The original tree of this variety was grown from a nut planted by the late Duminie Mire, of Union, St. James Parish, La., in 1836. The tree still stands, thrifty and vigorous, bearing 200 to 300 pounds of nuts yearly. It was first widely distributed by the late Col. W. R. Stuart, Ocean Springs, Miss., who gave it the name Van Deman. Previously, it had been propagated and distributed locally by the late Emil Bourgeois.^[H]

VALSIES. Reported by Ladd Bros., Stonewall, Miss., and listed in "Nut Culture in the United States," U. S. Dept. Agr., Div. Pomology, 1896, 64.

WILLINGHAM. Illustrated in Farm and Ranch, Vol. 23, No. 49, Dec. 3rd, 1904, p. 1.

YOUNG. Medium to large, 1-5/8 x 1 inches; ovate cylindrical, rounded at the base; color grayish-brown, splashed with purplish-brown markings from center to apex; base rounded; apex sloping rather abruptly, nipped; shell brittle, thin, .76 mm.; cracking quality very good; partitions thin; kernel full, plump, slightly wrinkled with broad and shallow sutures; texture fairly solid; flavor sweet, rich, nutty; quality very good.

Obtained from Chas. E. Pabst, Ocean Springs, Miss. Originated by and named for B. M. Young, Morgan City, Louisiana.

HYBRID PECANS.

The pecan appears to inter-pollinate freely with some of the other species of hickory, particularly *H. minima*, *H. laciniosa* and *H. alba*. A number of what are believed to be well-marked hybrids of the pecan with these different species have been found, the most noteworthy of which, perhaps, are given below:

McCALLISTER. (Syn.: *Floyd*.) Received from O. L. McCallister, Mount Vernon, Ind. This is probably a hybrid. It is the largest nut among all the hickories received at this office. The hull is about one-fourth of an inch thick when dry, and opens readily to the base with four valves. Nut 2-1/8 inches long, 1-5/16 inches wide, and 1-1/16 inches thick; base broad, rounded; apex broad, blunt, angular. In compressed form, in color of nut, also in the angularity and thickness of shell, it is quite similar to shellbark hickory. The kernel of a well-filled specimen is in color, consistency and flavor more like a shellbark of high quality than a pecan. The tree is reported to be "so similar to pecan in bark and leaf that it would be impossible to detect the difference," yet the buds and young wood more closely resemble shellbark. The tree was found many years ago on a farm now owned by Mr. McCallister. The nuts have little pomological value, as grown on the original tree some years, the kernel being shriveled and not filling more than one-third of the space within the shell; yet nuts from the crop of 1893 have been received at the Division of Pomology which were well filled with a kernel of very pleasant flavor. Possibly it may become more uniform in maturing fruit in Mississippi or Texas, where the season is longer than in Indiana. It is well worth a trial by experimenters in those States. Sargent gives a short description of this nut under the name Floyd, and accredits the points of his description to A. S. Fuller in New York Tribune, weekly edition, July 9th, 1892, and says it is perhaps a hybrid. (Nut Culture in the United States, 1896, p. 63-4.)



(Photo by Dr. Wm.
Trelease.)
Fig. 14.
The Nussbaumer
Hybrid.

NUSSBAUMER. In the American Agriculturist for 1884, p. 546, fol., A. S. Fuller published an account of a supposed hybrid between this species and the pecan, which has been called the Nussbaumer hybrid, after J. J. Nussbaumer, of Okawville, Ill., who first brought it to the attention of Judge Samuel Miller, of Bluffton, Mo. Mr. Nussbaumer writes me that the original tree, which stands in the bottom between Mascoutah and Fayetteville, Ill., in general appearance resembles *laciniosa*, though the bark is intermediate between that of the Pecan and Mockernut. Prof. Sargent states (Silva, vii, 158) that a small tree grown from this in New Jersey, by Mr. Fuller, cannot be distinguished from *laciniosa* of the same age; and I should hardly be able to distinguish an imperfect twig from a small tree, cultivated by Judge Miller, from *laciniosa*. The nut, however, is very peculiar, being more elongated than is usual in that species, and widened upwardly, less acutely angled "as if the ridges had been sandpapered down," and so thin-shelled that it can be crushed easily by pressing two together in the palm of the hand. A somewhat similar nut, originally from Indiana, was described by Mr. Fuller in the New York Weekly Tribune, July 9, 1892

(Sargent's Silva, l.c.), as cultivated by R. M. Floyd, of Cedar Rapids, Iowa. And in the autumn of 1895, Dr. J. Schneck sent me ample fruit, twig and leaf specimens of a similar hickory from Posey county, Indiana. The nut of this last is almost identical with a specimen of the Nussbaumer nut in the Englemann herbarium, while its twigs closely resemble those of *laciniosa*, and the leaves are decidedly of the pecan type. I am led to the conclusion, therefore, that these several forms really represent hybrids between *H. pecan* and *H. laciniosa*. In size, quality, and thinness of shell

they appear to be the most valuable of American nuts. (Trelease, Wm., 7th Report Mo. Bot. Garden, 1896, pp. 40-41.)

POOSHEE. Size small, 1-1/4 x 7/8 x 3/4 inches; ovate, flattened wedged, sutures prominent; color dull brown with a very few dark lines at the apex; base rounded; apex flattened abruptly, short pointed; shell medium in thickness, 1.5 mm.; partitions thin, 4-celled at base; kernel rounded in outline, light yellow in color, sutures broad, shallow, halves indented at base; surface much wrinkled and corrugated; flavor sweetish.

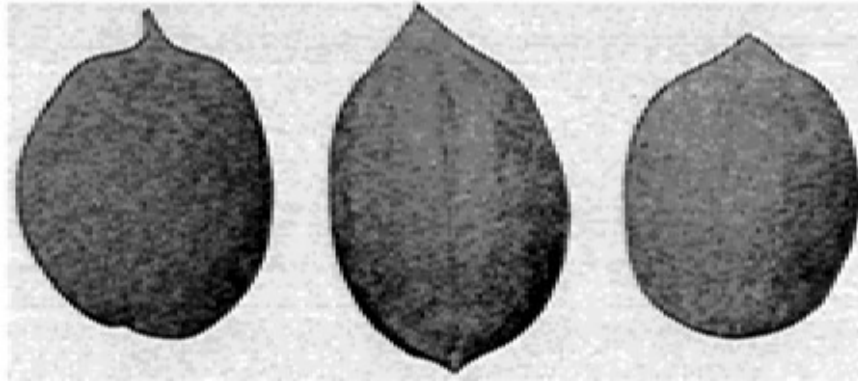


Fig. 15. *H. minima* and two of its hybrids, Westbrook and Pooshee.

Specimens of this nut were secured from Dr. J. F. Wilson, Poulan, Ga., who received them from Prof. Burgess, Clemson College, S. C. The nut presents exactly the same characteristics as the Westbrook, except in flavor and color of kernel. It, too, is doubtless a hybrid, *H. minima* x *H. pecan*. The original tree of this variety stands by or in the old Ravenel cemetery, near Pinopolis, Berkely county, S. C.

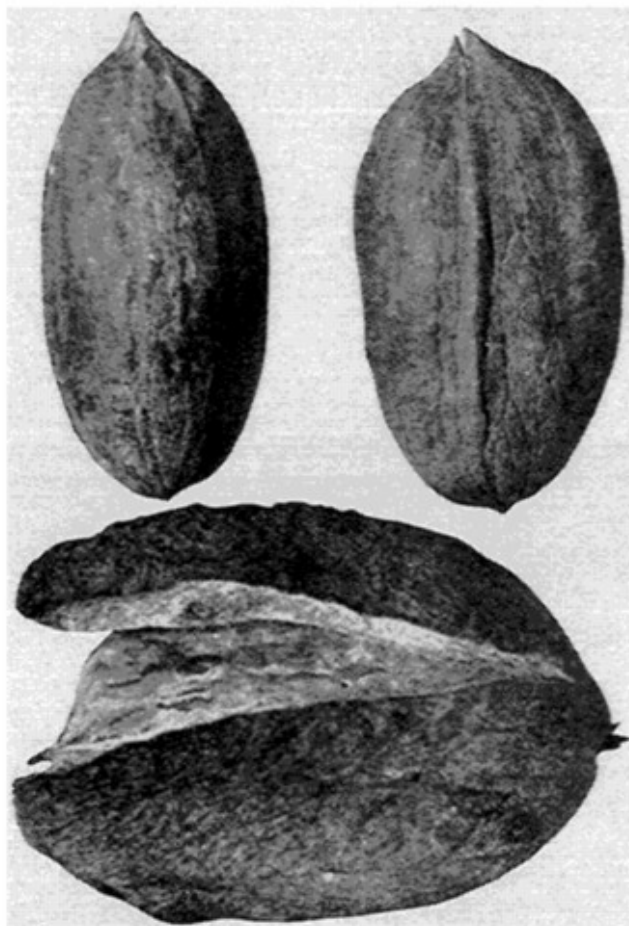


Photo by Dr. Wm. Trelease.

Photo by Dr. Wm. Trelease.

Fig. 16.

The Schneck Hybrid.

SCHNECK. In the autumn of 1894, Dr. J. Schneck, of Mt. Carmel, Ill., and F. Reppert, of Muscatine, Iowa, sent to the herbarium twigs and fruit of bottom-land trees that appear to be hybrids of this species with the pecan. The bark of the Iowa tree is described as being much like that of the Mockernut, while the tree of Dr. Schneck is smooth-barked, resembling the pecan. So far as I have seen them, the twigs of both might pass for those of *alba*, except that the outer scales of the terminal buds are persistent, while the foliage, though intermediate, is strongly suggestive of that of the pecan. The fruit is oblong, almost 2 inches long, the husk 6 mm. thick, parted nearly to the base, with strongly elevated margins to the segments, and rather persistent on the tree. The nuts are nearly as pale as in the Shagbark, conspicuously brown striped, slightly 4-

celled at the very base, and with a wall only 1 mm. thick. As is usual in ALBA, they are upwardly attenuate, and frequently the kernel is abortive. (Trelease, Wm., 7th Report Mo. Bot. Garden, 1896, pp. 44-45.)

WESTBROOK. Size small, 1-3/8 x 7/8 inches; ovate, flattened, prominently sutured; color brown with a few indistinct brownish streaks close to the apex; base rounded; apex wedge-shaped, ridge, abruptly-pointed; shell rough and irregular, thin, 8.5 mm.; partitions rather thin, 4-celled at base; kernel reddish-brown, much wrinkled, sutures of moderate width and depth, halves divided at the base, much corrugated in cross section; flavor decidedly bitter and puckery.

The parent tree is one standing in the yard of J. H. Westbrook, Mt. Olive, N. C., and grew from what, to all appearances, was a pecan nut. The foliage and general aspect of the tree closely resembles the pecan, though the serrations on the leaves are coarser and larger. The fruit resembles, in many respects, that of *Hicoria minima*, and, in short, it appears to be a well-marked hybrid between that species and *Hicoria pecan*.

FOOTNOTES:

[C] Taylor, Wm. A., Yearbook, 1904.

[D] Letter from Mr. S. H. Graves, dated June 19th, 1903.

[E] The Nut Grower, p. 119, March, 1904.

[F] Taylor, Wm. A., Yearbook, U. S. Dept. Agriculture, 1904.

[G] Taylor, Wm. A., Yearbook, 1904.

[H] Wm. A. Taylor, Yearbook, 1904.

CHAPTER V.

PECAN JUDGING.

Every grower of the pecan should be a judge of pecan nuts, and the ideas of growers, while they may differ on certain minor points, should agree on the more important characters of the nut. To enable growers, nurserymen and judges to work on a common standard of merit, a scale of points, in which each individual characteristic of the nut may receive a certain fixed number of credits, is indispensable.

At the second annual meeting of the National Nut Growers' Association, held in New Orleans, the following scale of points for judging pecans was adopted:

Pecan Nuts.	
<i>External characters.</i>	<i>Points.</i>
Size,	20
Form,	5
Color,	5
<i>Shell characters.</i>	
Thinness,	10
Cracking quality,	20
<i>Kernel characters.</i>	
Plumpness,	20
Color,	5
Quality,	15
Total,	100
 <i>Tree.</i>	 <i>Points.</i>
Vigor,	10
Habit,	10
Toughness,	10
Resistance to disease, insects,	10

Precocity,	10
Uniformity of ripening,	10
Productiveness,	40
Total,	100

The rating of a variety to be determined by averaging the rating of nut and tree.

EXPLANATORY NOTES, CHARACTER AND CONDITION OF SAMPLES.

All samples submitted for judging shall be fair average samples of the crop and not selected specimens. They should be tree-ripened, and should be thoroughly cured before judging. Polishing, coloring or other manipulation to disqualify:

Size—The nuts should be large and reasonably uniform in size; nuts running smaller than 100 per pound, to be disqualified.

Form—The nuts should be symmetrical in form and reasonably smooth of surface.

Color—The shell should be bright and clear in color without excess of surface markings.

Thinness—the shell should be sufficiently thin in proportion to size of nut to crush readily.

Cracking Quality—The shell should be brittle and should separate readily from the kernel leaving it clean and in perfect halves.

Plumpness—The kernel should fill the shell and must be smooth, externally, with solid meat of fine and uniform texture, free from internal cavities and with high relative weight of kernel to shell.

Color—The kernel should be uniformly bright and attractive in color.

Quality—The flavor should be sweet and rich, free from bitterness or astringence of either meat or skin.

PART III.

Cultural.

CHAPTER VI.

PROPAGATION OF THE PECAN.

The pecan tree is difficult of propagation by budding or grafting. Skillful propagators are satisfied with seventy-five per cent. of living buds or grafts, while very many have to be content with less. The difficulty is due, in part, to lack of skill; in part to lack of judgment in selecting good material with which to work; but in some regions it is due to the attacks of the bud-worm, *Proteopteryx deludana*, more than to anything else. The buds are eaten out and destroyed by this insect at the time they start into growth. In certain sections spring working of pecans has been abandoned entirely owing to the destruction wrought by this pest. But notwithstanding all the drawbacks, pecan trees can be, should be and are propagated in large numbers by budding and grafting, and the seedling is becoming more and more a thing of the past.

SEEDLING VS. GRAFTED TREES.

It is a fact worthy of note that the beginning of every tree-fruit industry is marked by the use of seedling trees. In the later stages of the development of the industry the seedling, owing to a more intimate knowledge of its failings and shortcomings, gives way to the grafted^[1] tree. This stage has already been reached in pecan orcharding.

It has been stated that a certain percentage of pecans would produce nuts identical with those of the parent tree. The author has yet to find the first instance in which this was the case. This truth is borne out by the observations of others.

In view of the fact just stated, if a planter desires to secure a certain definite fixed variety of pecan, it can only be done by planting grafted trees. Even though all the seedlings were of good size, yet the variation in time of ripening, quality, prolificness, form and size would be against them. Take a certain quantity of each of a number of our largest pecans—Stuart, Van Deman, Centennial and Frotscher for instance—mix them together, and under average circumstances the mixed lot will sell for less money in the open market than

the same varieties and the same nuts would if marketed separately. Mixed nuts, no matter how good the quality, cannot compete successfully in the market with a single uniform sample of the same or nearly the same quality.

Grafted trees will come into bearing at an earlier age than seedlings. In the case of seedlings it is very difficult to say when they will begin to bear, while grafted trees of the more precocious varieties may be expected to bear quite a little fruit in six or eight years from the time of planting.

The great objection to grafted trees is the first cost, and yet, in the face of that objection, it is best to plant grafted trees even if fewer of them are planted. If grafted trees are out of the question, then plant seedlings and top-work them. Grow the seedlings from nuts if necessary; but to those who live in sections where pecans can be grown, let me say, *plant pecan trees*; plant budded or grafted trees if you can—but plant pecan trees.

PECAN STOCKS.

Nursery trees are propagated entirely on pecan stocks, and in the present state of our knowledge, it is the best stock to use. It may be that the pecan will grow and thrive as well on a number of different species of hickory, but definite information bearing on this point is lacking. *Hicoria tomentosa*, *H. alba*, and *H. aquatica* have been used for stocks in North Carolina, Florida, and other States, the pecan being top-worked upon them. But for the present, at least, until our experimental knowledge is farther advanced, the safest advice is to use pecan stock only.

Too little attention on the part of propagators has been given to the kind, source and quality of the seed used to raise stocks for propagation work. The main object held in view in making a selection for seed purposes is to get just as many nuts as possible in the pound. The result of this policy is, that, without question, inferior seedlings are often used for stock; they lack stamina and vigor. Frequently in a nursery of budded or grafted stocks, or in a young pecan orchard, a wide variation in the size and vigor of the trees can be noticed. No satisfactory explanation has ever been offered, but there seems little reason to doubt that it is due to the use of heterogenous lots of seed for stock purposes. *The point must be emphasized, that greater care should be exercised in the selection of the seed used in nursery work.* Nuts from rapid-growing, vigorous, healthy trees only should be used. It is best to plant in spring only nuts which

matured the previous autumn. Preferably these nuts should be of fair or medium size for the variety to give the young seedling a fair start in life.

As already pointed out in regard to pecan shade trees for more northerly regions, so in the case of pecan nuts for use in raising stocks in northern sections. It is best to secure nuts from trees near the northern limits of nut production.

STORING AND PLANTING SEED NUTS.

If pecan nuts, intended for seed purposes, are stored and kept as nuts ordinarily are kept, they become dried out. Before they will germinate the following spring they must absorb all the moisture lost and considerably more; in consequence of which they are slow in starting. If too thoroughly dried out, many may fail to germinate.

To obviate this, and to insure better and more prompt germination, it is best to keep the seed nuts in moist sand or clay during the winter months. Procure a sufficient number of shallow boxes or trays; three feet by one and a half feet by six or eight inches will answer nicely. These are to be used in stratifying the nuts. The earth to be used should preferably be good clean sand, free from organic matter, or, if this cannot be secured, clay will answer. Place a layer of the earth about one inch deep in the bottom of the boxes, then a single layer of nuts, then a two-inch layer of earth, and so on in alternating layers until the boxes are filled. These should then be slightly moistened and set aside in a sheltered place and covered with pine straw, leaves or straw.

In spring, when germination has just begun in the nuts and the tiny sprouts are beginning to appear, they should be planted in rows. The ground should be deeply plowed, well broken up, pulverized, and made moderately rich. Ground which produced a heavy crop of cowpeas, velvet beans or beggarweed the previous season is excellent for the purpose. Farm-yard manure, well decomposed and plowed in the autumn previous, is one of the best manures to use. The ground should be lined off in perfectly straight rows four feet apart, running east and west, that, the buds may be inserted on the north side. The nuts should be planted four or five inches deep, depending upon their size and the character of the soil. Large nuts should be planted deeper than small ones, and in heavy soils nuts may be planted somewhat nearer the surface than in light sandy ones. The rows may be opened with a small turning plow, or, for

lesser areas, with a shovel. Place the nuts, a foot apart, carefully in the bottom of the furrow, cover with a hoe, roll the ground if the weather is dry, and then scarify the surface with a weeder or a light harrow to prevent evaporation of the soil moisture. Or the ground may be mulched with pine-straw, grass, leaves or other suitable material. If no mulch is applied, then the surface of the ground should be cultivated shallow from time to time.

Some propagators have adopted the plan, with good results, of planting the nuts in the nursery rows, in late fall.

CULTIVATION OF NURSERY SEEDLINGS.

From the time the young shoots begin to appear above the surface frequent shallow cultivation should be given. Once every ten days or two weeks is not too often, and the ground should be broken to a depth of one inch or so after every shower of rain. During dry weather more frequent cultivation, once every week, will be well repaid in the additional growth and vigor of the seedlings. A good commercial fertilizer, analyzing 5 per cent. phosphoric acid, 6 per cent. potash and 4 per cent. nitrogen, may be applied to advantage at the rate of fifteen hundred or two thousand pounds per acre. By the following autumn, the better seedlings will have ten or twelve inches of top, and two and a half or three feet of taproot. The following spring some may be whip-grafted at the crown, and by June, July and August of the same year many of them should have attained sufficient size for budding. Those which are not of sufficient size at this time can be worked the following spring and summer.



Plate IV. A Pecan Nursery. Photo by J. F. Jones.

THE NECESSARY MATERIALS AND TOOLS.

The materials and tools used in grafting and budding are: a grafting iron, a mallet, budding knives, grafting wax, strips of waxed cloth and twine.

Of grafting irons there are a number of different kinds, but one after the general type shown in Fig. 17, works very well. It will be noticed that the blade is curved at the corners, and the edge instead of being straight is curved downward in the center. This type of blade in some measure prevents the bruising of the bark when splitting the branch in cleft-grafting. Such a grafting iron may be made by almost any blacksmith. However, a good stout knife may be used instead.

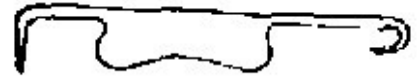
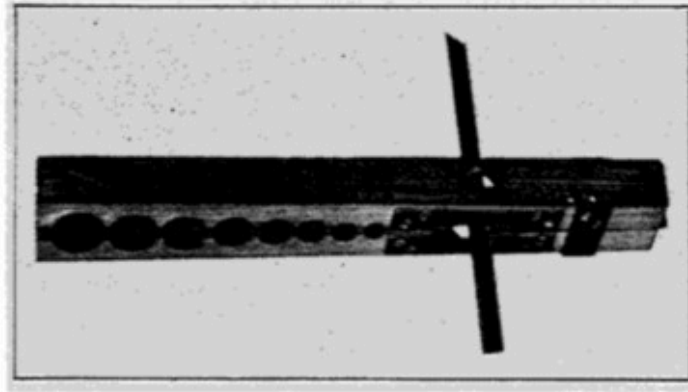


Fig. 17. Grafting Iron.

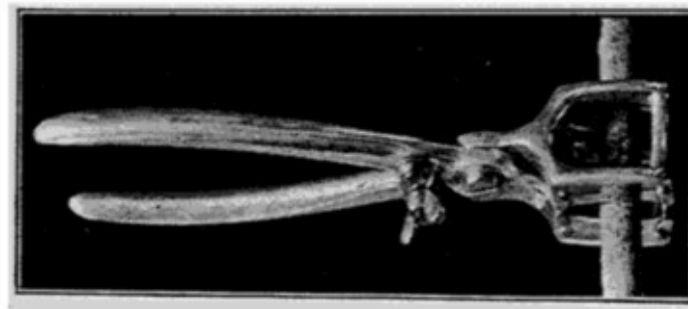


Fig. 18. Common Budding Knife.

For use in grafting, an ordinary budding knife, one of which is illustrated in Fig. 18, is well nigh indispensable. No other knife is so well adapted to making the smooth, sloping cuts on the scions.



White's Budding Tool.



Galbreath's Budding Tool.

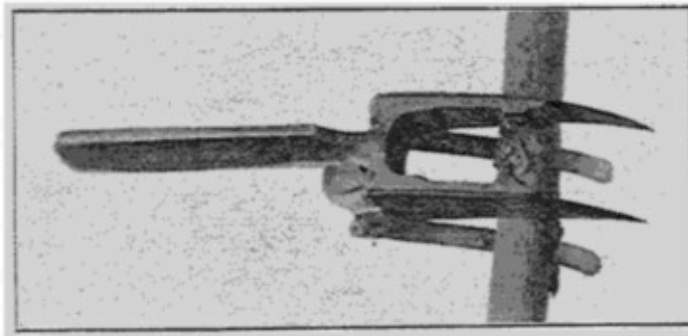


PLATE V. Nelson's Budding Tool.

Plate V.

Some persons can insert annular and veneer shield buds rapidly and well with nothing but an ordinary budding knife. In general, however, a budding knife having two blades, placed parallel with a space of three-quarters of an inch or an inch apart, is best. A very satisfactory knife may be made by fastening the blades of two ordinary budding knives on the sides of a piece of wood seven-eighths of an inch square and four inches in length. The blades can be firmly held in place by means of rivets and a piece of wire wound about the whole.

Three special budding knives, for use in pecan budding, have been introduced, one by Mr. Herbert C. White, of DeWitt, Ga., one by Mr. D. Galbreath, of New Orleans, La., and the other by Mr. Wm. Nelson, of New Orleans, La. In these knives the blades are fixed seven-eighths of an inch, one and one-eighth inch, and three-fourths of an inch apart, respectively. These make it possible to cut the buds and the place where they are to be inserted on the stock exactly the same size, an essential point in pecan budding. They have not yet come into general use, although well recommended by some who have used them. The White budding tool is particularly well adapted for use in top-working trees.

A good grafting wax may be made according to a number of different formulas. Either of the following will be found satisfactory:

- {Resin 6 pounds.
- I. {Beeswax 2 pounds.
- {Linseed Oil 1 pound.
- II. {Beeswax 1 pound.
- {Linseed Oil 1 pint.

Break the resin and cut the beeswax into small pieces. Place in an iron vessel, pour the oil over them and melt over a slow fire. Stir slightly to insure their being well mixed together, pour out into a bucket of cold water, grease the hands, and as soon as the mass is cool enough to handle, pull until it becomes light yellow in color. The wax may be made up in quantity and stored in greased tin or wooden boxes for future use.

To prepare waxed cloth, cut the cotton cloth into pieces of convenient size, say eighteen inches square, dip them down into the melted wax, remove them with a couple of sticks and stretch them out until cooled. For use, the cloth may be torn into strips of desired width and wound about a stick eighteen inches or so in length. Use a little grease to prevent the grafting wax and grafting cloth from sticking to the hands.

For waxed twine, procure No. 18 knitting cotton and drop the balls into the melted wax for a minute or two or until the wax penetrates them.

SELECTION OF SCIONS.

Great care should be exercised in the selection of scions for use in budding and grafting. Much of the immediate success of the work depends upon the character of the scions, while the health and longevity of the future tree may be materially influenced by the kind of wood used in propagating work.

The practice of taking scions and buds from young trees which have never borne, or from nursery stock, must be strongly condemned. They should be cut only from thrifty, vigorous, prolific trees. Even trees of the same variety differ in these things, and a thorough knowledge of what a tree will do and has done is the only true guide in the selection of scions. It is a well-known fact that desirable qualities can be reproduced and perpetuated by grafting.



From Bul. 57, Florida Exp. Sta.

Fig. 19.

Scions: 1-3, Curtis; 4-6, Van Deman; 7-8, Stuart. 1. Poor Scions—long, slender, pithy. 2,4,5,7,8. Scions from one year's growth. 3. Scion, partly one, partly two years old. 6. Scion with cut, back of tip. 8. Scion which bore fruit at a.

Grafts should be selected from well-matured branches of one year's growth. Fig. 19, No. 1, shows an undesirable scion. The wood is angular, small, the internodes long, and the pith large in proportion to the diameter. Either terminal portions of twigs may be used or portions back of the tip, but the buds

should always be well developed, full and plump—Fig. 19, Nos. 2 to 6. For this reason grafts should not be cut from wood far back from the tip of the branch. As stated, twigs of the previous season's growth are generally used, but scions composed partly of two-year-old wood may be used, provided the growth is not too large. Fig. 19, No. 3, shows one of these. Grafts are generally cut about five or six inches long, and should be from one-quarter to three-eighths of an inch in thickness.

It is best that the grafts be cut while still in a dormant state, and inserted in the stock just before the growth starts. The scions may be kept for a considerable length of time by placing them, loosely packed, in damp moss or sawdust, in a box. The box should be covered over with earth and the scions kept sufficiently moist to prevent drying out.

For bud sticks, well developed one-year-old branches, one-half to seven-eighths of an inch in diameter, and on which the buds are well formed, may be used. Such sticks frequently show three buds at a node, and if some misfortune should overtake one or two of these, there is still a chance of success, though the upper one being the strongest is generally the one which starts, provided it is uninjured and the bud takes. The degree of maturity of the bud is important, and care should be exercised that only those which are plump, full and well developed, are used. As soon as removed from the tree all bud sticks and grafts should be wrapped in damp newspapers to prevent drying out.

TIME.

Grafts should be inserted in spring just before or at the time growth starts. Buds may be inserted any time during the period when the bark will slip readily. Last year's dormant buds may be inserted early in the season, or buds of the current season's growth may be used during the latter part of July and the month of August, at which time they have become fully matured in the southernmost parts of the Gulf States. The time may even be extended into September. Very many of these late-inserted buds remain dormant during winter and begin growth in spring.

BUDDING.

Annular Budding.—A ring of bark about one inch in length is removed from the stock. A bud stick of the same size is selected, and from it a similar ring

with a good bud on it is removed by cutting around the bud stick and slitting down the back or side opposite the bud. This bud is then placed in position on the stock. After the buds are in place, a piece of stiff wrapping paper should be tied around the stock just above the bud and allowed to flare out over the bud to protect it from the sun and wind. Preferably all buds should be inserted on the north side.

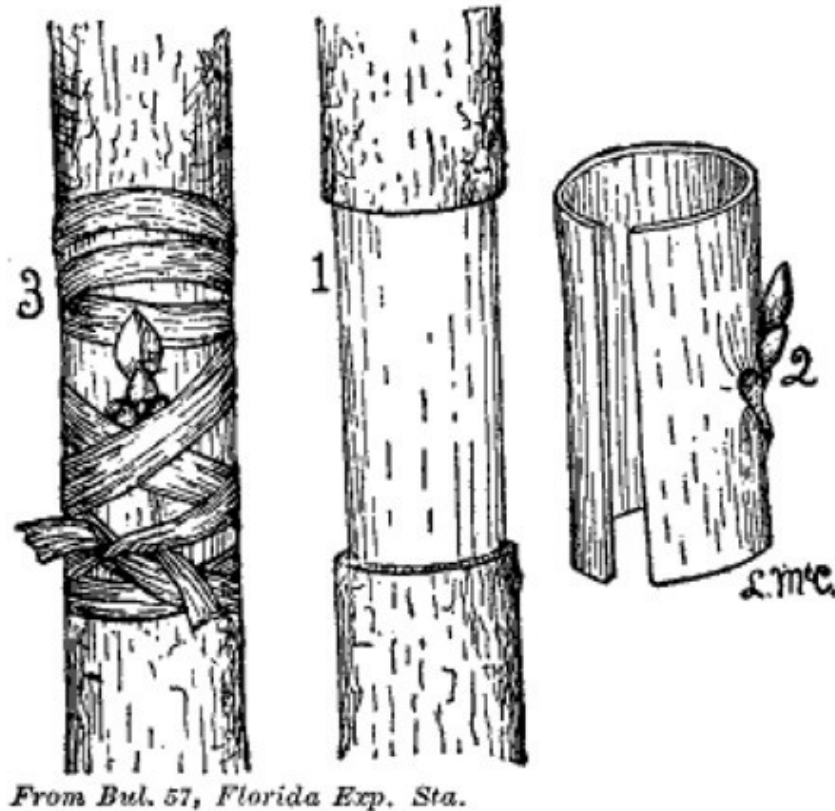


Fig. 20. Annular Budding. 1. Stock prepared for bud. 2. Bud. 3. Bud in place and tied.

Stocks from three-eighths to three-quarters of an inch may be worked by this method.

Veneer Shield-Budding. (Patch Budding). This method differs from the last only in that the piece of bark removed from the stock and the piece with the bud attached are not complete rings, but only parts. A rectangular or even a triangular piece of bark is taken out of the stock, a similar piece with a bud in its center taken from the bud stick is fitted in its place and wrapped in the usual way.

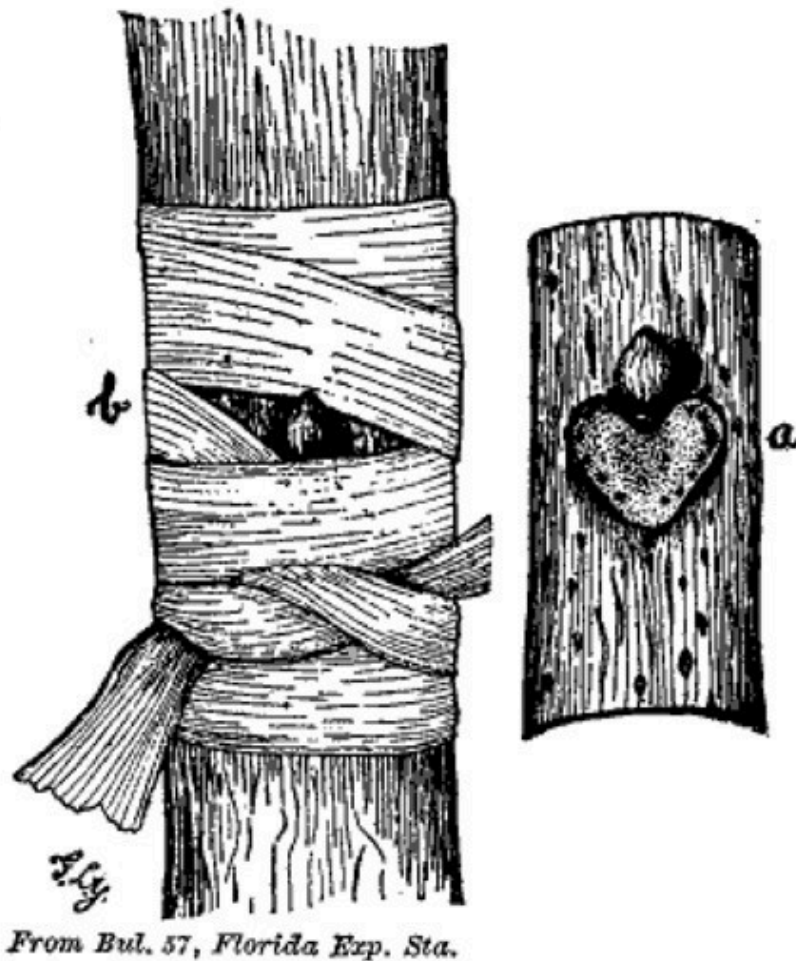


Fig. 21.
Veneer Shield-Budding.

Mr. George W. Oliver, of the Bureau of Plant Industry, Washington, D. C., has described^[1] a modified method of veneer shield-budding, which has given good satisfaction in his hands. Instead of removing the patch from the stock, it is slit down the center from top to bottom and the edges are lifted back, the buds inserted beneath and the side flaps are then tied down over it. He has also found that dormant buds of last year's growth give better results than buds of the current season.

The use of these buds has not, however, come into general use; first, because of the large amount of wood which must be destroyed to secure them; and second, because in those sections where bud-worms are prevalent, their larvæ are to be found clustered about the buds until quite late in the season and make their attack as soon as the bud starts to grow.

Mr. E. W. Kirkpatrick,^[K] McKinney, Texas, described a method successfully used by him, as follows:

"We prepare the stock to receive the bud by cutting out a section of bark and wood as shown in Fig. 22. The bud is cut from the scion in the same way the cut on the stock is made. It should be about the same length, width, thickness and shape of the bark removed from the stock (see Fig. 22), so that the bud will fit the stock. * * * * The bud should be firmly tied until growth begins, usually about twenty-five days, when the string should be cut and the stock also cut just above the bud. * * * All shoots must be kept rubbed off so as to give the buds the right of way. The small buds about the base of the scions or those on the two-year-old wood are preferred. Where the buds are small and in a cluster, several may be included in one set and the thinning done after the growth starts."

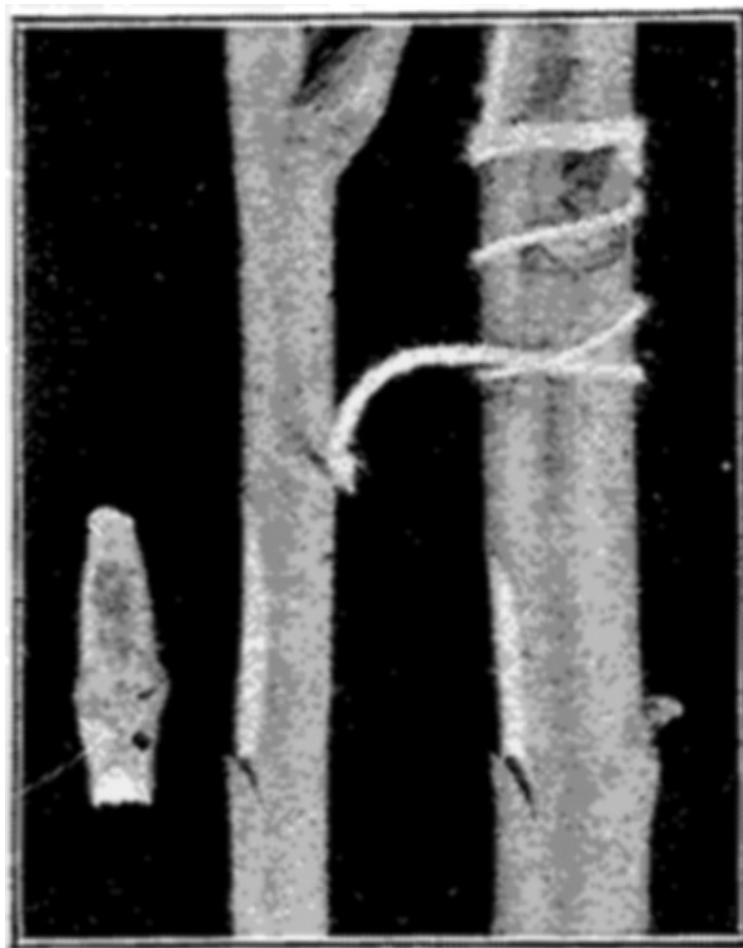
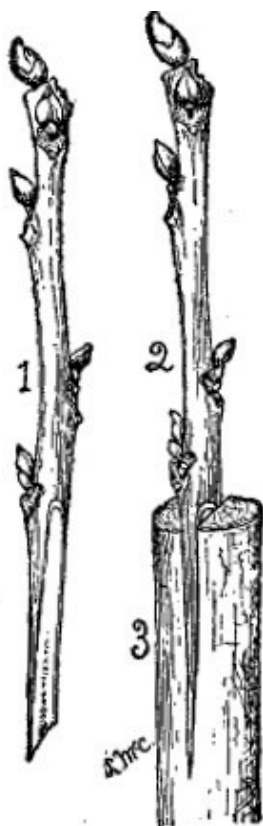


Fig. 22. Chip Budding. Bud cut; Incision made; Bud in place.

GRAFTING.

Cleft Grafting. Having selected the branch for cleft grafting and the point at which the scions are to be inserted, the branch should be carefully and smoothly cut off. The limb is then split by using the grafting iron. If rapid work is to be done, grafts should be prepared beforehand and carried to the field, wrapped in damp paper. In preparing the scion, a sloping cut should be made about one and one-half inches long, cutting into the pith from a point one-half way up the cut down to the lower end. On the opposite side, the cut should not be made to touch the pith, but should be confined to woody tissue throughout its whole length. The knife should have a keen, sharp edge. The cut should be clean, smooth and straight, and the scion should be left wider on the outer side. Start the cuts on each side of, and just at a bud, as shown in Fig. 23. Having made the cleft, it is opened with the wedge on the end of the grafting iron and the scion is placed in position. The cambium layers should be in contact. Slip the scion well down until the whole of the cut surface is within the cleft. If the stock is large enough insert two scions. After inserting the scion it should be firmly held in place by binding the stocks with strips of waxed cloth, after which a covering of wax may be placed over the cloth. The cut end of the stock should be covered, and if the scion be other than a terminal shoot, its distal end should be waxed also.

Whip Grafting. Branches, which are to be worked by whip grafting, should be less than one inch in diameter. The method is illustrated in Fig. 24. A sloping cut, an inch and a half long, is made diagonally across the stock. A corresponding cut is made on the scion, a tongue is raised about the center of each cut by making another cut with the budding knife held almost parallel to the sides of the wood. The tongue is raised a little on both stock and scion and the two are shoved together. They should be securely bound with a strip of waxed cloth, and a layer of wax should be spread over the whole, covering up all the cut surfaces to the exclusion of water, air and the germs of decay.



Bul. 57, Fla. Exp. Sta.

Fig. 23.
Cleft Grafting. 1.
Scion. 2. Scion
inserted ready for
tying. 3. Stock
showing cleft.



Bul. 57, Fla. Exp. Sta.

Fig. 24.
Whip Grafting. 1.
Stock showing
cut. 2. Scion. 3.
Stock and Scion
ready for
bandage.



Photo by J. F. Jones.

Fig. 25.
One year Pecan in fruit. Unusual; due to
bearing wood being used as a scion.

The scion and stock are preferably chosen of nearly the same size, but a scion somewhat smaller than the stock may be used, in which case the cambium layers along one side of the surfaces in contact should be placed opposite each other, and the projecting portion of the stock trimmed off.

AFTER-CARE.

In from ten days to three weeks, the buds should unite. They should be examined, and if union—indicated by the full, plump condition of the buds or the commencement of growth—has taken place, the wrappings should be removed. If growth has started, the stock should be cut off or lopped just above the insertion of the bud, in the case of budded trees. From time to time the trees should be examined, and all sprouts which might rob the bud of sap, thereby preventing its growth, should be rubbed off.

FOOTNOTES:

[I] The term grafted, as here used, embraces budded trees as well.

[J] Bulletin 30, Bureau Plant Industry, U. S. Dept. Agr., 1902.

[K] Farm and Ranch, Dec. 3, 1904.

CHAPTER VII.

TOP-WORKING PECANS.

Many of the pecan trees planted in groves have not fulfilled the hopes of their planters. These trees, raised from large selected nuts, for which the planters paid a dollar or more per pound, have not come true to seed. Some are irregular, shy bearers; others, when they do condescend to produce a few nuts, bear small-sized, inferior nuts, not larger than ordinary playing marbles, while some produce a large crop of good marketable nuts. Some are unhealthy, subject to the attacks of scab and rosette. It goes without saying that such pecan plantings are unprofitable. What is to be done with them? Briefly, this: Cut out and destroy, root and branch, those affected by rosette, those which are unhealthy, and top-work the remainder of those which do not produce a sufficient quantity of marketable nuts of good quality.

Top-working may be profitably applied to another class of trees—pecan trees in their native woods and thickets, and in some cases hickories, viz.: *Hicoria tomentosa*, *H. alba*, and *H. aquatica*, may be top-worked. Our knowledge is not sufficiently advanced in regard to top-working on hickory to warrant us in making any very strong recommendations, but the author has seen a large number of pecans worked on hickory, a few of which were in bearing, and all appeared healthy and vigorous. There is no good reason why hickories cannot be top-worked to advantage, and the delights of amateur efforts along this line will amply repay the attempt.



Plate VI.
A two year top-worked Pecan tree. Variety, Van Deman.

Again, seedling trees may be grown or purchased and set out in orchard form. When these have grown to an inch or so in diameter and have developed several branches, they may be top-worked. This method of securing a pecan orchard is somewhat slow, and is open to the objection that the buds or grafts frequently fail to take, and in consequence the task of top-working extends over a number of years, resulting in trees of irregular size and shape. But by this plan a planting of desirable varieties can be secured at little expense, and provided time is not a consideration, the plan will prove quite satisfactory indeed.

METHODS OF OPERATION.

It is best to insert both buds and grafts in parts having smooth bark, though grafts can be placed in rough barked parts as well. Frequently trees are in a very undesirable condition for top-working, and it should be borne in mind that those branches nearest the center of the tree will give the most satisfactory result in the rapid growth of buds inserted in them. If the tree is not in good shape for

working—i. e., if no branches of desirable size and age are found in convenient places—the tree should be partially trimmed to a pollard, cutting some of the main branches back to stubs, and when shoots have started from these they may be grafted or budded. In from six to twelve months from the time buds have started from the branches thus cut back, under average conditions the new shoots will have grown to sufficient size to permit of their being budded or grafted. The best time to prune back trees to start new shoots for top-working is early in the month of March. In removing large branches there is always danger of splitting, because of the weight of the heavy branches. This may be entirely obviated by sawing upward from the under side of the branch as far as possible, then cutting from the upper side downward. A branch will split off and drop without injury to the remaining parts. All cut surfaces should be well covered with white lead paint to prevent decay.

The method of procedure depends upon the size and age of the tree and whether the tree is to be budded or grafted.

In top-working old trees, only a portion of the branches should be worked at one time. If the whole top be removed at once, the tree suffers a severe shock. Two or three years are necessary to top-work a large tree, a half or a third of the top being worked each year. If the trees are of small size, the whole top may be removed at one time.

CARE OF TOP-WORKED TREES.

For several months after the new top has commenced to grow, the scions have but a slight hold upon the stock. The leaf surface is often so large that a slight wind may twist them off. To prevent this, a number of branches may be tied together, or they may be fastened to stubs of branches left temporarily. Posts may be driven into the ground close to the growing scions, to which they may be tied. Use soft bandages and burlaps.



Plate VII.
An old Pecan tree Top-worked in the
branches.

CHAPTER VIII.

SOILS AND THEIR PREPARATION.

The pecan succeeds on such a wide range of soils, that it is really easier to list those on which it should not be set than it is to enumerate those on which it may be planted. Of the soils not adapted to it, deep sandy lands, soils underlaid with quicksand close to the surface, soils with hardpan subsoil, wet, sour, poorly-drained lands, and stiff, pasty clays, may be mentioned particularly.

If pecans are planted on land with a quicksand subsoil, the roots are unable to make their way downward through the quicksand. So far as being able to take a downward direction is concerned, they might as well be planted on top of a plate of metal. The writer once planted a few nuts on such a soil, to see what they would do. At the end of three years the tops were about two feet in height; the taproot, while thick and stocky, was not more than six inches long. It stopped abruptly after numerous efforts to penetrate the quicksand. In normally developed trees of the same age, the taproot would have been three or four feet long. The same objections hold against soils underlaid with a hard, impervious layer.

While the pecan is at home on rich, alluvial river bottoms subject to overflow, yet it will not grow successfully on damp, soggy lands. It should not be planted on such soils unless they can be well drained, and not then until they have been limed and cultivated for some time to counteract the acidity of the land. We can definitely say that the pecan will do well on alluvial river bottoms, on sandy, loamy soils with a clay or sandy-clay foundation, on sandy-clay lands with clay predominating, on the flat woods sandy lands so common in the southeastern Gulf States, and on the higher uplands where hickory, dogwood, holly and oak abound.



Fig. 26.

Pecan Tree grown on quicksand. Note the taproot.

It is a fact worthy of note, however, that on extremely rich soils, the pecan will make wood growth at the expense of fruit, while on lands containing less fertility, less growth is developed with a proportionately large amount of fruit.

Choose not the poorest soil by any means, but a good, sandy loam in which there is a considerable amount of humus. A subsoil containing a very considerable amount of clay is to be preferred, by all means, for such a soil, with intelligent management, will gain rapidly in fertility.

PREPARATION.

The preparation of the soil should be complete and thorough. It may be stated, as an axiomatic truth, that the soil cannot be prepared for trees as well after they are planted as it can before, and nothing is to be gained by planting the trees in

poorly prepared land. Better by all means to spend a year or more in getting the land in shape.

If the land is covered with a growth of timber, this should be cleared away and the ground cultivated for a year at least before the trees are set. Corn is probably the best crop to grow on new land, and at the last working cowpeas should be sowed. On fairly good land this will be sufficient, but on poorer ground the land should be continued in cultivation another year, sowing it down in beggarweed, cowpeas, soja beans, or velvet beans. These crops should be plowed into the soil in autumn or early winter, after they are dead and dry.

On lands which have been cultivated for some time, these same crops should be sowed for one season previous to planting, at least. Every effort should be made to insure a good stand and a good growth. Inoculation of the seed with nitrogen-gathering germs will help, and a good fertilizer, such as the one recommended for these crops elsewhere, should be applied. Nothing will insure a good growth in the young trees so well as the nitrogen and humus added to the soil by leguminous crops. Stable manure may also be used to advantage.

The ground should be deeply and thoroughly broken with a two-horse plow. In many cases the soil conditions will be greatly improved by the use of a subsoil plow, running it after the ordinary plow so as to break and loosen the soil to a depth of twelve or fifteen inches, or even more.

CHAPTER IX.

WHAT VARIETIES TO PLANT.

What varieties shall I plant? An easy question to ask—a difficult one to answer; for, though the one attempting a reply may know something of varieties, their size, quality and prolificness, there is always an unknown personal equation entering into the problem.

Every variety of importance has its advocates. If a man has a preference for a certain variety, and is interested in it, let him plant that variety largely. He will be likely to give it better care and attention than he will a variety for which he has no particular liking or for one which he may regard even with disfavor.

The question of adaptation of varieties to certain localities is an extremely important one. A variety which may do well in a certain state or region, may not succeed in another; and on the other hand, some varieties may be grown almost anywhere. To answer questions of this sort, one must have an intimate knowledge of varieties in their local adaptations.

Two of the worst faults which a variety may have are partial barrenness or shy bearing and poor filling quality. In this last respect the worst sinners are the larger varieties, and in point of filling quality, medium and small-sized varieties will, in nearly all cases, be found to have the greatest range of adaptability. The larger varieties are more likely to succeed on rich lands where the rainfall, particularly during the summer months, is great.

Again, all varieties are not equally hardy, and some may not ripen their wood and fruit early enough in autumn to avoid late killing frosts. Such varieties should not be selected for planting in sections where there is danger of such injury, viz: principally along the more northerly outskirts of the pecan area. In such regions, early varieties should be planted, for early ripening of fruit and wood usually go together in the pecan.

Many varieties are late in coming into bearing; others begin to bear while quite young. This difference in precocity is worthy of consideration. Other things being equal, those varieties which begin to bear early and are prolific, should by all means be given the preference.

In addition to setting out an orchard of what he believes to be the best varieties for his section, or which experience has taught to be the best, the grower should, if he is thoroughly interested in his work, plant a tree or two of a number of other different kinds to test their merits and to learn something of their characteristics.

VARIETIES RECOMMENDED FOR DIFFERENT SECTIONS.

The following recommendations have been made by growers and others in different parts of the South. These may be changed with the knowledge which time alone will bring; but they represent the best, most accurate and up-to-date knowledge which can be given at this time:

VIRGINIA, NORTH CAROLINA, SOUTH CAROLINA.—In these States the different named varieties have not been grown and fruited long enough to warrant any very strong recommendations, Pabst, Stuart and Jewett have been planted in southeastern North Carolina and have succeeded, but on the whole, for the entire region of these three States, the most satisfactory and staple progress in pecan culture will probably come from the introduction of local varieties of merit.

GEORGIA.—Dr. J. F. Wilson, Secretary National Nut Growers' Association, Poulan, Ga., has selected his varieties for that section as follows: Stuart, Schley, Van Deman, Georgia and Frotscher.

Herbert C. White, horticulturist, G. M. Bacon Pecan Co., DeWitt, Ga., says that Georgia and Stuart are the best of the varieties thus far tested.

J. B. Wight, Cairo, Ga., believes in planting Frotscher principally in his section.

FLORIDA.—Prof. H. K. Miller, Monticello, Fla., believes in planting Schley, Dewey, Louisiana, Frotscher, Stuart, Russell, Pabst, Van Deman and Sweetmeat.

James A. Bear, Palatka, Fla., reports that Frotscher, Stuart, Van Deman, Curtis and Money-maker are doing well for young trees, while Rome and Centennial have not proved satisfactory.

Dr. J. B. Curtis, Orange Heights, Fla., recommends Curtis, Frotscher and Van Deman, these having proved most fruitful in his orchard.

J. H. Girardeau, Monticello, Fla., regards Van Deman, Frotscher, Pabst, Clarke and Schley as good varieties.

S. H. Graves, Gainesville, Fla., says: "Curtis, Stuart, Van Deman, Dalzell, Louisiana, Bolton and Frotscher are adapted here, and have proven good fruiters. From study and observation I would supplement this list with James, Money-maker, Success, Russell, Robson and Schley."

J. F. Jones, Monticello, Fla., recommends Stuart, Van Deman, Frotscher and Schley, emphasizing the first as a commercial variety, and the last-named as an excellent variety for the "Fancy" trade.

ALABAMA.—Prof. R. S. McIntosh, Auburn, Ala., believes Stuart, Van Deman, Pabst, Centennial and Schley to be good varieties for Alabama.

MISSISSIPPI.—Theo. Bechtel, Ocean Springs, Miss., says: "My selection at present for this section would be in the order named—Success, Stuart, Pabst, Frotscher, Russell and Van Deman."

Chas. E. Pabst, Ocean Springs, Miss., recommends Stuart, Pabst, Russell, Success, Van Deman and Rome.

Stuart Pecan Co., Ocean Springs, Miss., recommends Stuart, Van Deman and Russell.

Prof. H. E. Van Deman recommends Stuart, Van Deman, Money-maker and Pabst for the Lower Mississippi Valley.

LOUISIANA.—S. H. James, Mound, La., has found Money-maker, Stuart, Van Deman and Pabst, in the order named, best for his section. He says that Money-maker is extremely hardy, having withstood 20° below zero in Illinois, without injury.

Wm. Nelson, New Orleans, La., strongly recommends Frotscher and Centennial for his section.

B. M. Young, Morgan City, La., is planting Stuart, Russell and Young for commercial orchard.

TEXAS.—E. E. Risien, San Saba, Tex., says that San Saba is more in demand than any other variety he has. It succeeds well in his section. He recommends as well, Texas Prolific, Colorado, Kincaid, Atwater, Concho and others.

E. W. Kirkpatrick, McKinney, Tex., President National Nurserymen's Association, regards Stuart, Russell, Pabst and Money-maker as valuable for his section. Good results have been secured with Hollis and Wolford.

GENERAL RECOMMENDATIONS.

From careful observations, we believe that a list of varieties comprising Stuart, Georgia, Money-maker, Pabst, Success, Frotscher, Van Deman and Russell of the larger sorts will be found to contain varieties which will prove satisfactory in most locations. To these we must add Schley, San Saba, Curtis and a number of other medium-sized or small varieties of unsurpassed quality.

When about to plant in a given region, study the local conditions, visit the local trees or orchards, and upon these a conclusion may be based which is not likely to lead the prospective planter into very great error.

CHAPTER X.

PURCHASING AND PLANTING PECANS.

Since, in most cases, the trees are to be set in late autumn and early winter, the trees should be purchased in late summer and early autumn. Do not leave the purchasing of the trees until the last week, or the last minute, before planting, but buy in good season, i. e., several months before planting time. Too many forget about the trees until the time for setting them out has come, and not infrequently the matter is forgotten until after the season for planting is long since past.

The number of varieties in the commercial orchard should not be large. No greater mistake can be made than that of planting a few trees each of a large number of different varieties. Four or five, at most, are sufficient; got fewer varieties, rather than more.

Trees can be purchased in two ways: They can be secured direct from the nurserymen (usually by catalogue), or they can be purchased from agents. By far most of the pecan trees are bought from the nursery, and by many this method is preferred. If trees are secured from agents, be certain that they are responsible persons, representing responsible firms; be certain that they are properly accredited, i. e., have certificates to show whom they represent, and if they have not these, then send them off down the road, and the dog with them for company, if necessary. This may seem to be harsh advice, but had it been followed by many purchasing pecan trees in recent years, it would have been much to their advantage. Plenty of seedling trees have been bought and planted in the belief that they were good grafted or budded stock.

But agents, with all sorts of credits, have represented firms which were not honest. Budded and grafted trees of certain well-known varieties of pecans have been sold, which were not those varieties.

There is every reason to believe that scions have been taken from ordinary seedling trees of any kind, inserted in stocks and sold for the best varieties, and that a large number of trees have been substituted and sold for what they were not. The prospective planter must depend upon the honesty and integrity of the nurseryman, and should inform himself on this point.

The National Nut Growers' Association has done no greater service to the pecan industry than that which they have rendered in protecting the public from fraudulent agents and nurserymen. Happy is the nurseryman whose reputation for square dealing merits the trust and confidence of tree-planters throughout the country.

COST OF NURSERY STOCK.

At present, the prices quoted for one and two year old stock of standard varieties varies from 75 cents to \$2.50 per tree, in small numbers, with considerable reduction for trees in lots of one hundred or one thousand. It is not improbable that these prices may be somewhat reduced within the next decade, as greater efficiency is gained in propagating.

DETECTING BOGUS TREES.

How may budded or grafted trees be distinguished from ordinary seedlings or from "doctored" seedling trees? Many people have purchased seedling trees at a dollar or so per tree, under the supposition that they were budded or grafted stock. It is well to know something of the distinctions between them.

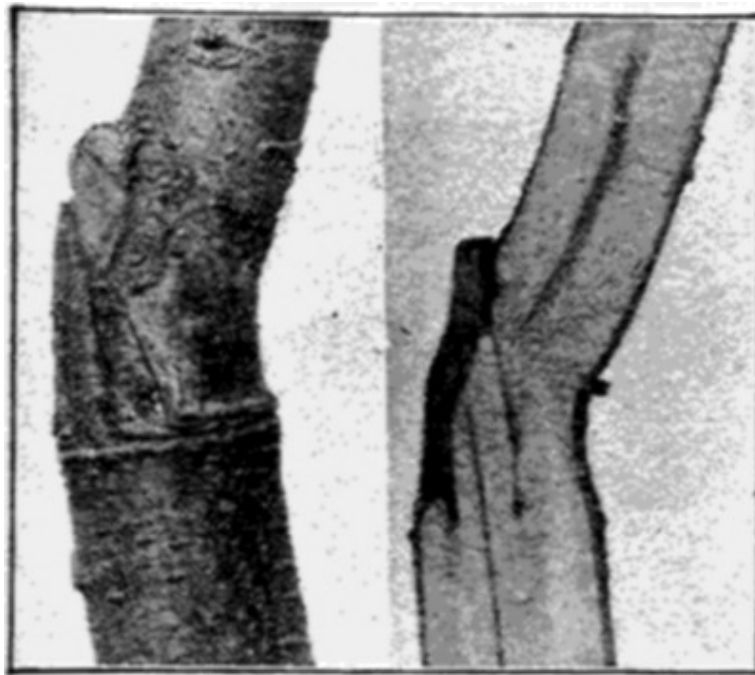


Fig. 27.
External and Longitudinal Interior View of

Bud Union.

If the trunks are straight and smooth, with bark uniform in appearance throughout, the trees have not been budded or grafted, unless the point of union is at the ground, and the trees having been grafted, and a terminal bud on graft has grown. If the young trees have been budded, the trunks will not be straight; a bend will be seen at the point where the bud was inserted (see Fig. 27), and the scars of the union of the veneer-shield or annular bud and the point at which the stock was cut off will be distinctly noticeable. The bark above the point of union on the grafted or budded stocks will be different from that below. There is something characteristic about the color and appearance and the number, size and shape of the lenticles of each variety of pecan, and while it is impossible to describe this difference in appearance (it can only be learned after a large amount of experience and observation), yet the very striking difference between the seedling stock and the wood of the variety worked upon, will serve as a useful index to the genuineness of the trees in question.

If the trees have been grafted instead of budded, the same statement will be true of the appearance of the bark. But the tree will be more nearly or quite straight, and the marks and scars at the point of union will be different. If the trees have been propagated by whip-grafting, the scar will be shaped like the letter N, the scar on young trees coming nearly or quite the whole distance across the stock. If the trunk of a whip-grafted tree is split through the point of union, the N-shaped mark in the form of a dark line may be distinctly made out, as shown in the illustration. In trees propagated by cleft-grafting, the union scar will be long, slim and V-shaped.



Fig. 28.
External and Longitudinal Interior
View of Whip-graft Union.

But to make the similarity between the bogus and genuine trees more striking, the practice has been resorted to of scarring the stocks so as to make them resemble the genuine article. This we have known to be done, more particularly in the case of budded trees. Incisions were made in the trunks of seedling trees to resemble those made in inserting a veneer-shield or an annular bud. The incisions were made so as to include a bud, and the top of the seedling tree was then cut off just above the bud. A tree doctored in this way makes a very close imitation of the real article, and the buyer needs to be on his guard. But the appearance of the bark, as already noted, will serve as a guide. If in doubt, it may be well to sacrifice a few trees and cut them carefully open down to the pith just through the point of union. *If the trees have been doctored, the tissues of the wood and the pith will be continuous; but, if the trees are genuinely budded or grafted, the tissues and pith will not be continuous.*

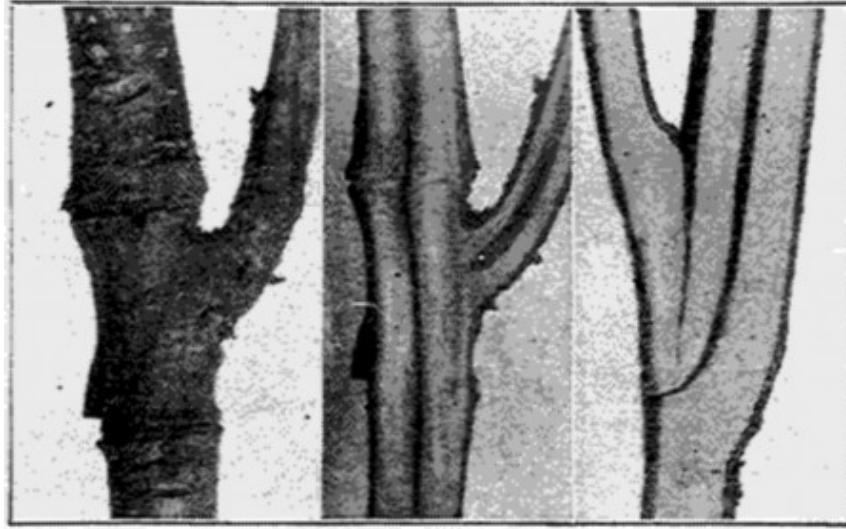


Fig. 29.
Annular bud growing (left). Split through same
union (center). A normal branch union (right). Pith
non-continuous (center); continuous (right).

Finally, if still in doubt, send two or three trees to the botanist or horticulturist of the Experiment Station of your State, and ask his opinion.

PLANTING PECAN TREES.

Too often but slight attention is given to this important piece of work. There is too frequently a disposition on the part of the person setting trees of any kind to do the work as rapidly as possible without consideration for the future welfare of the trees. Few realize that time spent in careful, intelligent preparation of the soil and in setting the trees is time well spent, and well paid for in the after development of trunk and branch. Better a month spent in preparing the future home of the young tree, than years of its life spent in an unequal struggle for existence. More than that, the tree may die outright, and a year must elapse before it can be replaced. It is generally stated that the pecan is a slow grower, and yet I have seen trees from twelve to fourteen years old which measured from thirty-five to fifty-seven inches in circumference at the base, while under less favorable circumstances others stood still for a period of six or seven years, or until they had accumulated sufficient energy to overcome the untoward conditions of their environments.

Time. The best time to plant pecan trees is during the months of December, January and February. Planting should not be delayed

until late in spring, as the percentage of loss will be very materially increased. Preference must be given to the earlier portion of the planting season, as the wounds on the roots will have had time to callous over, and the ground will be firmly packed about the roots by the winter rains. Then, with the opening of the growing season in spring, the trees will be ready to make a good, vigorous start.

Distance Apart. The distance apart at which the pecan trees should be set must depend upon the character of the soil and the amount of fertility and moisture it contains. If planted too close, the trees may become their own worst enemies. Too close planting will not prove satisfactory. It is doubtful whether the trees should ever be planted closer than forty feet apart even on light lands, while on heavier soils this distance should be increased to sixty, seventy-five or eighty feet.

TABLE OF DISTANCES		
DISTANCE.	NO. OF TREES	NO. OF TREES
	Rectangular System.	Hexagonal System.
40 x 40 feet.	27	31
40 x 50 "	21	
40 x 60 "	18	
50 x 50 "	17	19
50 x 60 "	14	
60 x 60 "	12	13
60 x 70 "	10	
70 x 70 "	8	9
80 x 80 "	6	
100 x 100 "	4	

To find the number of trees that can be set on an acre for any distance, not given in the above table, multiply the distance apart in feet together and divide the product into 43,560, the number of square feet in an acre. The result will be the number of trees which can be put on an acre of ground.

PLANTING SYSTEMS.

For setting orchards a number of different systems may be used, but the two best adapted to the pecan orchard, are the square or rectangular and the hexagonal or septuple. If mixed plantings, such as pecans and peaches, are to be made, then the quincunx system should be used and a peach tree set in the center of the square or rectangle formed by every four pecan trees.

Square or Rectangular System. In this system is included only the methods of setting trees in rectangles, either square or oblong. It is by far the most commonly used of all the systems, and the ease with which a field can be laid off in rectangles, is greatly in its favor.

The rows of trees intersect each other at right angles, and cultivation may be carried on conveniently either crosswise or lengthwise of the orchard. The planter has the choice of placing the trees the same distance apart both ways, or of planting them closer together in the rows than the distance between the rows.

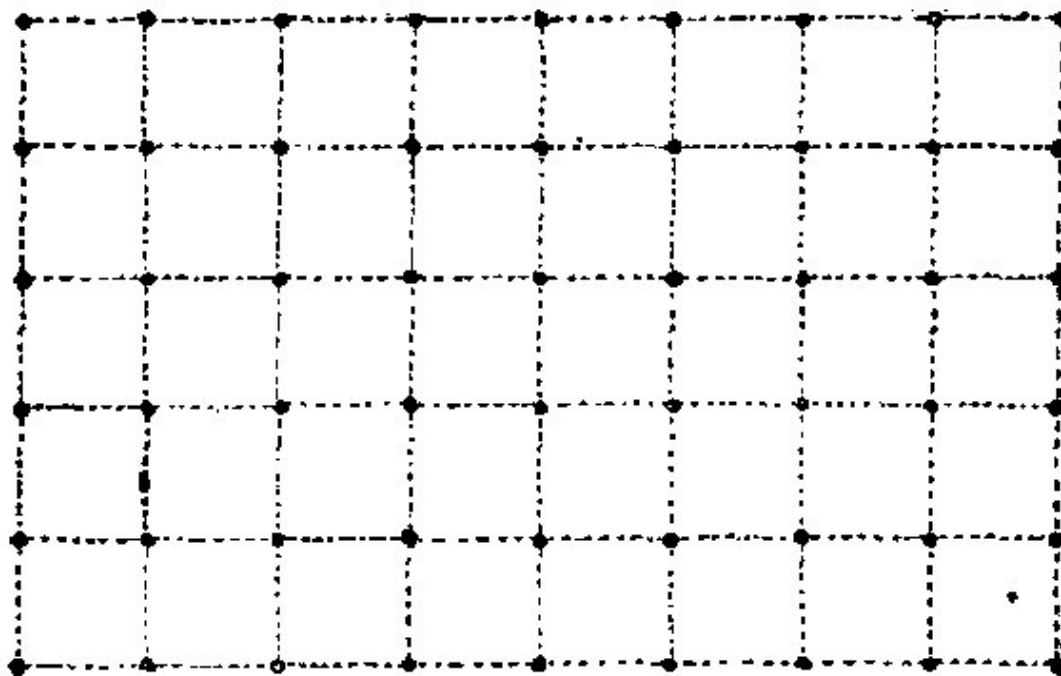


Fig. 30.
Rectangular Planting System.

It has been argued that space is not equally divided among the trees, and while this is apparently true, yet, on the other hand, the roots of pecan trees, in most cases, penetrate and permeate all the space allowed in ordinary distances. The roots will certainly secure all the food and moisture in the top two or three feet of soil.

When trees are to be planted by this system, the stakes must be set so as to be exactly in line, whether viewed from the end or from the side of the field.

Hexagonal, Septuple or Equilateral Triangle System. By this system, six trees are set equidistant from a seventh placed in the center. The basis of the system is not the square, but the circle, since the radius of the circle is approximately equal to one-sixth of the circumference of the circle. The name septuple, sometimes applied to this system, refers to the fact that the number of trees in each group-unit is seven. Equilateral triangle system refers to the planting of the trees in equilateral triangles, but is identical with the hexagonal or septuple.

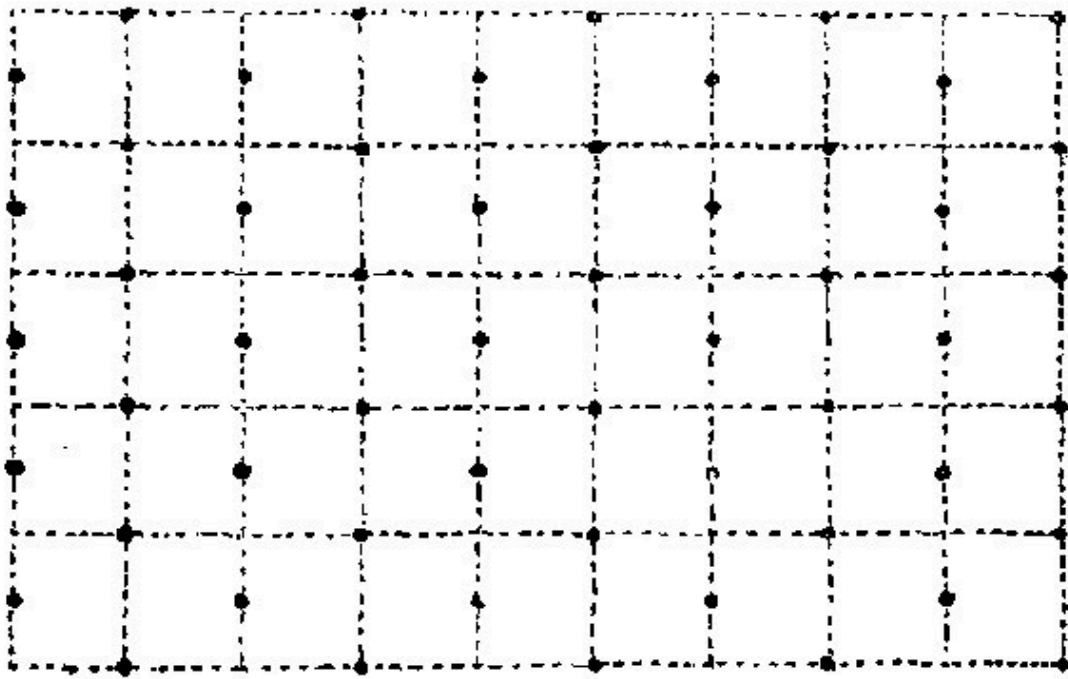


Fig. 31.
Hexagonal Planting System.

It is the only system whereby each tree is placed equally distant from each of its adjoining neighbors, and the only system which equally divides the space among the trees. By this method about fifteen per cent. more trees can be set per acre than by the rectangular.

For permanent plantings, at regular distances, this system and the rectangular should be recommended before other systems.

LAYING OUT BEFORE PLANTING.

Level and smooth the ground, harrow and pulverize thoroughly, then proceed to stake the ground off, placing a stake for every tree.

Laying Out Squares or Rectangles with the Plow. If a good plowman can be secured, very satisfactory work can be done with the plow. In some cases a man can be found who needs nothing in the way of a guide, except two or three stakes. But with a sufficient number of stakes and a marker attached to the plow, good results can be secured by almost any plowman.

Furrows should be run both lengthwise and crosswise of the field, their intersections marking the place where the trees are to stand. At each one set a stake.

It is essential that a true, square corner should be secured. This may be done by sighting with an ordinary carpenter's square set upon three posts.

Laying Out in Rectangles with a Wire. A wire, long enough to reach down one side of the field, should be provided. Stretch this straight out between two posts and mark off the distance which the trees are to stand apart, upon it. At each point marked, firmly twist a piece of small wire about the larger one. These should then be soldered in place. It will not do to have them shift. This wire may be rolled upon a roller when not in use.

Measure off along both ends of the field and set small stakes on the tree rows, at the marked places on the wire. Tightly stretch the wire down the first tree row, attaching it firmly at the ground level to a pair of good, stout posts. Then plant a lath stake at each mark on the wire. Set all of them on the outside of the wire, so as not to interfere with moving it. When this row is completed, lift the end stake with the wire attached, stretch on the second row, set the stakes as before and repeat the operations until the work is completed.

Laying Out in Hexagons. Stretch the wire down one side of the field and firmly set the tree stakes, or stake out the base line by any method, firmly setting a stake for each tree. Then procure two pieces of wire with rings at each end, the length of each wire and ring to be exactly the distance between the stakes as set on the base line. Stretch these wires out toward the side where the next tree row is to stand. At the point where the rings overlap set a stake for a tree. Remove wire number one and set it on the third stake in the base line, stretch the two tight and set a tree stake. Repeat as often as necessary. In setting the third row of stakes, use the second as a base line, and so on.

Planting the Trees. After setting a stake for each tree, the ground is ready for digging the holes and setting the trees. A planting board, such as is shown in the accompanying illustration, should be provided. It is made of a piece of inch board, four or five inches wide and five feet long. The ends may be notched or holes may be bored in them. In the center of one side, a notch, one and a half inches deep, should be cut. Provide a large number of small wooden pins or sticks, about one foot long and well sharpened.

When ready to dig a hole, place the planting-board so that the notch in the side fits against the tree stake. Then place one of the small pins in each of the holes or notches at the ends of the board. Allow these to remain in the ground. Remove the board and the tree stake and dig the hole.

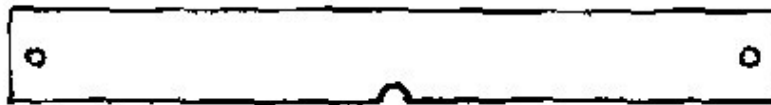


Fig. 32.
Planting-Board.

The hole should preferably be dug just before setting the tree. In some cases, however, it may be necessary to have all the holes dug in advance. Make them wide and deep, six or eight inches wider than the extended lateral roots and eight inches deeper than the length of the taproot.

In setting the tree, place the planting-board back on the pegs and place the tree at the right depth, against the notch in the side. It will

then stand exactly where a stake stood, and if the stakes were in line, the trees will be also, if they are kept perpendicular while the earth is being filled in. The earth should be packed about the roots by hand, the tree being set no deeper than it stood in the nursery.

To start the trees off well, one pound to one pound and a half of a good fertilizer, analyzing about six per cent. potash, five per cent. phosphoric acid and four per cent. nitrogen, should be thoroughly mixed with the earth that is used in filling in the hole. Preferably, only surface soil should be used to place about the roots.

When the hole is filled in about three-fourths, water may be applied to advantage, particularly if the weather is dry. A good application should be given after the work is completed, so as to establish the capillary movement of the water in the soil.

The greatest care should be taken to prevent the roots from becoming dry, if they do, the chances of their living, after planting, are very greatly reduced.

From the time the trees are lifted from the nursery row until they are set in the orchard, the sun should never be allowed to shine on them. Neither should they be exposed to hot or drying winds. Should it happen that the trees are received before everything is ready for planting them, they should be unpacked and healed in, in a shady place.

The roots of the trees must be pruned before planting, but this should be done under a shed. All broken parts of roots should be carefully cut off, leaving good, smooth surfaces, and the taproot cut or pruned back, as described in the chapter on pruning. When the pruning is finished, *the trees should be wrapped in a damp blanket or in damp sacks and taken to the field.* When needed for planting, they should be removed one by one and set out.

CHAPTER XI.

CULTIVATION AND FERTILIZATION.

Too many of our ideas of fruit culture are borrowed from the woods, from the trees in the pasture lands and uncultivated places generally. As the pecan is a forest tree in many sections of the country, the inference is, that it needs no cultivation, no fertilizer, in short, is amply able to take care of itself. So it is, but not able to yield, at the same time, the large crops of nuts that are the object of its being planted.

From the woods, there is one lesson which it would be well for everyone to learn; a lesson, not of the trees, but of the soil, of the dense mass of mold, of partially decayed leaves, of vegetable matter, of humus that covers the forest floor. The soil in the pecan orchard needs humus, vegetable matter; so does the soil in any other kind of orchard, and to obtain results it must be provided.

Now, it is a well-known fact that a number of years (ten or twelve) must elapse before a pecan orchard will begin to give any adequate returns for the time and care bestowed upon it and the money invested in it. During this period, if rightly handled, the ground may be made to produce something else than pecan trees, and that, too, without injury to them. But in growing a crop in the orchard, bear in mind that the trees need, and are benefited by, cultivation, and that fertilizer will make them grow.

But, as already noted, humus is needed, and since this is the case, corn or cotton or clean-culture crops, which leave little behind them to make humus after they are removed, should not be grown every year. Some of the legumes should be brought in. Cowpeas, soja beans, beggarweed, velvet beans, alfalfa and melilotus can all be grown in the pecan area. Not all of them in every locality, but some one or more of them in every section. To keep up the supply of vegetable matter, grow one of these leguminous crops every two or three years, or oftener, and after they have died and dried on the surface, plow them into the soil. And when corn is grown, sow cowpeas at the last working of the crop, to enrich the soil. These legumes will add nitrogen to the soil and help to reduce the fertilizer bills, for nitrogen is the costliest of all the fertilizer materials which we buy.

Sometimes, it will not do to crop the orchard. A condition may have to be met, in which there is not enough water to supply both the trees and the growing crops and one or the other will suffer—the trees, usually. In such a case the advisability of cropping is questionable unless, of course, water in sufficient quantity can be supplied by irrigation.

Small grains, oats, wheat, etc., should be rigidly excluded. When corn or cotton is planted, leave out a row or two of the crops where the tree row is. Let the trees have feeding space, but cultivate all the ground.

If the season is dry, then give cultivation just as often as can be done. Every week or ten days, between the first of April and the first or middle of July, the ground should be stirred in young orchards. Shallow cultivation is all that is necessary after the first plowing. A weeder or light harrow will do the work. This shallow cultivation will preserve a dust mulch, a couple of inches or so in depth, and the loss of soil moisture by capillary action and evaporation will thereby be prevented; more moisture will be retained in the soil and the trees will be benefited accordingly.

Whether the orchard is planted in a crop or not, cultivation should begin about the time growth starts in spring. The ground should be plowed and leveled with a cultivator. After that, frequent shallow cultivation should be given with a light harrow or weeder. Once every week or ten days, if the weather is dry, will result in much good to the trees. If a shower should fall during one of these dry periods, the ground should be cultivated just as soon as it can be worked. A light harrow, which will break up the surface crust formed by the rain and leave instead a shallow mulch of pulverized soil, will go a long way toward conserving and holding the water which has been added by the recent rainfall.

The cultivation of old orchards may vary somewhat from that given younger ones. Some recommend that the old orchard be seeded to grass (Bermuda or Johnson grass) and used as a pasture. This may answer in some cases, particularly on very rich, alluvial soils, but, in general, it will not do as a definite policy year in and year out. Those orchards planted in grass which the author has had an opportunity to examine, have usually shown a large percentage of trees with branches dead at the tips, "stagheaded," with yellow leaves and a general appearance of unthriftiness. It may have been that these orchards were planted in grass while the trees were too young. The better treatment, and the safer one to follow in old orchards, is to cultivate the ground in spring and sow down in cowpeas or some other legume. Beggarweed, velvet

beans or soja beans will answer well in many localities. Allow these to make what growth they will, and, when dead and dry, plow them back into the soil. It may seem strange to cultivate a forest tree, but it is the plan to follow to get results. Good results could doubtless be secured by seeding the pecan orchard in alfalfa and using it for a hog pasture up to the ripening season.

Cultivation should not be prolonged too late. If it be, the trees will continue to grow later than they should. Enough time will not be left in many sections before the coming of the first frosts. If the immature, sappy wood is caught by an early frost, severe injury may result. In the more southern extension of the pecan area cultivation can be carried on later than toward the northern limits of the region. Ordinarily, it is safest to cease cultivation not later than July the first to July the fifteenth.

FERTILIZATION.

On deep rich, alluvial soils the trees may not need to be fertilized, but many of the soils on which pecans have been set in orchard form, require to be fertilized to secure the best results. The three important plant foods required by plants and most frequently deficient in soils are nitrogen, phosphorus and potash. One or two or all three of these substances may have to be supplied.

Nitrogen, which is used by the trees largely in making growth of leaf and wood, may be supplied from a number of different sources, viz: stable manure, cotton seed, cotton-seed meal, dried blood, fish scrap, sulphate of ammonia and nitrate of soda. These substances are the principal commercial sources of nitrogen. Large amounts of nitrogen are gathered by leguminous crops; cowpeas, vetch, beggarweed, velvet beans, alfalfa and others may be planted to advantage, resulting in a great saving in fertilizer bills, and besides, adding the necessary vegetable matter and humus.

The most common source of phosphorus, usually referred to as phosphoric acid, is acid phosphate. Some is obtained from bone, and bone meal is a good fertilizer to use among pecan trees. The results obtained from its use are not immediate, but since the bone does not decay rapidly, they extend over a considerable period. On the whole, acid phosphate is as satisfactory as any material as a source of phosphoric acid, and the goods with the highest percentage are usually the most economical in the end. A good grade is that analyzing fourteen per cent.

Potash may be purchased, as kainit, the raw salt, or as muriate of potash, low grade sulphate of potash and high grade sulphate of potash. Of these the sulphates are usually given the preference in fruit growing. Of the domestic sources of potash, woodashes are important.

The amount of fertilizer which it is best to apply is difficult to decide upon; much depends on the character of the soil, what crops are cultivated and whether a crop of legumes is grown or not.

If legumes are grown for the benefit of the orchard, they should be fertilized, and if the crop is turned back into the soil, this may be sufficient for the trees, particularly while they are young. For the legumes, a good fertilizer to use per acre is:

Kainit, 100 lbs.; Acid Phosphate, 200 lbs.

or, High-Grade Sulphate of Potash, 50 lbs.

Acid Phosphate, 200 lbs.

In any case some allowance should be made for the amount of nitrogen collected by the legumes. When corn, cotton or some other crops are grown in the orchard, fertilizing may simply consist in distributing an additional amount of the crop fertilizer for the benefit of the trees.

For the growth of the young trees, a larger amount of nitrogen and a relatively smaller amount of phosphoric acid and potash are required, while for older trees, the reverse is true. Phosphoric acid and potash are required by bearing trees for the formation of fruit. Consequently, when the pecan orchard comes into bearing, these materials should be increased in the fertilizer applied. If the soil is not very rich at the time of planting, good results will follow the use of a pound of good commercial fertilizer at the time of planting.

A good fertilizer for young trees should analyze five per cent. phosphoric acid, six per cent. potash and four per cent. nitrogen. For bearing trees, one analyzing eight per cent. phosphoric acid, ten per cent. potash and four per cent. nitrogen will give good results. If so desired, well-known brands of commercial fertilizers, having approximately the above analysis, can be purchased in the markets, but if preferred, the several materials may be purchased separately, then mixed and applied.

APPLYING THE FERTILIZER.

The roots of young trees do not extend to any great distance away from the trunk. In distributing the fertilizer this fact should be remembered. A safe rule for all small-sized trees is to commence just outside an imaginary circle of two feet radius and apply the fertilizer in a circular band extending out some distance beyond the spread of the branches. Old trees, or those having a considerable spread of top, when planted in orchard form, should be fertilized by broadcasting the fertilizer over the ground. In the northerly pecan sections, all the fertilizer should be given in one application, about the time growth starts in spring, and plowed in, while farther south, two applications may be made, one at the time mentioned above, the other from the first to the middle of June.

CHAPTER XII.

PRUNING.

The pruning of the pecan is neither difficult nor complicated. In short, after the top of the tree is well started, little need be done except to cut back a branch here and there that the trees may develop well-rounded, symmetrical tops. A splendid type of tree is shown in Plate VI.

HIGH VS. LOW-HEADED TREES.

Frequently trees are so pruned that their first branches are eight or ten feet from the ground. Even young trees are pruned to slim stems, surmounted by a small umbrella-like top. Such trees frequently have to be tied to a post to keep them upright until such time as they attain sufficient size to support themselves. Such pruning should not be countenanced. The trees will make a much more rapid and satisfactory growth, and their trunks will be less affected by the hot sun, if the branches are allowed to develop lower down. Sometimes the system of pruning pecans with tall, bare trunks is adopted to allow of crops being grown under the trees, or because it is desired to use the ground as a cattle pasture. These considerations should not weigh against the welfare of the trees. As much ground can be cropped around low-headed trees as is good for them, and, in brief, the cows should be pastured elsewhere.

Ordinarily the top of the tree should be so shaped that the lower branches will be about four feet from the ground. The trunk will be shaded and protected, the crop will be nearer the ground, and the low tops will be less subject to the destructive force of heavy winds, so injurious to both fruit and branches.

To start the trees at four feet, the tops must be cut back to that height at the time the trees are set, or, if smaller, when they have grown to that height. Three or four buds nearest the top should then be allowed to develop and form the main framework of the tree. After this the trees will need little or no pruning, except the cutting back of straggling branches, and the removal of dead or broken ones.

Some writers have advised the persistent and severe cutting back of the tops, from time to time, so as to keep them small, compact and low, but such a system

of pruning must be put into practice on a considerable scale for a number of years before it can be recommended. Such a plan might prove valuable where the trees are subject to the force of strong winds, but otherwise it is of doubtful value.

TIME TO PRUNE.

Pruning may be done at any convenient time, but the best period is probably either just before the flow of sap in spring, or just after the trees have fully developed their leaves in spring. Following the removal of branches of any considerable size—three-quarters of an inch and upward—the wounds should be carefully painted over with white lead paint to prevent decay.

CARE OF BROKEN TREES.

When trees are broken or injured by wind-storms, the broken branches should be cut off and the resulting wounds carefully trimmed and painted. If the branches are only partly split off, the injury may be repaired, in many cases, by pressing the branch back into place and bolting it there, so as to hold it firmly in place. Trees with forked trunks should be protected by passing a bolt through the two branches some distance above where they divide to prevent splitting.

NURSERY ROOT-PRUNING.

Too frequently the root system of pecan trees, intended for planting, is but poorly developed. The root consists almost entirely of one large taproot destitute of laterals. Such trees are slow in starting and are hard to transplant. Figure 33 shows an excellent root system on a nursery tree. Such a tree should be almost as easily transplanted as an apple tree. A little more care on the part of nurserymen would insure good root systems.

In a former publication it was suggested that the young seedlings intended for stocks be root-pruned "in the fall, after the trees are one year old. It could easily be accomplished by running the tree-digger down the row at a depth of nine or ten inches. The taproots could thus be severed, and the following spring, or summer, the trees could be worked (budded or grafted). This course of treatment would insure greater success in transplanting, as it would have a tendency to develop the lateral roots; and in addition to that, it would, in all probability, induce earlier fruiting."

ROOT TRIMMING BEFORE PLANTING.

Two year old taproots should be cut to eighteen or twenty-four inches; larger ones, in proportion. The old idea that transplanted pecan trees, the taproots of which have been cut back, will not live and bear, is not borne out by experience. They are in no-wise injured by its partial removal, and it might all be removed were it not that so many would die in transplanting.



Fig. 33.
A nursery tree with a good root system.

Figure 34 shows two pecan trees at two years. The one on the right was carefully lifted so as to preserve as much as possible of the taproot, while the one on the left had the taproot cut when it was transplanted at one year. In the latter, six small roots from four and one-half to eight inches in length had grown out to replace the taproot, these doubtless having supplied the tree with as much nourishment as would have been given by its single taproot. Furthermore, without doubt, one of these roots would have grown so as to replace the taproot.

The advice has been given to cut the taproots back to five or six inches, but under general average climatic conditions throughout the pecan region anyone who follows this advice will have reason to regret it. Our experience in transplanting pecan trees has been such as to indicate the necessity of having a well-branched, well-developed root system, and a taproot, when present, should be left at least as long as indicated above.

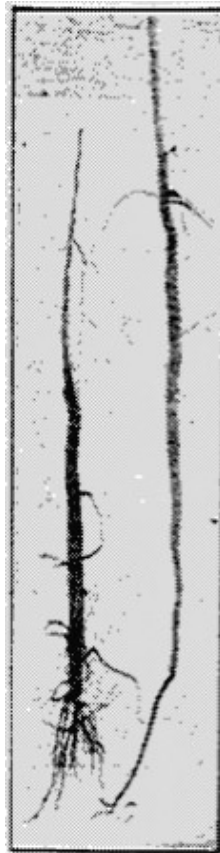


Fig. 34.
a. Taproot
cut at 1 yr. b.
Taproot not
cut.

A long taproot is objectionable on account of the additional cost and labor entailed in digging holes of sufficient depth for planting. To shorten the length of the taproot, Mr. E. E. Risien, of San Saba, Tex., has patented a method which has given satisfactory results. The nuts from which the stocks are grown are planted over strips of mosquito netting, the netting being some distance below the level of the nuts. When the taproots have penetrated to the netting, their growth is stopped, and the lateral roots develop better in consequence.

PART IV.

Harvesting. Marketing.

CHAPTER XIII.

GATHERING, STORING AND MARKETING PECANS.

While, in preparing a crop of pecan nuts for market, such extreme care need not be exercised as in handling a crop of peaches, plums or oranges, still there are a number of details which require careful attention to secure the best results. Careful attention to these few points is quite as necessary as in handling any other fruit crop, though it might appear otherwise.

Time to Gather. As a rule the bulk of the nut crop must be disposed of before Thanksgiving, and there is in consequence a strong disposition to gather the crop anyway, whether ready or not. Much might be said on both sides of the question, but in general it must be granted that gathering the crop while still somewhat immature, and beating the trees to cause the nuts to drop, cannot be commended.

When the great majority of nut husks are open, the crop of the tree is ready to be harvested. It will not do to wait until every burr is open (some varieties never open, but such are extremely undesirable), for it will usually be found that by far the most of those which do not open, on trees which open their burrs uniformly, are faulty, and it will not pay to wait for them. Neither should such be left on the tree, but the whole tree should be stripped at the time already indicated. It will be necessary to use light bamboo poles to remove the nuts with closed burrs.

Picking. The nuts must either be picked by hand or knocked off the trees onto the ground with sticks. From whatever standpoint we may regard the gathering of the crops, in orchards of good varieties, the best plan for the removal of the nuts is to take them off, in so far as possible, by hand. Men should climb the trees and collect the nuts in sacks. Men provided with sacks can, with the help of a good extension ladder, reach the most of the nuts on ordinary trees, up to forty or fifty feet in height. A good man will pick one hundred pounds of the shelled nuts in a day, at a cost of one dollar—or one cent per pound.



Fig. 35.
After the Harvest.

In gathering the crop, the product of each individual tree, in the case of heavy-bearing seedlings, or of each group of trees of a single variety of grafted trees, should be kept in a single pile or lot. It will not do to mix nuts of different sizes, shapes and colors, if the best price is to be hoped for.

Curing. As soon as removed from the trees the nuts should be carried to the curing house. This house should be absolutely rat-proof. Here they are to be picked from the hulls, the unopened burrs being placed apart by themselves. If they open later, well and good; some good nuts may be found among them, but usually they are inferior and should be kept strictly apart from the other portion of the crop. The cost of removing a hundred pounds of nuts from the hulls is about fifty cents.

As soon as the nuts have been separated from the hulls, they should be spread out in shallow trays for curing. These trays should be two and one-half or three feet wide and four or five inches deep. The bottoms are best covered with wire netting with meshes about one-half inch square. They may be arranged around the walls of the curing room, one tier above another. The room should be provided with good ventilation so as to give a free circulation of air. In the trays

the nuts may be placed two or three layers deep; if placed too deep there is danger of their moulding. They should be turned over from time to time, and, under average conditions, two weeks will be sufficient to cure them thoroughly.

Grading. Before packing for market, the nuts should be carefully graded. Too much attention cannot be given to this detail. Rigid grading pays—it pays handsomely, and the more abundant the supply, the better it pays.

It will not do to mix together nuts of all sizes, shapes, and colors—some small, some large, some pointed, some blunt, some dark, some light, some streaked, and then expect to get the full value of the crop. It cannot be done with a good grade of pecans.

Perhaps in no kind of fruit which is placed on the market can a more nearly absolutely uniform grade be made (see Frontispiece). The variety should be the basis of the grade. In gathering the crop, each variety should be put by itself as it is gathered. In most varieties the size is quite uniform, and little else need be done; but if there is any considerable variation in size, the small ones should be removed from the first grade of nuts.

Polishing and staining should not be done. It is always best to let each variety retain its own individual marks and characteristics. These are a part of the market quality of the variety and should, by all means, be retained. Mixed lots of seedling nuts may be polished to render them more uniform, but the staining is an abomination, though some people would rather have it, not knowing, perhaps, what a pecan looks like without it.

Shipping Packages. The package should be strong and light, and should afford ample protection to the product. We have known pecans to be shipped by mail, freight or express, in bags, and losses have occurred. Barrels for larger shipments, and wooden boxes for smaller ones are best, and afford the necessary protection. Gift packages, holding ten or twenty pounds or even more, should be made of half inch stuff at least, with ends three-quarters or one inch thick. Grocery boxes may be cut up, planed off, and made over. In all cases the packages should be neat and clean, and in perfect keeping with the contents. The name and address of the grower, the name of the variety, and the number of pounds should be neatly stamped on the outside.

Marketing. As it is at present, so will it be for many years to come, strictly first-class pecans will be handled almost entirely by or through a private trade. We know of several growers who dispose of their crops of several thousand pounds

annually to private customers who have learned the value of good nuts. So greatly has the demand increased that in no single instance is anyone of these men able to supply the demand of the natural outgrowth of his own work, and orders are usually booked a year or more in advance. This is the ideal method of handling the crop, and the one method which enables the grower to secure the best price for his product.

In building up such a private trade, advertising must be resorted to, either through the newspapers, magazines and other channels, or by distributing samples of nuts. "Once a customer, always a customer" should be the motto for the grower to hold in mind, and every effort should be made and every precaution taken to *see that the nuts, from year to year, are absolutely uniform in size, shape, and quality*. Do not send a customer one size, shape, or quality one year, at a certain price, and the next year vary it. Such treatment will tend to make customers dissatisfied, and the grower may lose them entirely. This point cannot be too strongly emphasized.

Strictly first-class nuts may be disposed of to advantage to the first-class grocery or fruit trade in the larger cities. In cities of any considerable size, there will always be found a grocer or fruiter who is willing to take a first-class article at a price considerably above the usual market price of ordinary nuts. The writer once submitted samples of nuts of medium, but uniform size and good quality, to a grocery firm in New York. They replied that they would take nuts like the samples at twelve and a half to fifteen cents a pound in carload lots, when the common run of pecans could be purchased at four or five cents per pound.

As the output of high-grade pecans is increased, they may be disposed of through the usual nut trade channels—the commission men. The bulk of the product in the country to-day is handled by commission men, either being purchased direct or sold on consignment. If sold for cash in the home market, well and good, but if sold on consignment, choose one reliable commission house in each city in which the product is to be marketed—never two in the same city—and ship to it right along.

Storing. During the cold weather following the gathering of the crop, little or no change takes place in the flavor of the kernels. During the heat of summer, however, they deteriorate. The natural amount of moisture in them is reduced, the air enters, oxidation takes place and the flavor becomes rancid.

These changes can be prevented if the nuts are kept in cold storage, say at a temperature of from thirty-five to forty degrees. When nuts are kept in the

house, they should be stored in the coolest possible place, in sealed jars or tight boxes.

PART V.

Diseases. Insects.

CHAPTER XIV.

FUNGOUS AND OTHER DISEASES OF THE PECAN.

The fungous diseases attacking the pecan have not been thoroughly investigated. They have not, however, become so numerous or common as to cause serious damage except in a few instances. The true fungous diseases are usually propagated and disseminated by means of spores, and the most effectual method of control usually consists in spraying with Bordeaux mixture or some other fungicide. For all fungous diseases of the pecan which may be controlled by spraying no substance will give better results than Bordeaux mixture, and directions for preparing it are given at the end of this chapter. Paris green, at the rate of four ounces to each fifty gallons of liquid, may be added to the mixture for the destruction of biting insects. For effectual work in spraying large trees, a platform should be erected on the wagon-bed to make it possible to reach the tops with the spray.

PECAN LEAF BLIGHT (*Cercospora Halstedii*): This disease of pecan leaves causes them to turn brown, wither up and drop prematurely. At first, small brown spots are noted. These become larger, and at length the whole leaf is destroyed. When attacked by this disease the tree makes no progress. An examination of the discolored areas, under a microscope, shows the presence of tuft-like growths of spores upon short conidiophores. As they become matured the spores are scattered by the rain or wind and so the disease is spread. It probably lives over from one season to another on the diseased leaves.

The most effective remedy is to spray thoroughly three times with Bordeaux mixture. The first application should be given just when the young leaves are expanding, followed by two others at intervals of two or three weeks. The fallen leaves should, if feasible, be gathered and burned.

Pecan Scab (*Fusicladium effusum*): This disease attacks the fruit, leaves and twigs. The husks of the diseased nuts become covered with dark spots or specks. They become hardened and crack open in

places. As a result of the attack, growth is stopped, the fruit does not fill out and mature, but drops prematurely or, in some cases, remains attached to the trees long after the leaves have fallen. Round, black spots form on the leaves when attacked by the fungous. These become dead and brown and in most cases the whole leaf is destroyed. When attacked, the trees are usually so badly injured that they make little progress. Not all varieties are subject to the disease in the same degree and some appear to be entirely exempt.



Photo by H. A. Gossard.

**Fig. 36.
Spraying Pecan Trees.**

Those varieties which are not attacked should be given preference in propagating work. The disease may be further controlled by spraying with Bordeaux mixture, as directed under leaf-blight.

PECAN ROSETTE: [L]"The earliest symptoms are a peculiar crimping of the leaves at the ends of the branches. These leaves are smaller with crimped margin, and when held to the light show light green or yellow streaks between the veins. The leaf tissue in these light-colored areas is thin and undeveloped and often breaks away leaving angular holes in the leaves. A tree usually shows the disease over the whole top at once, though sometimes only a single branch is affected at first. As the disease progresses, the foliage assumes a bunched appearance, due to the formation of tufts of leaves at the

ends of the branches. This characteristic has led us to use the term "Rosette" as a name for the malady.

"The next stage of the disease which is observed the second year or later, is a dying-back of the branches from the tips. This is followed by the development of numerous small, lateral branches from adventitious buds. These are short, producing thick clusters of small, unhealthy leaves, sometimes reduced to mere skeletons, so that the rosetted appearance of the tree is intensified. This goes on from year to year. The growth of the tree is checked and these abnormal branches are formed only to die back each year. Trees in the earliest stages of rosette have been observed to have light crops of nuts, but, when badly diseased, are barren and unsightly or worse. Rosette has been found in all ages, from nursery stock to trees forty feet high.

"The cause of the disease remains a mystery. No fungous or other parasite can be detected in the earliest stages. The appearance of the trees leads us to infer that the trouble is internal, due to some derangement of the nutritive or assimilative functions of the plant, but we are unable to correlate this with any corresponding external conditions. That is to say, that so many cases have been observed on fertile soil, when cultivation, drainage and plant food had all been provided, that it is impossible to conclude that the disease could be due to starvation or to the lack of any single element in the soil, nor can it be due to over-feeding, since it occurs in light soils and in neglected orchards.

"It seems probable that it will be classed by the plant pathologist with peach rosette, peach yellows, and related diseases, the causes of which still remain unknown after years of investigation. The indications are that it is contagious, though a complete demonstration of this point remains to be made; at any rate, it must be regarded with concern until more knowledge is available."

The best recommendation that can be made in regard to pecans affected by this disease is to dig them up and burn them.

Bordeaux Mixture.

Copper sulphate, 5 pounds.

Lime (unslacked), 5 pounds.
Water, 50 gallons.

Dissolve the copper sulphate in two gallons of water, place it in barrel No. 1 and add water to make twenty-five gallons. Slack the lime, reduce it to a very thin paste, place it in barrel No. 2 and add water to make twenty-five gallons. To mix the solutions of lime and copper sulphate, dip a bucketful from each barrel, and pour together into the barrel of the spray pump. *The two mixtures should flow together as they are poured into the barrel.* This is one of the secrets of making a first-class mixture. The best arrangement is to have the barrels, Nos. 1 and 2, elevated, and use a piece of rubber hose to run the liquids into the pump barrel.

If a large amount of spraying is to be done, a somewhat different policy should be pursued. Too much time would be taken up in preparing the ingredients in small quantities. Instead, large amounts of copper sulphate should be dissolved and large quantities of lime slacked beforehand. This may be done as follows:

In a fifty-gallon barrel place about forty gallons of water. Put one hundred pounds of copper sulphate in a sack and suspend it in the water. As soon as dissolved, fill up to the fifty-gallon mark. When well stirred, each gallon will contain two pounds of copper sulphate. Each time some of the solution is dipped out, the height of the remaining portion should be marked on the inside of the barrel. Before taking more of the solution out of the barrel, any amount of water lost by evaporation should be made good by filling up to the mark last made.

As soon as procured the lime should be slacked, placed in a barrel and kept covered with an inch or two of water. In this way it can be kept indefinitely.

To prepare Bordeaux mixture from these stock solutions, dip out two and a half gallons of the copper sulphate solution, place it in barrel No. 1 and dilute to twenty-five gallons. From the slacked lime take fifteen pounds, or thereabouts, to allow for the water it contains, reduce to a thin paste, place it in barrel No. 2 and add

water to make twenty-five gallons. Pour the contents of barrels Nos. 1 and 2 together, as already directed.

Tests: If free copper be present, severe injury may be done to the foliage or other tender parts of the plants. Sufficient lime should be added to neutralize it.

Dip out a small quantity in a porcelain saucer or shallow bowl, and holding it on a level with the mouth, blow the breath gently into it. If a thin pellicle forms on the surface, more lime must be added. Add and test until it does not form. An excess of lime will not hurt.

Another test is to dip the blade of a clean knife into the mixture. If a thin film of copper forms on it after holding it there a minute or so, more lime must be added.

Use good materials and prepare the mixtures thoroughly.

In making up the various mixtures, never use iron vessels, but use glass, wood or crockery receptacles instead.

Strain all mixtures thoroughly into the spray pump to prevent clogging of the pump or nozzles.

Spray thoroughly and in good season. *Be in time.*

Do not use mixtures which have been leftover and allowed to stand for some time.

FOOTNOTES:

[L] Orton, W. A., proceedings second annual convention National Nut Growers' Association, 1903, p. 82. 1904.

CHAPTER XV.

INSECTS ATTACKING THE PECAN.

Some time ago the statement was occasionally made that the pecan had no known enemies. This, to thinking and observing persons, was too good to be true, and fortunately the words, "no known," were inserted, for later investigations, particularly on the part of Profs. Gossard and Herrick, have revealed the fact that the pecan, in common with all other fruit trees, is subject to the attacks of insect and other enemies. But the outlook is hopeful, for we know of the abandonment of no fruit industry because of the attacks of insect pests, and the pecan industry is in no wise in danger of being abandoned because of their inroads.

FEEDING HABITS OF INSECTS.

If an insect is to be successfully controlled, the grower must know something of its life-history, and particularly of its feeding-habits. Careful observation of the insect, while at its work of destruction, will frequently give a clue to the method of control. Many insects, like the caterpillars of the pecan, bud-moth and case-worm, obtain their food by biting off pieces of the leaves or other parts of the tree and swallowing the solid particles. On the other hand, a number of insects, such as the scales and plant-lice, obtain their food by thrusting their small, bristle-like sucking tubes into the tissues of the leaves and sucking out the juices contained in the cells.

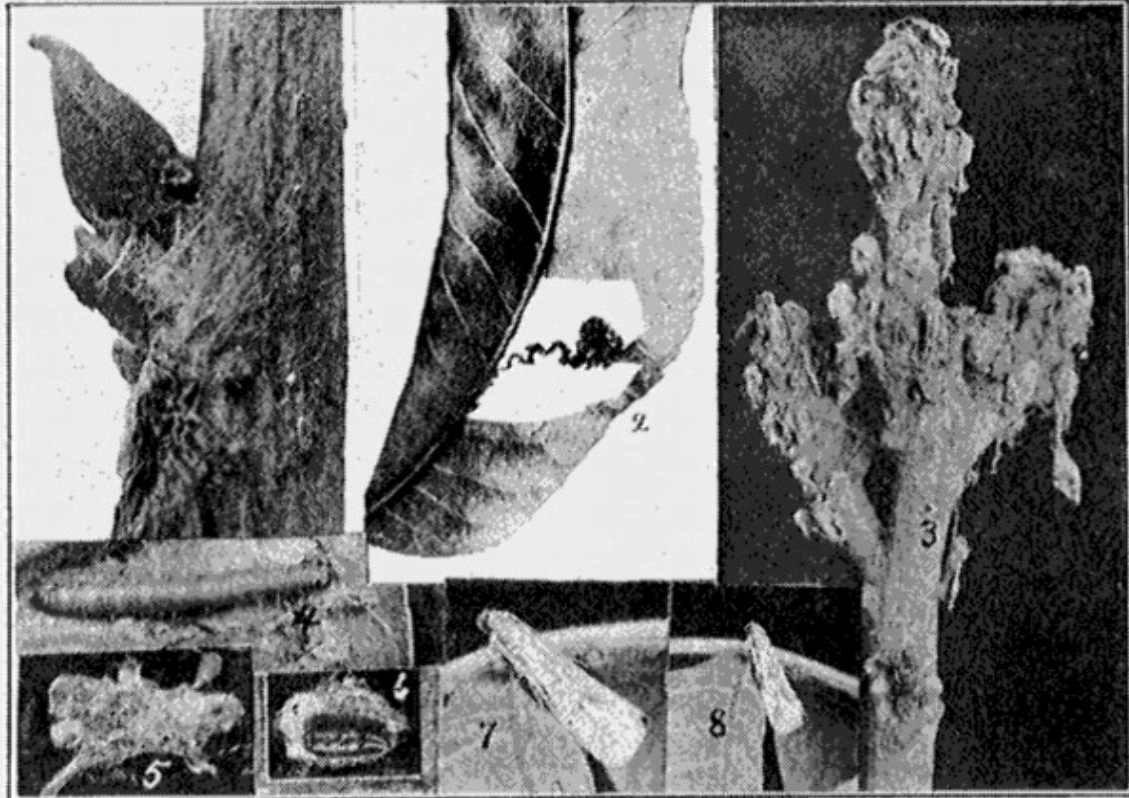


Plate VIII.
The Pecan Bud Moth (*Proteopteryx deludana*).

- 1. Winter stage on bud, enlarged. 2. Tube made in leaf. 3. Work of bud destruction. 4. Caterpillar, enlarged about twice. 5. Cocoon, enlarged. 6. Chrysalis, reduced. 7. Moth, enlarged. 8. Moth, about natural size.**

It is quite obvious that these two classes of insects cannot be controlled or destroyed in the same way. Those which eat solid particles of food may, in most cases, be destroyed by applying some poisonous substance, such as arsenate of lead or Paris green, to the food which they eat. But those which obtain their food by sucking cannot be killed in this way. They can be destroyed, however, by spraying over their bodies some substance, such as kerosene emulsion, which will penetrate their bodies and so kill them. Or, they may be killed by suffocating them with a gas or by stopping up their breathing pores with some powdered substance, such as pyrethrum. Some insecticides, such as resin wash, act both as a caustic application and a suffocating covering.

For convenience in referring to insects which attack the pecan, we have grouped them as follows: (1) Insects attacking buds and leaves; (2) Insects attacking the trunk and branches; (3) Insects attacking the fruit.

INSECTS ATTACKING BUDS AND LEAVES.

THE BUD WORMS: At least two species of caterpillars are known by this name. The moth of one has been called the bud-moth. The caterpillar of the other has been called the case-worm. Prof. Gossard writes, that he unexpectedly found adult moths of *Proteopteryx deludana*, November 28th, 1905, and therefore believes, from this observation and other circumstantial evidence, that he was "mixed" regarding the autumn life-history of these insects, as set forth in Bulletin 79 of the Florida Experiment Station. He furnishes the following paragraph as a summary of what he can say of the bud worms:

"The Bud Moth, *Proteopteryx deludana*, is a serious pest, especially in young orchards. Sometimes, in such orchards, even when large, scarcely a tree can be found during the month of May that does not contain one or several nests. The caterpillars are usually found singly, each with one side of a leaf folded over it and fastened to form a tube, or sometimes two leaves are fastened together with silken bonds and the caterpillar feeds between them. As fast as the leaves it has attached become brown and die, it draws fresh leaves to the dead ones and fastens them there, thus gradually making a very conspicuous nest. The caterpillar is full grown during the last of May and the first of June when they transform into moths. Their pupæ cases are formed of silk and excrement, smoothly lined with silk and snugly hidden away in a nest of leaves. In about two weeks from the time of pupation, the moths appear. Early specimens have sometimes been hatched from buds, only partially expanded. They are small, about five-sixteenths of an inch in length and five-eighths of an inch across the expanded wings. In general color they are grayish, streaked and dotted with blackish-brown. A characteristic habit is to alight and rest on the tree trunk, head downward. The moths have again been observed in November, suggesting that there are two broods a year. Thorough, persistent spraying with arsenate of lead or Paris green, in April and May, ought to control this species."

THE CASE WORM (*Acrobasis nebullella*): This insect, often found associated with the bud moth, probably does more damage than any other pecan insect. The caterpillars are about five-eighths of an inch in length, a dirty brownish-green in

color, and live in silk-lined cases or tubes attached to the petioles of the leaves. From these they protrude themselves to feed. Frequently a pair of leaflets are tied together (Plate IX, Fig. 6), and between these the caterpillars live and feed upon the tips of the protecting leaflets. Opening buds, partially developed and full-grown leaves alike are destroyed. Earlier in the season, characteristic nests of partially eaten leaves, petioles and excrement are formed by several caterpillars tying the mass together with silk. In this nest they live and develop. The caterpillars pupate within their silken tubes, and the small gray moths (five-eighths to three-fourths of an inch in length) emerge about two weeks after pupation, chiefly in June. The small, hibernating "cocoons" found on and around the buds in winter and the tortuous tubes observed on the leaves in summer and fall, which have been referred to (*Proteopteryx deludana*), probably belong to this species. At least, caterpillars one-fourth grown and contained in cocoons apparently not essentially different from the smaller ones, contain worms having the characteristic appearance of the grown *acrobasis*. Spraying with arsenicals in April, May and June should destroy this pest. Spraying in late July and August would also promise results of value.

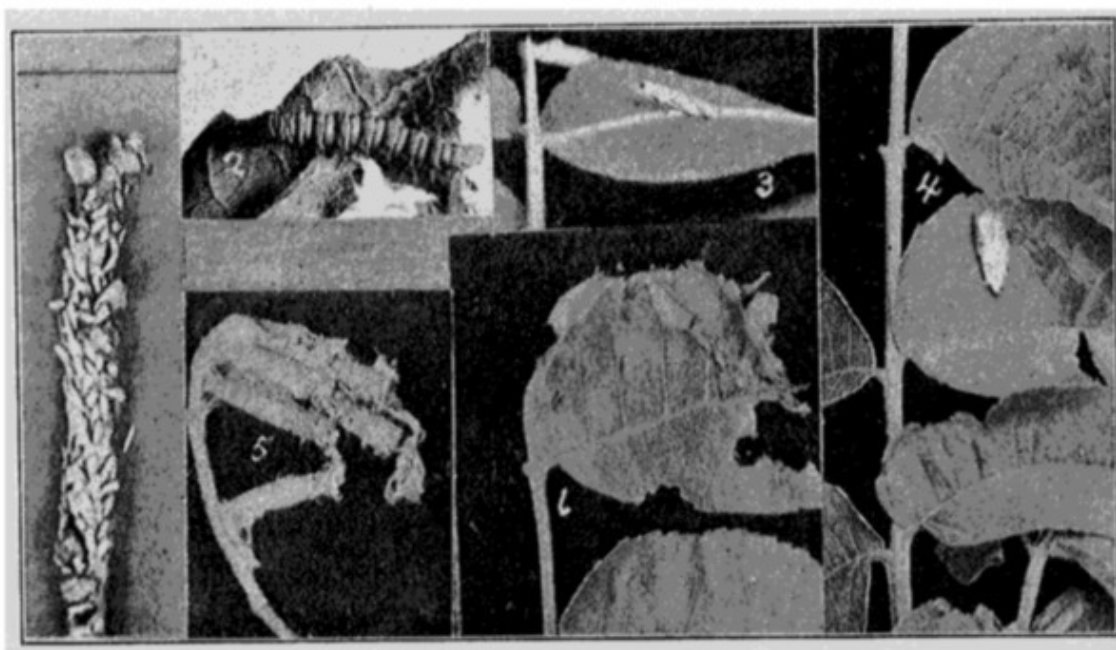


Plate IX.
The Case Worm.

- 1. Supposed winter stage. 2. Caterpillar, enlarged. 3-4. Moth, nearly natural size. 5. Cases. 6. Work on leaves.**

THE CATOTOCALAS (*Catocala piatrix* and *C. viduata*): The caterpillars of these insects are frequently found during April, May and June feeding upon the leaves of the pecan. They are ravenous feeders, and if present in sufficient numbers, considerable damage is done. The caterpillars are from two to two and a half or three inches in length when fully extended, gray and striped, leathery in appearance, very closely resembling the bark of the tree upon which they rest when not feeding. Having attained its full growth as a caterpillar, it ties together two or three leaves with strands of silk, thus making a loose cocoon within which it pupates. The pupa is dark brown, covered with a whitish or bluish-white bloom. In about one month the moths emerge. They are large in size, the body being one to one and one-fourth inches long and the expanded wings two and one-half to three inches across. When at rest they are dull gray in color, more or less marked with irregular waving lines. The hind or under-wings are strikingly different from the fore-wings. In *C. piatrix* they are deep orange-yellow marked from side to side with two black bands. The hind-wings of *C. viduata* are dark brown and edged with a narrow white band.

The caterpillars may be destroyed by spraying with some one of the arsenical poisons, or they may be removed by hand and destroyed. Prof. Gossard recommends the tying of a piece of burlap around the trees. Beneath this the caterpillars hide during the night and they may then be destroyed.

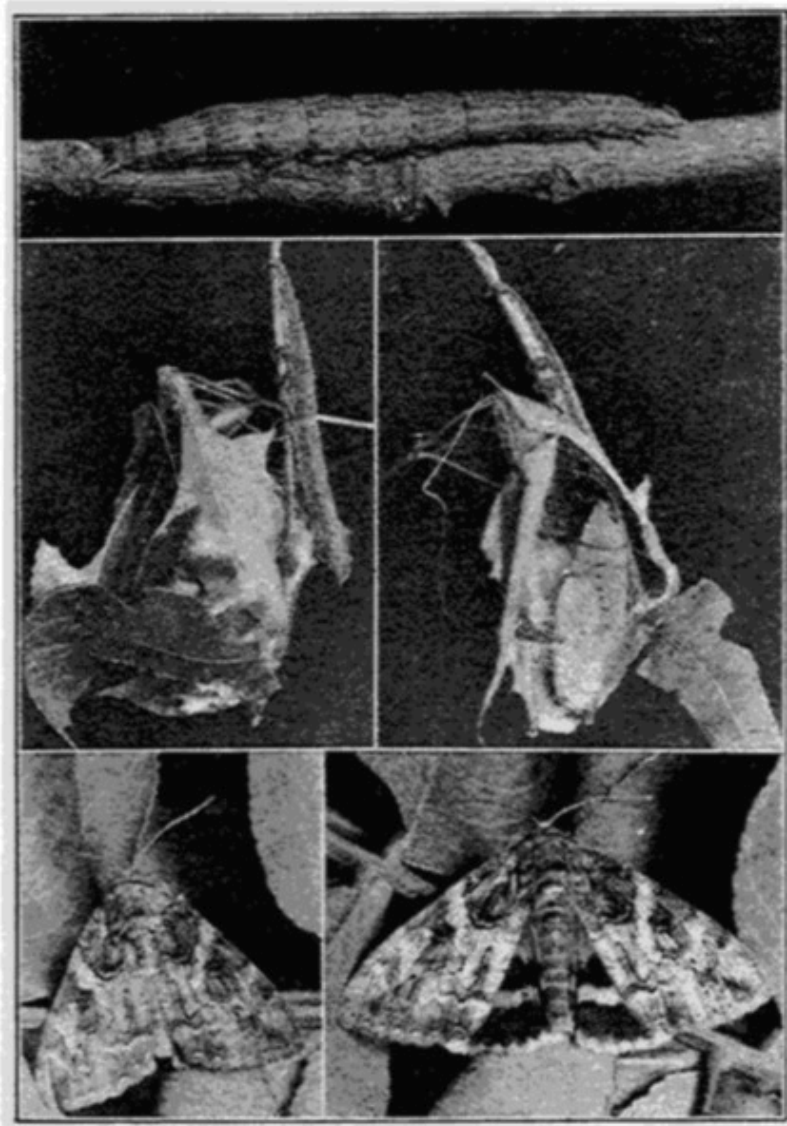


Plate X.
A Pecan Catocala. (C. Piatrix.)

**Caterpillar, Cocoon, Chrysalis, and Moths about
 one-half natural size.**

THE FALL WEB-WORM (*Hyphantria cunea*): The caterpillars of this insect begin work early in spring, shortly after the leaves are full grown. They work in colonies, and the leaves on which they feed are enclosed in a web, which is extended as the caterpillars grow or as they require additional leaves to feed upon. When full grown the caterpillars measure about one inch in length and are covered with hairs both long and short. The matured caterpillars leave the webs and crawl down the trees to hunt for places beneath the bark, under sticks,

weeds and trash in which to pupate. A light, flimsy cocoon, composed of silk and the hairs of the larva, is made. From this, in due time, a beautiful moth, an inch or an inch and a quarter across the wings, emerges. The wings are pure white or white spotted with black or brownish-black. The eggs are laid in masses of four or five hundred on the leaves. These hatch in about ten days, and the colonies of young caterpillars begin their work of destruction. There are two broods in the South each summer; the first appearing in May and June, the second in August and September. The fall brood hibernates in the pupa state.

The caterpillars may be destroyed on small trees by removing the webs and killing the larvæ. On large trees a torch of some sort may be used to burn the web and the caterpillars within it. They may be also held in check by applying a spray of Paris green or arsenate of lead at the time the broods are feeding.

THE PECAN CATERPILLAR (*Datana interrigma*): A buff-colored moth, having a body about one-half inch long and a wing expanse of one and three-fourths inches, with four transverse brown stripes on the front wings, lays its greenish or white eggs in clusters of five to twelve hundred on the underside of the lower leaves of the pecan trees. These eggs hatch in less than a week, and the colonies of young caterpillars at first feed upon the undersides of the leaves. They cast their skins four times, each time increasing in size and changing their color somewhat. The last moult, and sometimes the last two, take place on the trunk of the tree, and the clusters of discarded skins frequently remain for several months afterwards. After the last moult they ascend the trees, remain feeding for a short while, then go down to the ground to pupate. When disturbed, the larvæ raise both ends of their bodies from the twigs or leaves, on which they rest. They are easily recognized by this habit. When full grown they are one and one-half to one and three-quarters of an inch in length, covered with dirty white hair, and marked with two conspicuous longitudinal white lines, one on each side of the body. There are two broods, the last one hibernating in the ground in the pupa state.

The leaves on which the eggs are laid may be gathered and destroyed, or the colonies of young caterpillars may be gathered and burned. Later, they may be burned off with a torch, killed when clustered on the trunk during the last moult, or poisoned with an arsenical spray.

INSECTS ATTACKING THE TRUNK AND BRANCHES.

THE TWIG GIRDLER (*Oneideres cingulatus* and *O. texana*): These two insects frequently do considerable damage to pecan trees in late summer by cutting off the smaller branches. Branches from one-fourth to three-fourths of an inch are usually the ones attacked. The insect is a beetle, and the two species closely resemble each other. They are dark gray in color, one half to five-eighths inch in length, with antennæ longer than the body and provided with stout, powerful mandibles. The female insect cuts the branch by working round and round it until it is almost entirely severed. She then lays a number of eggs in it, usually one or two being placed near each bud. A small cut is made and the egg is inserted between the bark and the wood, and the opening is then sealed up with a gummy substance. As the insect moves along the twig a series of transverse cuts are made in the bark. The twigs usually drop to the ground. The eggs hatch as soon as the weather becomes sufficiently warm in spring, and the larvæ feed in the twigs, making tunnels through them as they grow. Later, they pupate within the tunnels and emerge during August and September as fully developed insects, having spent one year in their growth from egg to mature insect. It is believed that in some cases the life cycle lasts two years.

The best and most effective treatment is to gather and burn all the twigs which have been cut from the trees. This should be done, preferably late in autumn after the leaves have fallen, as there is greater certainty of getting all the severed twigs than if left until a later date.

THE OAK PRUNER (*Elaphidion villosus*): Sometimes^[M] pecan twigs, when smartly bent, will snap off with a clean, square cut across the branches, as if they were hollow-glass tubes, breaking at cracked or weakened places. An examination of such a broken stem shows "that its woody part, with the exception of a few fibers and the bark, has been cut across as if with a saw by a soft, yellowish-white grub, which can often be found in a burrow in the severed part. Since the uncut bark is the chief support left for the branch, any stiff wind or even its own weight will break it off as soon as it has become deadened. * * *

"The adult is a longicorn beetle, of slender, cylindrical form, over one-half inch in length and about one-eighth of an inch in width. It is of a dull, black color, tinged with brown on the wing covers, especially toward their tips. The underside of the body and legs are chestnut colored. Over all parts of the body can be found short, grayish hairs. Some small, gray spots on the wing-covers and a whitish dot on each side of the thorax are formed by dense

collections of gray hairs at these points. Coarse, round punctures are thickly sprinkled over the upper surface of the thorax and wing-covers.

"The larva, when grown, is about three-fifths of an inch long, tapering backwards from the neck. The body is divided by deep grooves into twelve rings or segments. There are three pairs of feet. The color is yellowish-white, the front of the head being blackish. Probably, about midsummer, with a possible variation of two months in each direction from this date, the parent beetle deposits her eggs, preferably on a small twig of the preceding year's growth. Upon hatching, the young larva commences to eat the tender wood just beneath the bark, and later enters the center of the twig and works toward its base. In this manner it works its way into the main limb, which may be of considerable size, and feeds within it for a period of about three years. The burrow thus becomes several inches in length, in many cases. Just before transforming to pupæ some, but not all, of the larvæ, cut the wood for the purpose of dropping the branches, as before described. Limbs in which the immature larvæ are working often break off with ragged end when bent with the hand.

"* * * Pick up and burn all fallen branches. Similar attention should be given nearby oak and hickory limbs, which have fallen."

THE PECAN TREE BORER (*Sesia scitula*): The moth of this insect is clear-winged and closely resembles the moth of the peach tree borer. Little is known of its life-history.

"It^[N] is probable that the eggs are deposited by the female moth on the bark of a tree near a fresh wound. For example, near newly set buds. The eggs hatch and the larvæ bore into the bark, and there live for a time, eating out the soft inner-bark and tender wood. It is certain that the borers live in these situations the over winter and change to pupæ in the spring, from which the moths emerge in April. The moths I reared appeared April 3rd, 4th and 6th. The pupæ are in cocoons, just under the bark. The cocoons are made from excrement and bits of bark that have been fastened together with silk similar to the cocoons of the peach tree borer. Whether these moths, that emerge in the spring, lay eggs and produce a brood in the summer, that in turn develops a fall brood of larvæ, I am unable to say."

"The^[O] young borer is apt to gain entrance to the sapwood through some wound in the bark, such as a graft-union, and here it feeds, sometimes completely girdling the sapwood above and below the wound. It is said to prefer to attack buds that have been budded on old, large trees. As a general rule the burrows ascend the tree in a spiral about the trunk, so complete girdling is unusual, but growth sometime ceases above the groove, new limbs being shot out from below."

The only satisfactory means of controlling this pest is to go carefully over the tree and dig out the borers. The trees should be examined from time to time in order to keep them free from borers.

INSECTS ATTACKING THE FRUIT.

THE PECAN WEEVIL (*Balantinus caryae*): In some localities considerable damage has been caused by the pecan weevil. The insect is a small, brownish-black snout beetle, somewhat less than one-half inch in length. The proboscis or snout is slender and as long as the body. With this proboscis the beetle bores a very small hole through the husk and shell of the immature pecan to the kernel, and at the bottom deposits an egg. This egg hatches into a larva, which feeds upon the kernel of the nut. In autumn the larvæ, when full grown, bore holes through the shells of the pecan and enter the ground in which they pass the winter. The next season they emerge from the earth as fully-matured insects, and about the month of August deposit their eggs in the nuts.

After the harvesting of the crop the hogs should be allowed to feed under trees in which the weevil is present, so as to devour any infested nuts which may have been left on the ground. Poultry may also be of assistance in destroying the insects after they have entered the ground to pupate. It is probable that the larvæ in the nuts may be destroyed by fumigating with carbon bi-sulphide. The nuts should be placed in a tight box, and one-half pound for each five hundred cubic feet of space used, allowing them to remain for forty-eight hours.

THE HICKORY SHUCK WORM (*Grapholitha caryana*): Sometimes pecan nuts are attacked, as they approach maturity, by a small, white caterpillar, which mines its way through the shucks of the nuts. This caterpillar is the hickory shuck worm, the larva of a small moth.

But little is known of its life-history, and until more is known of its habits, the best advice that can be given is to gather and destroy the infested nuts by

burning them.

FOOTNOTES:

[M] Gossard.

[N] Hedrick. (See index of literature).

[O] Gossard. (See index of literature).

PART VI.

Uses. Literature.

CHAPTER XVI.

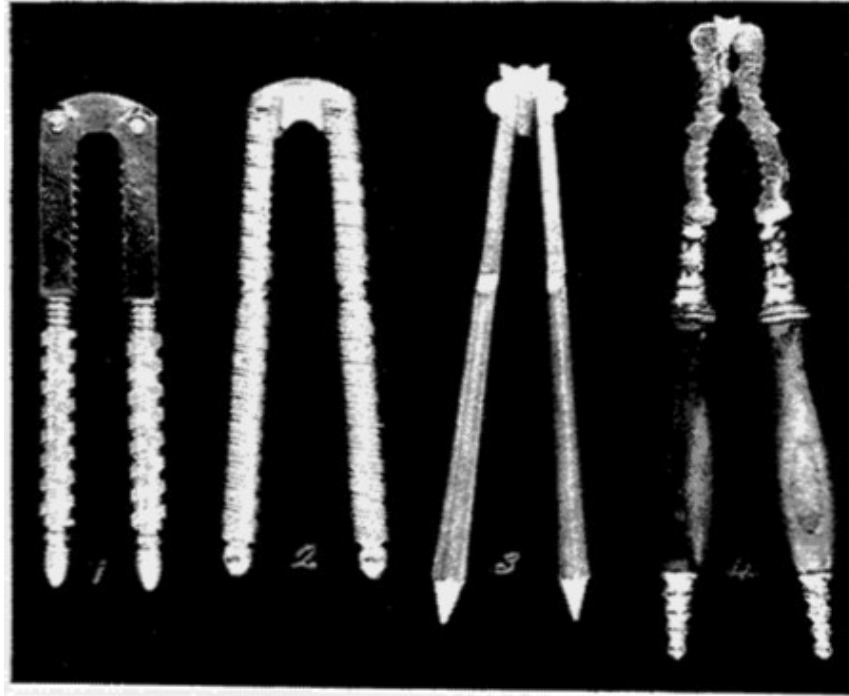
PECAN KERNELS.

Pecan nuts are used in a variety of ways. Not so very long since they were used almost entirely for dessert purposes, now they are largely used in making pastries and confections of different kinds. Based on these uses, new industries for supplying the kernels have been developed. The kernels are now put on the market in glass jars of different kinds and sizes, usually retailing at from 50 cents to 75 cents per pound. This is perhaps the most convenient form in which to buy them, but unfortunately, they are too frequently old and rancid. When stock is carried through the heat of summer in the ordinary jar, this is invariably the case, and some new method of packing them must be introduced if this way of disposing of the product is to increase in favor, as it should. Certain experiments now under way give promise that the kernels can be kept fresh and free from rancidity indefinitely.

For the present, at least, the only certain way of procuring good, fresh pecan kernels is to procure fresh nuts—those which have been kept over in cold-storage are good—and crack them at the time when they are needed. For the household, an ordinary pair of nut-crackers will answer, but they should be of a particular type. The jaws should be formed with sharp-cutting edges.

NUT-CRACKERS.

In the accompanying illustration, four kinds of nut-crackers are shown. The two at the right are reversible. The best pair is represented at the extreme left of the engraving. The bars are square, the grooves in them are curved inward leaving the teeth sharp and pointed out flush with the edge.



From American Nut Journal, Petersburg, Va.

Fig. 37.
Nut Crackers of different types.

To remove the kernels without breaking, grasp the nut with the crackers as close to the end as possible, and gently but firmly apply sufficient pressure to force the sharp teeth of the crackers into the shell. Revolve the nut and repeat the operation until the end is marked with a ring of indentations. Then apply a little greater pressure to start a slight crack, and follow the crack around until the end of the shell drops off. Treat the opposite end in the same way. Next, place the nut lengthwise between the crackers, so they will grasp the side, having the backs of the two halves of the kernel, not the space between the halves, towards the bars. This must be emphasized, because, if pressure is applied at right angles to the edges of the halves instead of against their backs, the chances are that they will be broken when the shell is broken. Having the crackers in position, apply sufficient pressure to crack the shell. Shift the crackers a little to one side of the crack, apply pressure again and a piece of the shell breaks out. A few gentle squeezes will remove the remainder of the shell and the kernel drops out intact.

A hand-power cracker, capable of quite efficient work, is manufactured by Thomas Mills & Bro., Philadelphia, Penn. It has a capacity of one hundred pounds per day, and is capable of giving ninety per cent. of perfect halves.

For factory use, two machines, for extracting kernels at a rapid rate, have been invented, one by Mr. Robert E. Woodson, St. Louis, Mo., and the other by Mr. Grim, New York city. These make it possible to extract pecans in large quantities for commercial purposes. The nuts are fed into a hopper and the machine then takes care of them. In regard to the Woodson machine shown in the adjoining illustration, the inventor says that "in cracking one hundred pounds of nuts there were obtained 39-1/2 pounds of perfect halves and 3-1/2 pounds of broken pieces. This test shows 92 per cent. of perfect halves. I do not claim that this result may be obtained at all times and under all conditions, for the hardness of the shell and the dryness of the nuts make a difference in the results."

Pecans which have become somewhat dry should be soaked in water over night. This renders them much more easily cracked.

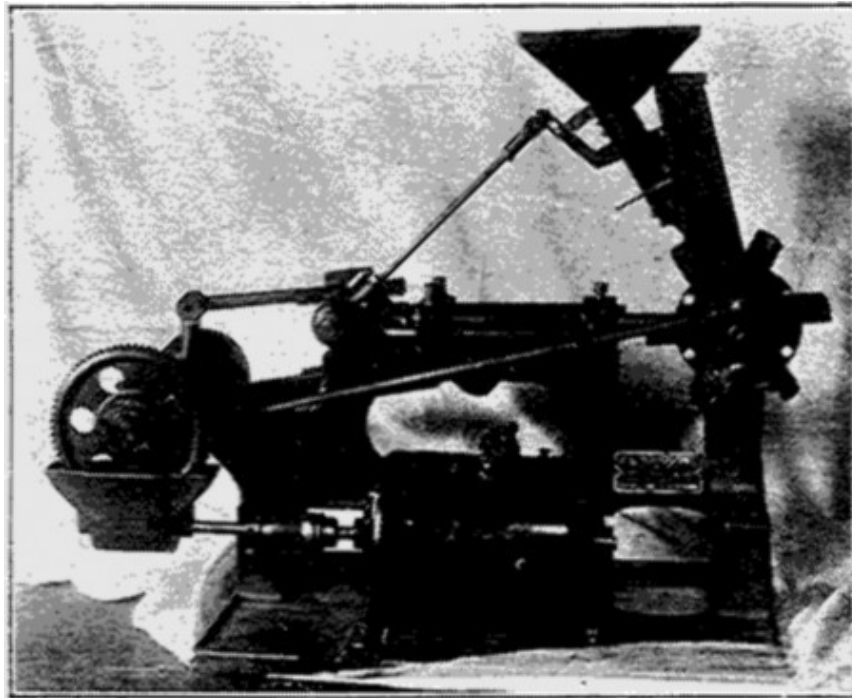


Fig. 38.
Woodson's Power Kernel Extractor.

PECAN OIL.

Oil extracted from almonds, peanuts, cocoanuts and other nuts is now used for various purposes, and at no distant time it is probable that pecan oil may also be placed on the market. Only the cheaper, inferior grades of nuts can be used in

oil-making, as the larger and better quality of nuts are worth too much for dessert purposes.

Ordinary nuts will run about fifty per cent. kernels, and these kernels analyze about seventy per cent. oil or fat. On this basis one hundred pounds would give approximately thirty-five pounds of oil. Of course the better grades of nuts will give sixty per cent. kernels, and would consequently yield more oil.

Pecan oil might be used as a salad oil. It might be put to other culinary uses, as well as finding a possible place among medicinal oils.

CHAPTER XVII.

PECAN LITERATURE.

But little has been written on the culture of the pecan. The following brief list of bulletins, articles or chapters in general works, comprises practically all that has appeared from the pens of American writers:

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